Topology-aware spatial attribution in traffic assignment: Structural guarantees, adaptive summaries, and response-supervised learning

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ARTICLE INFO

Keywords:
spatial attribution
traffic assignment
origin–destination demand
network topology
response-supervised learning
explainable artificial intelligence

ABSTRACT

Origin–destination demand changes alter network outcomes through spatially heterogeneous link-flow responses. Existing explanations are either a single system-wide value or a link-level map, leaving no principled intermediate spatial scale. We develop a topology-aware framework that represents each OD effect as a boundary-attribution profile and reports the coarsest profile satisfying a prescribed error tolerance, for both reference and learned traffic assignment. At a fixed flow state, the profile is constant over all two-sided connected regions for every conservative response if and only if the marginal outcome weights agree on the biconnected edge blocks and bridges traversed by the queried OD path; weights elsewhere are unrestricted. When a single value is insufficient, the divergence of the edge-attribution column yields the exact profile range, the minimax constant, minimum-segment summaries for ordered boundary families, and an error budget separating learned-model error from spatial compression error. On Anaheim, 83–211 boundary values are represented by 4–14 segments at a 5% family-scale tolerance. On 96 Sioux Falls test paths, supervising demand-to-flow responses reduces median attribution-column error from 21.18% to 14.19% and family-boundary error from 8.74% to 5.75%; at every matched segment count on the pre-specified tolerance grid the final error is lower, with paired intervals excluding zero. The framework provides physically valid, error-controlled, and learnable spatial explanations of traffic-assignment responses.

1. Introduction

An OD demand change may have a clear network-wide cost effect while producing different signed contributions at successive screenlines. For planning, the missing information is not only how much the demand change costs, but where the contribution changes and whether that change is produced by the studied traffic or by rerouting elsewhere; demand-management impact tracing, corridor and screenline impact assessment, regional network assessment, and the location of background-rerouting effects all ask this spatial question. A system-wide scalar is too coarse, whereas an edge map is often too detailed. We therefore represent OD demand effects at the coarsest spatial resolution that preserves their signed boundary attributions within a prescribed error.

The explanation object is the *boundary-attribution profile*: the signed aggregation of an OD-specific edge-attribution column over the links crossing each screenline or regional boundary, read as a function over the boundary family.

What the profile measures. For a region containing the origin, $\phi_j(U)$ is the signed OD-specific change in the selected link outcome across the region's boundary: moving the boundary reveals where that contribution is created, attenuated, or reversed by congestion and network-wide rerouting. It is not a cost inside the region, and it is not a count of the OD's own vehicles; it is the signed attribution of the chosen outcome on the boundary's crossing links. The network-wide contribution C_j, attribution on roads within a region, and boundary attribution $\phi_j(U)$ answer distinct questions.

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The objective is not to force distinct boundaries to agree, but to identify the coarsest spatial explanation that preserves the original attribution within a stated error.

For learned assignment models the framework supplies what accurate prediction alone cannot, and each of its three parts plays one role for learned explanations: the topology theorem is the *structural certificate* that a learned attribution can admit one transferable value; the node-divergence representation is the *exact validation tool* that measures learned-reference error over every boundary at once; and response supervision is the *training mechanism* derived from that error representation. Exact conservation of the learned map transfers directly to flow-valued integrated gradients; our analysis identifies the additional topological and outcome-weight conditions required for this conservation to persist after outcome weighting, and quantifies the spatial profile when those conditions are not imposed. Trustworthiness in this paper therefore has three concrete meanings: physical validity, through exact flow conservation of the learned assignment map; representational reliability, through explicit spatial approximation tolerances on every reported summary; and model fidelity, through exact or bounded disagreement between learned and reference explanations. Structural compatibility and compression error are certified from the learned model and the reported profile alone, whereas learned-reference fidelity is an offline validation guarantee requiring reference assignments. Physical penalties, output projections, and conservative parameterizations constrain predicted flows (Raissi et al., 2019; Shi et al., 2021; Valente et al., 2025; Richter-Powell et al., 2022); Section 7 measures how much each conservation mechanism contributes to attribution-level guarantees under matched pretraining.

Transportation sensitivity analysis asks how changes in origin–destination (OD) demand affect link flows and system outcomes (Tobin and Friesz, 1988; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). For a differentiable assignment model or learned surrogate, integrated gradients allocate a finite outcome change across OD demands (Sundararajan et al., 2017). Resolving this allocation through individual link flows additionally locates the contribution on the network — and raises the spatial question above, because different aggregations of the same edge attributions need not agree.

The question differs from tracking physical OD vehicles through screenlines or cordons (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002): an OD-indexed sensitivity describes the network-wide response, including other ODs' route changes. Flow conservation transfers to flow-valued attribution, but outcome weighting introduces additional topology-dependent conditions (Section 3).

We take the boundary-attribution profile as the primary spatial explanation, and the paper asks three questions about it. *When does this profile collapse exactly to a single transferable value?* A topology theorem answers with a query-dependent structural condition. *When it does not, what is its least-complex representation under a prescribed error?* A node-divergence representation returns the exact boundary range, the minimax common summary, and error-controlled segmentations of ordered boundary families. *How can a learned assignment model be trained to reproduce the resulting profile accurately?* Response supervision acts on the discrepancy identified by the error bound. The same order organizes the paper; Figure 1 traces the full pipeline from a demand change to a certified, error-controlled summary, for reference and learned assignment alike.

Contributions. This paper makes one overarching contribution: it turns the spatial attribution of OD demand effects from an arbitrary boundary aggregation into a mathematically controlled representation for both reference and learned traffic assignment. First, we characterize exactly when a boundary-attribution profile admits one transferable value. For two-sided connected regions, this occurs for every conservative response if and only if the marginal outcome weights agree on the biconnected edge blocks and bridges traversed by the queried OD path; weights outside that path are unrestricted — the principal structural result of the paper. Second, when one value is insufficient, node divergence yields the exact profile range and the coarsest ordered summary satisfying a prescribed approximation tolerance, together with separate model and compression errors — the operational and reporting consequence of that result. Third, the same error representation identifies demand-to-flow response mismatch as the trainable source of learned explanation error and leads to response supervision, which improves spatial attribution on both Sioux Falls and Anaheim without sacrificing flow accuracy — its learning consequence.

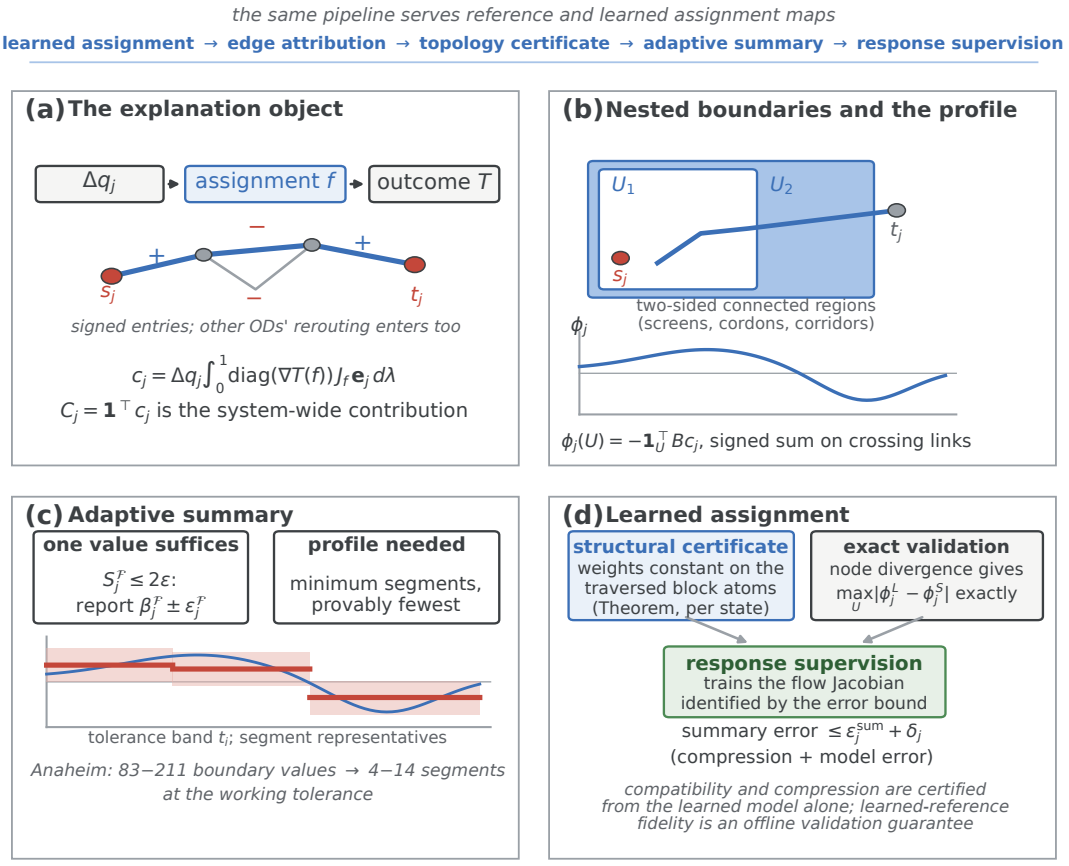


Figure 1: The paper as one pipeline. (a) An OD demand change is propagated through an assignment map — a reference SUE model or a conservative learned surrogate — and weighted by the outcome's marginal weights, giving the signed edge-attribution column c_j; rerouting of other ODs enters the same column. (b) Nested two-sided connected boundaries turn the column into the boundary-attribution profile $\phi_j(U)$. (c) At a prescribed tolerance the profile is reported either by one transferable value (when its certified range is small enough) or by the minimum-segment summary, each value carrying its explicit error. (d) For learned assignment, the topology theorem supplies the structural certificate, node divergence validates the learned column against a reference exactly, and response supervision trains the discrepancy the error bound identifies; the reported summary carries a total budget of compression plus model error. Compatibility and compression are certified from the learned model alone; learned-reference fidelity is an offline validation guarantee.

2. Related work

Attribution and cost allocation. Our edge-level object is input-specific conductance (Dhamdhere et al., 2019; Lundstrom et al., 2022): demand is the input, link flow is an intermediate variable, and the traffic outcome is the output. Its sum over links gives the integrated-gradient contribution of an OD, and continuous marginal cost allocation underlies Aumann–Shapley sharing (Aumann and Shapley, 1974; Billera et al., 1978). In transportation applications, attribution typically explains a prediction model by ranking its input features (Sarıyer et al., 2024; Guo et al., 2024), whereas the object attributed here is a link flow that must satisfy node balance. Existing scalar attribution methods do not characterize how a fixed OD contribution changes across physical network boundaries; that additional property is what we analyze.

Conserved attribution flows on graphs. Graph-based explanation methods construct attribution flows with boundary-consistency or conservation properties on directed acyclic graphs: Shapley Flow makes edge attributions consistent across different choices of explanatory boundary, and the recursive Shapley value

recursively constructs edge-attribution flows that conserve at internal nodes (Wang et al., 2021; Singal et al., 2021). Those works design new attribution rules satisfying such axioms on computational or causal graphs; the present problem instead asks when an already specified traffic-outcome attribution is boundary-consistent on a cyclic physical network, uniformly over flow-conserving responses.

Traffic sensitivity and learned assignment. Equilibrium sensitivity analysis studies demand-to-flow derivatives and their aggregation (Tobin and Friesz, 1988; Yang and Bell, 2007; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). Learned assignment maps, implicit optimization layers, and logit equilibrium metamodels make such derivatives available to explanations (Liu et al., 2023; Amos and Kolter, 2017; Lassen et al., 2025); training on derivative targets is Sobolev training (Czarnecki et al., 2017; Yuan et al., 2022); fully differentiable formulations propagate these gradients end to end (Li and Nie, 2026; Seo, 2026), graph-attention surrogates embed node-based flow conservation in the training loss (Liu and Meidani, 2024, 2025), and expressivity and learning guarantees for end-to-end user-equilibrium models are established in Liu et al. (2025). These studies primarily evaluate flows, efficiency and conservation; physical OD tracking through assignment and cordon counts concerns commodity flows (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002). None quantify whether OD-indexed spatial explanations are transferable or how their boundary error should be controlled; the present work supplies both.

Conservation and graph structure. Physics-informed losses and conservative architectures constrain physical outputs (Raissi et al., 2019; Shi et al., 2021; Richter-Powell et al., 2022); projections can enforce constraints exactly (Valente et al., 2025; Baez et al., 2024). We examine their transfer to flow and outcome attributions; for approximate conservation, derivative bounds of Landau–Kolmogorov type are classical (Shadrin, 2014), and Appendix A.4 transfers such control to attribution residuals. The relevant circulation space and its decomposition into circuit classes are classical (Oxley, 2011; Hopcroft and Tarjan, 1973). Classical block and circulation results characterize graph structure; they do not characterize when an outcome-weighted OD response has one transferable value over a specified family of traffic regions. The new classification couples the classical decomposition to queried OD directions and their transitive equality constraints.

3. Boundary attribution of OD demand effects

3.1. Edge attribution and the spatial aggregation question

Let $\mathcal{G} = (V, E)$ be connected, with incidence matrix $B \in \{-1, 0, 1\}^{n\times m}$ (negative at a link's tail), OD vectors $\mu_j = e_{t_j} - e_{s_j}$, $s_j \neq t_j$, and $M = [\mu_1, \ldots, \mu_K]$. Reciprocal and parallel arcs remain distinct edges when orientations are forgotten. Let $f : \Omega \subset \mathbb{R}^K \to \mathbb{R}^m$ map demands to flows on an open demand domain Ω, and $T : \mathbb{R}^m \to \mathbb{R}$ be a specified edgewise outcome. Assume both maps are C^1 on open neighborhoods of the relevant paths. The construction reads as follows: a demand change is first propagated to link-flow responses, then weighted by the marginal contribution of each link to the selected outcome; summing over all links gives the OD's system contribution, and summing with boundary orientation gives its signed boundary attribution. With $\Delta q = q^1 - q^0$ and $\gamma(\lambda) = q^0 + \lambda\Delta q$, define

$$g_j = \Delta q_j \int_0^1 J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \tag{1}$$

$$c_j = \Delta q_j \int_0^1 D_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \quad D_\lambda = \operatorname{diag}(\nabla T(f(\gamma(\lambda)))), \tag{2}$$

$$C_j = \mathbf{1}^\top c_j, \qquad \sum_j C_j = T(f(q^1)) - T(f(q^0)). \tag{3}$$

The vector g_j attributes flow changes; c_j resolves the outcome contribution through links; C_j is its scalar OD contribution. Both vectors are signed system sensitivities, including other ODs' route adjustments, rather than physical commodity flows. Entry $c_{e,j}$ is input-specific conductance through link e (Dhamdhere et al., 2019).

If $Bf(q) = Mq$, differentiation and integration give

$$BJ_f(q) = M, \qquad Bg_j = \mu_j \Delta q_j. \tag{4}$$

The same conclusion holds for $Bf(q) = Mq + b$ with constant background demand balance b. Outcome attribution is *OD cut consistent* when $Bc_j = \beta_j \mu_j$ for some $\beta_j \in \mathbb{R}$. The scale β_j is a boundary value, generally different from C_j: on two series links with unit weights, $C_j = 2\Delta q_j$ but $\beta_j = \Delta q_j$. Outcome weighting is what makes cut consistency a genuine restriction: on two series links, a conservative unit response $(1, 1)$ with weights (w_1, w_2) balances at the intermediate node only if $w_1 = w_2$, although the response itself conserves exactly.

For $s_j \in U \subseteq V \setminus \{t_j\}$, write $b_j = Bc_j$ and define the boundary attribution $\phi_j(U) = -\mathbf{1}_U^\top b_j$. Any two separating sets satisfy

$$\phi_j(U') - \phi_j(U) = -\sum_{v \in U' \setminus U} b_{jv} + \sum_{v \in U \setminus U'} b_{jv}; \tag{5}$$

for nested sets $U \subseteq U'$ the second term vanishes, leaving $\phi_j(U') - \phi_j(U) = -\sum_{v \in U' \setminus U} b_{jv}$. The difference between two boundaries is thus the enclosed imbalance, with incoming and outgoing regions entering with opposite signs. For a separable outcome $T(F) = \sum_e g_e(F_e)$, the attribution itself also has a direct reading. With $\delta^+(U)$ and $\delta^-(U)$ the arcs leaving and entering U,

$$H_U(F) = \sum_{e \in \delta^+(U)} g_e(F_e) - \sum_{e \in \delta^-(U)} g_e(F_e), \qquad \phi_j(U) = \mathrm{IG}_j[\, H_U \circ f \,], \tag{6}$$

because each entry $c_{e,j}$ of Eq. (2) is the integrated gradient of the scalar outcome $g_e \circ f$, and the incidence signs turn $-\mathbf{1}_U^\top b_j$ into the signed boundary sum of H_U. For TSTT, $\phi_j(U)$ therefore attributes the demand change to the excess of the total travel cost on the region's outgoing arcs over that on its incoming arcs: a boundary-related outcome in its own right, neither the region's internal cost nor a vehicle count.

Proposition (cut-consistency equivalence)**.** *OD cut consistency holds exactly when every separating cut has one common signed value $\phi_j(U) = \beta_j$.*

Indeed, balance implies this identity. Conversely, comparing $\{s_j\}$ with $\{s_j, v\}$ forces zero divergence at every interior node; the two endpoint entries are then opposite because incidence columns sum to zero. The three spatial summaries of one attribution — the system-wide value $C_j = \sum_e c_{e,j}$, attribution $\sum_{e \in E_R} c_{e,j}$ on a road set E_R specified before aggregation, and the boundary value $\phi_j(U)$ — answer the three questions of Figure 1(a): how much, where on specified roads, and what signed aggregation on a specified boundary.

The road partition is part of the attribution object: subdivision changes which crossing-link costs a boundary includes (Example A.8).

Throughout, $\mathcal{U}_j^{\mathrm{all}}$ contains all sets separating s_j from t_j; $\mathcal{U}_j^{\mathrm{conn}}$ restricts both induced sides to be connected in the underlying undirected network; and $\mathcal{U}_j^{\mathrm{geo}}$ is a specified finite family of straight screenline regions, filtered for two-sided connectivity. The unrestricted extrema below are closed-form exact values. Sioux Falls connected extrema use complete enumeration. Geometric extrema and Anaheim's selected connected-region ranges are exact only over their stated finite families, and do not determine extrema over all connected regions.

3.2. Boundary families, response classes, and explanation accuracy

A common boundary value is a property of two objects at once: which boundaries are allowed, and which demand responses are quantified over. We therefore state the transferability questions along these two axes. *Boundary families*: all separating regions $\mathcal{U}_j^{\mathrm{all}}$; two-sided connected regions $\mathcal{U}_j^{\mathrm{conn}}$; or a prescribed finite family such as a screenline set. *Response classes*: every conservative response at fixed weights; the path-feasible response envelope of one assignment model; or a single realized attribution column. The graph-topological classification of Section 4 covers the first two families under all conservative responses; prescribed finite families admit an exact linear-algebraic test of the same universal kind (Proposition 1 below), and model envelopes substitute for the conservative class inside any of them. Four levels result, ordered by what varies.

The *structural* level asks whether one value is guaranteed for all flow-conserving responses and is answered by the topological condition of Section 4; the *model-specific* level asks the same for the path-feasible response envelope of one assignment model and is answered by the linear-algebraic test of Proposition 1; the *realized* level asks how much the boundaries of one attribution column differ and is answered by the exact range and profile of Section 5; and the *learned-reference* level quantifies the distance between two realized columns, reported separately from the consistency properties (Section 5.5).

Proposition 1 (exact family criterion)**.** *Fix a query j, a nonempty boundary family $\mathcal{F} \subseteq \mathcal{U}_j^{\mathrm{all}}$, and an affine class of edge responses $x = x_j^{\mathrm{ref}} + \mathcal{S}$ with basis $N_{\mathcal{S}}$ ($\mathcal{S} = \ker B$ recovers all conservative responses). Let $A_{\mathcal{F}} = [-\mathbf{1}_U^\top B]_{U\in\mathcal{F}}$ stack the oriented attribution functionals. All responses in the class give one common summary α_j on $\mathcal{F}$ — that is, $-\mathbf{1}_U^\top B D x = \alpha_j$ for every $U \in \mathcal{F}$ and every $x \in x_j^{\mathrm{ref}} + \mathcal{S}$ — if and only if*

$$A_{\mathcal{F}} D N_{\mathcal{S}} = 0 \qquad \text{and} \qquad A_{\mathcal{F}} D x_j^{\mathrm{ref}} \in \operatorname{span}\{\mathbf{1}\}, \tag{7}$$

in which case α_j is the common entry of $A_{\mathcal{F}} D x_j^{\mathrm{ref}}$.

Substituting $x = x_j^{\mathrm{ref}} + N_{\mathcal{S}} z$ makes the attribution affine in z; the identity holds for every z exactly when the z-coefficient block $A_{\mathcal{F}} D N_{\mathcal{S}}$ vanishes and the constant block has constant rows. On the full separating family the criterion reproduces the classification theorems, which characterize when (7) holds for every weight profile without enumerating $\mathcal{F}$; on a prescribed screenline set it is the exact universal test, computable from one sparse product per family member. For a fixed assignment model, the envelope $x_j^{\mathrm{ref}} + \mathcal{S}$ collects the path-feasible responses that the fixed path sets can express; the tests are then sufficient for the actual assignment map, and necessity for a specified equilibrium map would require a separate attainability argument, which we do not claim. On Sioux Falls, whose full-background fixed-path model allows path exchanges spanning the complete circulation space ($T_{\mathrm{swap}} = \Delta \ker A_{\mathrm{OD}}$ of dimension $m - n + 1$), the path-exchange envelope coincides with the affine space of conservative responses: for this path-feasible response envelope, the model-specific criterion coincides with the unrestricted conservative-response criterion, so the universal condition is tight rather than conservative for that path structure; restricted path systems separate the two levels (Appendix B.1, Table C.4). A realized attribution can balance without satisfying the universal conditions; pointwise balance along a demand path implies integrated balance, and the converse can fail through cancellation. Appendix A.3 collects these examples and the directionality conditions under which weaker balance notions coincide with the classified property.

4. Topological conditions for transferable spatial attribution

Take the block-cut tree of the underlying undirected multigraph. Its non-bridge blocks, together with each bridge as a singleton, are the *block atoms*. They are the classical graphic-matroid circuit classes, with each bridge retained separately. The *OD block path* $\mathcal{P}(s,t)$ is the atom sequence on the unique block-cut-tree path between the OD endpoints. For several queries, merge all atoms traversed by each $\mathcal{P}_j = \mathcal{P}(s_j, t_j)$ and take the transitive closure. The resulting edge sets $\bar{E}_1, \ldots, \bar{E}_{K'}$ are the *OD-compatibility classes*; atoms on no queried path remain separate.

4.1. Main result: two-sided connected traffic regions

Traffic analysis typically aggregates on screenlines, cordons and regional boundaries whose two sides are each spatially connected. The main theorem shows that this restriction strictly localizes the condition for a common boundary attribution to the queried OD block path; the magnitude of the resulting freedom is topology- and query-dependent. Let

$$\mathcal{U}_{st}^{\mathrm{conn}} = \{U \subset V : s \in U,\ t \notin U,\ \mathcal{G}[U] \text{ and } \mathcal{G}[V \setminus U] \text{ are connected}\},$$

where connectivity uses the underlying undirected multigraph. These are separating bonds: both sides of the boundary form connected regions.

Main implication. For two-sided connected traffic regions, transferable OD boundary explanations require uniform marginal weighting only along the block atoms traversed by the queried OD paths. Heterogeneity elsewhere in the network is immaterial to this guarantee.

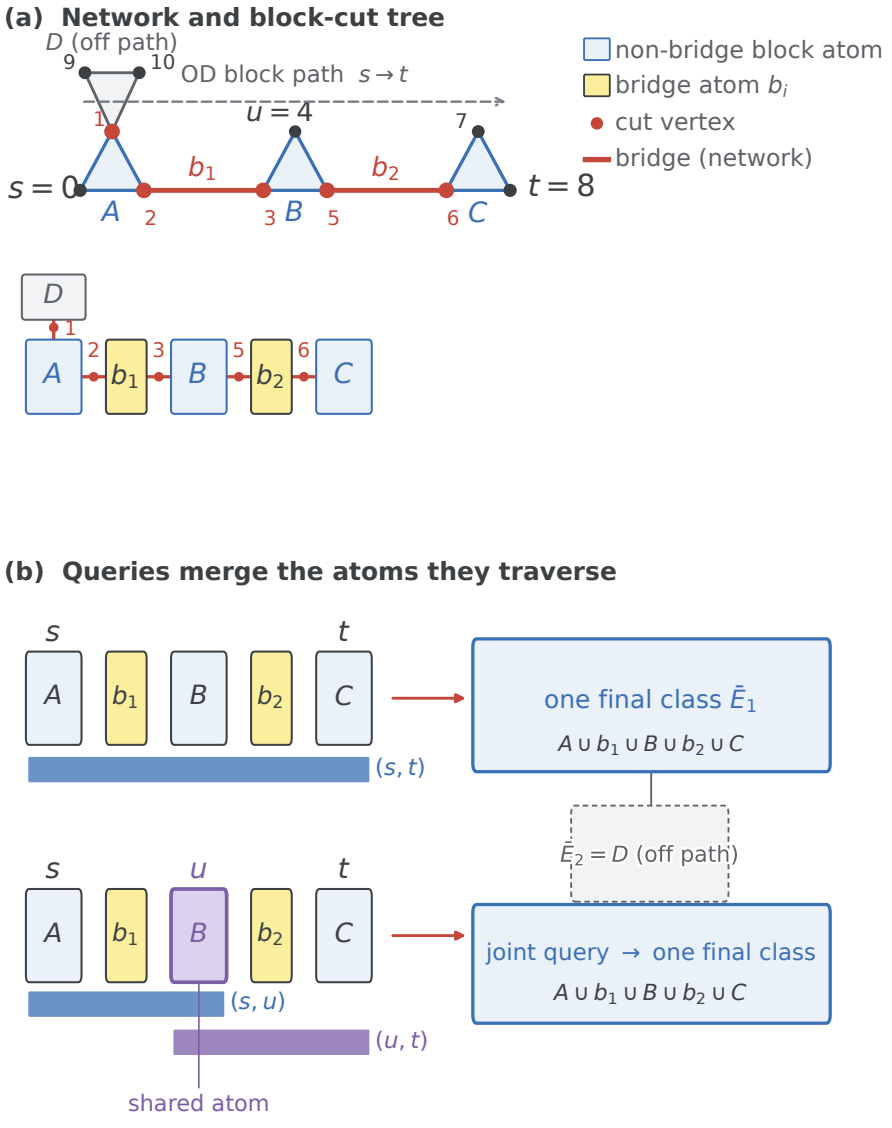


Figure 2: (a) A two-bridge network with its block-cut tree: non-bridge block atoms A, B, C, D and bridge atoms b_1, b_2 joined at shared cut vertices; the queried OD path runs $s \to t$ and the pendant atom D hangs off the path. (b) OD queries merge the atoms they traverse: the single query (s, t) merges $A \cup b_1 \cup B \cup b_2 \cup C$ into one class while D stays separate; two queries (s, u) and (u, t) sharing only atom B merge the same full chain transitively. In the connected-region example below, atom D carries heterogeneous weights $3, 4, 5$ — invisible to the queried connected boundaries but violating unrestricted transferability; the unrestricted-family counterpart with uniform pendant weights appears as Example B in Appendix A.2.

Theorem 2 (two-sided connected-region classification)**.** *For a connected loopless multigraph, $\mu = e_t - e_s$, and $D = \mathrm{diag}(w)$, the property*

$$\exists \alpha \in \mathbb{R} \quad \forall x \text{ with } Bx = \mu \quad \forall U \in \mathcal{U}_{st}^{\mathrm{conn}} : -\mathbf{1}_U^\top BDx = \alpha \tag{8}$$

holds if and only if every edge in the atoms of $\mathcal{P}(s, t)$ has one common weight α. Weights of edges outside those atoms are arbitrary, even within an off-path block atom.

The proof in Appendix A.6 uses connected prefix and suffix regions within each traversed atom. Each admissible boundary crosses edges in just one such atom, which explains why off-path heterogeneity is

invisible to this region family. This is a universal fixed-weight statement; agreement for one realized response need not imply its conditions.

Corollary 3 (several queries on connected regions)**.** *At one shared weight vector, Eq.* (8) *holds for all queried ODs if and only if edges in each group of atoms joined by the queried block paths have one common weight. An atom on no queried path has no weight restriction, including no within-atom restriction.*

Thus restricting attention to connected traffic regions permits additional outcome heterogeneity. It changes the compatibility classification, while the unrestricted extrema and error formulas of Section 5 retain their own stated boundary family.

4.2. Design implications

The theorem gives three design freedoms, each checkable before any model is fitted.

Query design. Only the atoms traversed by the queried OD paths enter the condition, and adding queries merges atoms transitively through shared blocks. On Figure 2 ($s = 0$, $u = 4$, $t = 8$, linear outcome $T_w(F) = w^\top F$), the query (s, u) joins A, b_1, B while b_2, C and D keep their own constants; adding (u, t) joins the remaining chain through the shared atom B; adjusting only b_2 and C then restores a common summary for both queries while the unrelated atom D keeps weight 3. Appendix A.2 tabulates the three stages with exact profile ranges.

A constructive distinction on Figure 2 (Example A). Keep weights 1 on A, b_1, B, weights 2 on b_2, C, but set the three off-path D weights to $3, 4, 5$. The first query (s, u) admits common boundary value 1 on every two-sided connected region for every conservative response. The second query (u, t) initially has connected boundary values spanning $[1, 2]$ and requires a segmented profile. Changing only b_2, C to weight 1 makes both queries compatible on connected regions, while D remains internally heterogeneous — permissible for connected transferability, since no queried connected boundary sees inside D. The model-specific diagnosis of Table C.4 verifies exactly this post-merge configuration in its synthetic row. The unrestricted theorem makes a stricter demand on the same network; that counterpart, with uniform pendant weights, is Example B below.

Outcome design.

Corollary 4 (linear outcome functionals)**.** *Let $f \in C^1$ be exactly conservative on an open neighborhood of the attribution path, and let $T(F) = a^\top F$ be a linear functional of realized flows. Then $\nabla T \equiv a$, so structural cut consistency over the unrestricted separating family holds if and only if a is constant on the compatibility classes, with common cut value $\beta_j = \Delta q_j \, a|_{\bigcup \mathcal{P}_j}$; over two-sided connected regions the weaker condition of Theorem 2 applies instead — a constant on the block atoms traversed by the queried OD paths, with no constraint on off-path atoms. Both are conditions on the outcome definition alone, checkable by inspection before any model is trained, and the pointwise classification holds at every λ.*

Nonlinear compatible outcomes also exist: if $T(F) = \Phi(\sum_{e \in \bar{E}_1} F_e, \ldots, \sum_{e \in \bar{E}_{K'}} F_e)$, its gradient is class-constant at every flow. More familiar traffic outcomes require equality of their actual marginal weights. The Beckmann potential has weights $t_e(F_e)$; total system travel time (TSTT) has weights $t_e(F_e) + F_e t'_e(F_e)$; a separable index $\sum_e g_e(F_e)$ has weights $g'_e(F_e)$. Heterogeneous link parameters and congestion typically violate these equalities. A single-link outcome has a nonzero common boundary value only when the queried OD block path is precisely that singleton bridge.

Region-family design. The admissible family is part of the attribution question, and the theorem quantifies what each choice costs: the connected family ignores off-path heterogeneity, the unrestricted family does not, and a prescribed finite family such as a screenline set admits the exact universal test of Proposition 1 in place of a graph-topological classification. On public road networks the size of that freedom is itself measurable from topology alone, and it is largest for corridor-type and articulation-adjacent queries and smaller in large biconnected cores. Across every positive-demand OD pair, the share of edges the connected condition constrains — against 100% for the unrestricted condition — has median 100% on Sioux Falls (one

atom), 95.4% on Anaheim, 95.3% on Winnipeg and 89.1% on Eastern-Massachusetts. The distribution is tightly concentrated on the first three networks (Anaheim 10–90% quantiles 95.4–96.1%), whereas Eastern-Massachusetts carries a genuine peripheral tail: 2.2% of OD pairs constrain fewer than 75% of edges, down to 0.8% for OD pairs localized in a single peripheral block (Appendix B.1, Table B.2). Where the release is small, the realized summaries of Section 5, not the structural classification, carry the reporting.

4.3. A stronger criterion for all separating sets

When every separating set — regular or irregular — must share one boundary value, the condition tightens and reaches beyond the queried paths.

Theorem 5 (single-OD classification, static form)**.** *Let $\mathcal{G}$ be a connected multigraph with one OD pair (s,t), $s \neq t$, let $w \in \mathbb{R}^m$, and let $D = \mathrm{diag}(w)$. The* response-independent OD cut consistency property

$$\exists\, \alpha \in \mathbb{R} : \qquad BDx = \alpha\, \mu \quad \text{for every } x \in \mathbb{R}^m \text{ with } Bx = \mu, \tag{9}$$

with the scale α independent of the response x, holds ***if and only if***

(i) *w is constant within every non-bridge atom; and*
(ii) *all atoms on the s–t block path $\mathcal{P}(s,t)$ share one common value.*

In that case every conservative response induces the same boundary vector: $BDx = w^*\mu$ for every x with $Bx = \mu$, where $w^* := w|_{\bigcup \mathcal{P}(s,t)}$ is the common value of (ii) and $\alpha = w^*$ is the scale in (9).

Condition (i) concerns every atom, including those off the OD path. Their constants may differ; only the traversed atoms must share the scale w^*. Off-path atoms are constrained because a signed circulation can be added to any conservative response, and heterogeneous weights can turn it into node divergence. Connected regions change this conclusion, as Theorem 2 shows. The proof adds a virtual edge from t back to s (Appendix A.2): conservative OD responses then become circulations on the extended graph, so the boundary vector is response-independent exactly when diagonal weighting preserves the circulation space — equivalently, when every weighted cycle has zero node divergence — which forces equal weights within each block atom and one shared weight along the atoms the OD block path connects.

Theorem 6 (multi-OD classification, static form at a shared realized flow state)**.** *Let $\mathcal{G}$ be connected, let each OD pair satisfy $s_j \neq t_j$, and let $w \in \mathbb{R}^m$, $D = \mathrm{diag}(w)$. The response-independent OD cut consistency property (9) — for each queried j, some $\alpha_j \in \mathbb{R}$ with $BDx = \alpha_j \mu_j$ for every x with $Bx = \mu_j$ — holds simultaneously for the queried ODs* ***if and only if*** *w is constant on each OD-compatibility class $\bar{E}_k$.*

The classification concerns all signed responses compatible with vehicle conservation; an equilibrium model generally realizes only a subset of these responses (Appendix A.3). All queries use the same realized weight vector. Queries with $\Delta q_j = 0$ have zero attribution columns and need not enter the closure. Requiring all ordered pairs of distinct vertices collapses the classes to one per connected component (Corollary A.4).

4.4. Pathwise attribution and certificates for learned assignment

Corollary 7 (path form: endpoint implication)**.** *Let $f \in C^1$ be conservative on an open set, $T \in C^1$, $\gamma(\lambda) = q^0 + \lambda \Delta q$ the straight-line path with $\Delta q \neq 0$, and $w(\lambda) = \nabla T(f(\gamma(\lambda)))$, $D_\lambda = \mathrm{diag}(w(\lambda))$, shared by all participating ODs. If the equivalent conditions of Theorems 5 and 6 hold at $w(\lambda)$ for every $\lambda \in [0,1]$, then, for every participating j simultaneously,*

$$Bc_j = \beta_j\, \mu_j$$

holds with

$$\beta_j = \Delta q_j \int_0^1 w^*_{k(j)}(\lambda)\, d\lambda,$$

where $w^*_{k(j)}$ is the common value on the compatibility class $k(j)$ containing the atoms of $\mathcal{P}_j$.

The specified edgewise outcome and the network representation are both part of the object. On two series links, an outcome whose marginal weights oscillate around unity can integrate to the same balanced column $(1, 1)$ as the linear outcome on one demand segment, yet fail to balance on another (Example A.7); subdividing a link leaves the composite map unchanged but creates interior divergence and even halves the original cut value at the mixed split (Example A.8). Response independence concerns conservative responses on a fixed network; it does not make the attribution invariant to road subdivision.

Corollary 8 (conservative learned assignment)**.** *Let f^L be an exactly conservative learned assignment map on an open domain. Then its flow-valued integrated gradients inherit OD conservation, $Bg_j^L = \mu_j \Delta q_j$. Write $w(\lambda) = \nabla T(f^L(\gamma(\lambda)))$ along the attribution path. For each λ, structural cut consistency of the learned explanations is equivalent to the weight conditions of Theorems 5 and 6 at $w(\lambda)$ for the unrestricted separating family, and to those of Theorem 2 for two-sided connected regions. When the unrestricted-family conditions hold at every λ, the integrated explanations are realized cut consistent, $Bc_j^L = \beta_j \mu_j$ (Corollary 7); when they fail at some λ, the node representation of Section 5 gives the exact spatial explanation range of the learned column over the unrestricted family and its exact maximum boundary disagreement from any reference assignment, with restricted families scanned through the same node sums.*

Proof. The conservation transfer is Eq. (4) applied to f^L. The pointwise equivalences are the two classification theorems applied to the fixed weight vector $w(\lambda)$; the integral claim is Corollary 7 with $D_\lambda = \mathrm{diag}(w(\lambda))$; the general-profile bound is Corollary 12. □

Conservation of predicted flows is thus the first step of a chain whose later steps are topological; Section 7 evaluates the whole chain empirically.

5. Adaptive spatial summaries and error guarantees

Procedure 1 summarizes the full pipeline from an OD demand change to a reported explanation; the subsections that follow derive each step in turn.

5.1. Obtaining attribution columns

For a separable outcome $T(F) = \sum_e g_e(F_e)$, a single changing OD gives

$$c_{e,j} = g_e(F_e^1) - g_e(F_e^0). \tag{10}$$

Along this demand path, $dF_e/d\lambda = \Delta q_j \partial_j f_e$; the fundamental theorem of calculus applies link by link, including endogenous background rerouting. Two endpoint assignments therefore recover the attribution column without sensitivity integration. The boundary attribution is $H_U(F^1) - H_U(F^0)$. For jointly changing ODs, endpoint differences determine only the sum of attribution columns. Sensitivity integration in Eq. (2) separates the individual OD contributions. Both cases then use the same node representation and spatial aggregation (Procedure 1) below. After preprocessing, all boundary values for a prescribed screenline family are obtained from one incidence multiplication and a linear scan; numerical verification and same-task timings against endpoint and vectorized alternatives are collected in Appendix B.6.

5.2. Node representation and exact profile ranges

Once the classes are constructed, structural cut consistency over the unrestricted family of all separating regions is tested by

$$\eta_{\mathrm{all}}(\lambda) = \max_k \max_{e \in \bar{E}_k} |w_e(\lambda) - \bar{w}_k(\lambda)|, \qquad \bar{w}_k = |\bar{E}_k|^{-1} \sum_{e \in \bar{E}_k} w_e, \tag{11}$$

and the joint condition of Theorem 6 is exactly $\eta_{\mathrm{all}} = 0$. For two-sided connected regions, let $E_Q = \bigcup_j \bigcup_{A \in \mathcal{P}_j} A$ collect the edges of all queried block paths and define

$$\eta_{\mathrm{conn}}(\lambda) = \max_{k:\, \bar{E}_k \cap E_Q \neq \varnothing} \max_{e \in \bar{E}_k} |w_e(\lambda) - \bar{w}_k(\lambda)|, \tag{12}$$

Procedure 1: From an OD demand change to a reported spatial explanation

Input: incidence B, query (s_j, t_j), edge attribution c_j, an ordered boundary family $\mathcal{F}$ with tolerance ϵ_j, selected regions U, W, and optionally a reference column c'_j.

0. For a single changing OD and separable outcome, obtain c_j from the two endpoint states using Eq. (10); for joint changes, input the separately integrated column.
1. Compute $b_j = Bc_j$; return its node distribution.
2. Return $\phi_j(U) = -\mathbf{1}_U^\top b_j$ and $\phi_j(U) - \phi_j(W) = -(\mathbf{1}_U - \mathbf{1}_W)^\top b_j$: the boundary-attribution profile over $\mathcal{F}$.
3. Scan interior nodes to return the extrema in Eq. (15) and the midpoint and half-range in Eq. (17), and their sets $\{s_j\} \cup \{v \in V \setminus \{s_j, t_j\} : b_{jv} > 0\}$ and $\{s_j\} \cup \{v \in V \setminus \{s_j, t_j\} : b_{jv} < 0\}$.
4. Summarize: if $\varepsilon_j^{\mathcal{F}} \le \epsilon_j$ return the common summary $\beta_j^{\mathcal{F}}$; otherwise apply Proposition 10 and return the minimum-segment summary with each segment's position range, representative value and worst error ($\le \epsilon_j$ by construction).
5. If c'_j is supplied, return $\mathcal{K}_j(c_j - c'_j)$ as the exact maximum absolute error over unrestricted separating regions, the family scan over $\mathcal{F}$, and, when step 4 produced a summary, the total budget $\varepsilon_j^{\mathrm{sum}} + \delta_j$ of Eq. (21).

The incidence product and scan cost $O(|E| + |V|)$ per column; selected-region sums cost $O(|V|)$; the segmentation is one linear pass over the family. The same node representation serves all these tasks. The columns must already have been computed; their reference-attribution cost remains separate.

which omits every class on no queried path. By Corollary 3, $\eta_{\mathrm{conn}} = 0$ is exactly the connected-region compatibility condition, and off-path atoms may remain internally heterogeneous, so $\eta_{\mathrm{conn}} = 0$ does not imply $\eta_{\mathrm{all}} = 0$. Class construction uses the block-cut tree and transitive merging; testing a weight vector then needs one edge scan. No fitted-model derivatives are required for a linear outcome.

For a realized integrated attribution, define

$$d_j = (I - P_{\mu_j})Bc_j, \qquad P_{\mu_j} = \frac{\mu_j \mu_j^\top}{\|\mu_j\|_2^2}. \tag{13}$$

Endpoint cut consistency over all separating regions holds precisely when $d_j = 0$; only then does $\mu_j^\top Bc_j / \|\mu_j\|_2^2$ represent a common cut value, whatever mechanism produced the balance (Section 3.2). For a connected family or another prescribed finite family, $d_j = 0$ is sufficient but not necessary; realized consistency is then read from the attribution range over that family.

Without a connectivity restriction, each interior node can enter the separating set independently. Therefore Eq. (5) gives

$$S_j := \max_U \phi_j(U) - \min_U \phi_j(U) = \sum_{v \notin \{s_j, t_j\}} |(Bc_j)_v|. \tag{14}$$

The extrema themselves have a closed form: writing $b_j = Bc_j$ (Eq. (2)), $(x)_+ = \max(x, 0)$, and summing over interior nodes $v \notin \{s_j, t_j\}$,

$$\phi_j^{\min} = \phi_j(\{s_j\}) - \sum_v (b_{jv})_+, \qquad \phi_j^{\max} = \phi_j(\{s_j\}) + \sum_v (-b_{jv})_+, \tag{15}$$

since the minimizing set retains exactly the interior nodes with positive divergence. Subtracting gives, for $I_j = V \setminus \{s_j, t_j\}$,

$$\phi_j(U) - \phi_j^{\min} = \sum_{v \in I_j \setminus U} (b_{jv})_+ + \sum_{v \in I_j \cap U} (-b_{jv})_+ \quad \text{and} \quad S_j \le \sqrt{n-2}\, \|d_j\|_2, \tag{16}$$

the first identity being the exact gap to the unrestricted minimum — the positive divergence the region omits plus the negative divergence it contains — and the second the Cauchy–Schwarz bound (the projector of Eq. (13) changes only the two endpoint entries). When $\phi_j(\{s_j\}) > 0$, the unrestricted family therefore contains a negative boundary value whenever the interior positive divergence $\sum_v (b_{jv})_+$ exceeds the origin-cut value: node imbalance, extremal location, and sign crossing are the same statement. $S_j = 0$ is equivalent to $d_j = 0$. For network regions whose induced subgraphs and complements are both connected, define S_j^{conn} by restricting the same extrema to that family. Then $S_j^{\mathrm{conn}} \le S_j$. We report both ranges as percentages of the absolute origin-cut value $|\phi_j(\{s_j\})|$, which remains explicit because the ratio depends on that denominator

(absolute ranges when it is zero). Within each region family, the four levels above apply with their family-specific form: the unrestricted column uses η_{all} and d_j, connected families use η_{conn} and the family's profile range, and a prescribed finite family $\mathcal{F}$ uses the exact criterion of Proposition 1; for maximum model difference, $\mathcal{K}_j(e_j)$ is exact and family scans give the restricted value.

Corollary 9 (best common boundary value)**.** *For the unrestricted family, the unique best constant in maximum absolute boundary error is*

$$\beta_j^\star = \frac{\phi_j^{\max} + \phi_j^{\min}}{2} = \frac{(b_j)_{t_j} - (b_j)_{s_j}}{2}, \qquad \min_{\beta\in\mathbb{R}} \max_{U\in\mathcal{U}_j^{\text{all}}} |\phi_j(U) - \beta| = \frac{S_j}{2}.$$

Hence all separating boundary values lie within a prescribed absolute tolerance ε of some common value if and only if $S_j \le 2\varepsilon$.

Proof. For any constant, its largest error at the two attained extrema is at least half their difference; their midpoint attains this bound for every boundary value. Equation (15) and $\mathbf{1}^\top b_j = 0$ give the midpoint formula. □

The coefficient agrees with the endpoint projection in Eq. (13). It approximates the original boundary values; it does not change their edge attribution or define a new system contribution.

5.3. Common values and minimum-segment summaries

For any nonempty prescribed finite family $\mathcal{F}$, the same midpoint argument gives

$$\beta_j^{\mathcal{F}} = \frac{\phi_{j,\max}^{\mathcal{F}} + \phi_{j,\min}^{\mathcal{F}}}{2}, \qquad \varepsilon_j^{\mathcal{F}} = \frac{\phi_{j,\max}^{\mathcal{F}} - \phi_{j,\min}^{\mathcal{F}}}{2}, \qquad |\phi_j(U) - \beta_j^{\mathcal{F}}| \le \varepsilon_j^{\mathcal{F}} \quad (U \in \mathcal{F}). \tag{17}$$

Ordered families — a corridor's nested screenlines, or the regions of one projection direction — additionally support piecewise common summaries. The construction is stated for any ordered family.

Proposition 10 (minimal contiguous boundary summary)**.** *Fix an ordered family of boundary values $a_1, \dots, a_N$ and a tolerance $\epsilon > 0$. Among partitions into contiguous intervals whose every interval has range at most 2ϵ and does not strictly mix signs, the longest-feasible-prefix procedure returns one with the minimum number of intervals. The rule is elementary and applies to any ordered sequence; its value here is the coupling to the boundary scan, the node-difference interpretation, and a stated tolerance — a single interval reproduces the common summary of Eq.* (17)*.*

When a reported family consists of several ordered chains — the 36-direction geometric family used on Sioux Falls induces 22–35 distinct chains per query — the procedure is applied within each chain and the reported count is the sum over chains; the minimality guarantee is per chain.

Proof. No feasible first interval can extend past the longest feasible prefix, so every optimal partition starts with a prefix no longer than the greedy choice; replacing the first interval of an optimal partition by the full greedy prefix cannot make the remainder infeasible, and the same argument applies inductively to the remainder. □

The tolerance is a reporting choice, and the greedy minimality of Proposition 10 extends from the constant tolerance to position-dependent tolerances, which control the relative error at small-magnitude positions.

Corollary 11 (position-dependent tolerances)**.** *Fix an ordered family $a_1, \dots, a_N$ and tolerances $t_i \ge 0$. Call an interval I feasible when some representative $\hat{a}$ satisfies $|\hat{a} - a_i| \le t_i$ for all $i \in I$. Then I is feasible if and only if*

$$\max_{i\in I}(a_i - t_i) \;\le\; \min_{i\in I}(a_i + t_i).$$

If the no-mixing rule of Proposition 10 is imposed as an additional reporting requirement, feasibility is inherited by subintervals, so the longest-feasible-prefix procedure again returns the minimum number of contiguous segments, and the segment representative may be taken as the midpoint of the feasibility intersection.

Proof. The intersection $\bigcap_{i\in I}[a_i - t_i,\, a_i + t_i]$ is nonempty exactly under the displayed condition, and every subinterval of a feasible interval retains a nonempty intersection; the exchange argument of Proposition 10 then applies verbatim. □

The two standard settings used throughout are instances of this corollary. The recommended *mixed* setting is the mixed absolute–relative tolerance $t_i := \epsilon_{\text{abs}} + \tau|a_i|$ with ϵ_{abs} at 1% of the family maximum and $\tau = 10\%$. The *absolute* setting is the pure family-scale absolute rule $\epsilon_j = 5\% \max_U |\phi_j(U)|$, which compresses further but permits large relative errors where the profile is small. On Anaheim, the mixed setting raises the median segment count only from 10 to 12 (per-query ranges 4–14 absolute against 6–17 mixed) while cutting the worst relative error at small-magnitude positions from 55% to 28% in the median over queries (94% to 45% at worst; Appendix C.11 reports the full sensitivity). Every query therefore receives a usable output under either setting, and both are reported in the summaries below (the learned-assignment experiments of Section 7 use the absolute setting). The reporting rule is stated once and organizes the numerical sections:

Adaptive reporting rule. For a specified boundary family, report the minimax common value $\beta_j^{\mathcal{F}}$ when its exact error $\varepsilon_j^{\mathcal{F}}$ meets the chosen setting's tolerance; otherwise report the boundary attribution profile itself, or its minimum contiguous segmentation, in which every boundary value a_i is reproduced within its tolerance t_i (mixed: $t_i = \epsilon_{\text{abs}} + \tau|a_i|$, representative from the per-segment feasibility intersection; absolute: $t_i \equiv \epsilon_j = 5\% \max_U |\phi_j(U)|$, interval midpoint; Proposition 10).

The absolute pair remains available regardless of the normalization, and each ordered boundary family is summarized at the coarsest resolution that meets the prescribed tolerance: the outcome is either an error-controlled common value or an error-controlled profile.

5.4. Exact scanning of predefined straight screenlines

Fix D directions before evaluating attribution. For each direction, sort nodes by their coordinate projection and group equal projection values; no screenline splits such a group. A candidate position is a gap between adjacent groups. Forward and reverse union–find sweeps record whether each prefix and suffix induces a connected subgraph. This geometry and topology preprocessing is reusable across queries, demand paths and models. For one attribution column, form $b_j = Bc_j$ (Eq. (2)) and prefix sums $P_{j,k} = \sum_{i\le k}(b_j)_{\pi(i)}$. If s_j lies in the prefix and t_j outside, its attribution is $-P_{j,k}$; for the reverse orientation it is $P_{j,k}$ because $\mathbf{1}^\top b_j = 0$. Retain only endpoint-separating gaps with both sides connected. Scanning these values gives exact attributions and extrema over this prescribed finite screenline family. Preprocessing takes $O(D(|V|\log|V| + |E|\alpha(|V|)))$; after preprocessing, a column takes $O(|E| + D|V|)$ using sparse incidence multiplication. Duplicate regions may be removed without changing extrema.

On Sioux Falls, 36 directions at five-degree intervals give 1642 admissible query–position boundary values across six queries (duplicates across directions retained for all algorithms). On exactly these oriented regions, the prefix scan, a vectorized baseline that precomputes the signed crossing matrix $A_{\mathcal{F}}$ (one sparse row per oriented region, +1 leaving and −1 entering) and aggregates every boundary value with one sparse product $A_{\mathcal{F}}C$, and direct crossing-edge summation all agree to 2.2×10^{-11} TSTT units once the reference columns are available. Appendix B.6 reports the same-task timings of the three aggregations and the amortization threshold of the precomputed matrix.

5.5. Certificates and exact error budgets for learned assignment

For a query j and any edge vector h, define the seminorm

$$\mathcal{K}_j(h) = \frac{1}{2}\left(\left|(Bh)_{t_j} - (Bh)_{s_j}\right| + \sum_{v\notin\{s_j,t_j\}} |(Bh)_v|\right). \tag{18}$$

Corollary 12 (exact maximum boundary error)**.** *Let $e_j = c_j^L - c_j^S$ be the difference between learned and reference attribution columns. Then*

$$\max_{U:\, s_j\in U,\ t_j\notin U} |\phi_j^L(U) - \phi_j^S(U)| = \mathcal{K}_j(e_j). \tag{19}$$

At a common demand q, write $h_j(q) = D^L(q)J_j^L(q) - D^S(q)J_j^S(q)$. For any subfamily $\mathcal{U}_j$ of separating boundaries,

$$\epsilon_j^{\mathcal{U}} := \max_{U\in\mathcal{U}_j} |\phi_j^L(U) - \phi_j^S(U)| \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda. \tag{20}$$

The proof in Appendix A.5 applies Eq. (15) to the error column. Evaluation requires one incidence multiplication and node scan, without enumerating boundaries. Connected regions and straight screenlines inherit an upper bound; their restricted maxima need not equal $\mathcal{K}_j(e_j)$. The endpoint term must be retained: the two explanations may each be cut-consistent but have different common boundary values, which internal imbalance alone misses.

The seminorm evaluates spatial explanation errors directly. Equation (20) identifies the weighted response discrepancy $h_j(q) = D^L(q)J_j^L(q) - D^S(q)J_j^S(q)$ as the direct local driver of boundary error, and $\mathcal{K}_j$ defines the corresponding reference-matching losses, whose complete specifications are collected in Appendix B.8. In the studied assignment setting the weight-shift component of h_j is small relative to the response component (Section 7), so the unweighted Jacobian loss is evaluated as a stable proxy for the weighted quantity; Section 7 also evaluates the bound chain of Eq. (20) directly on the test paths.

5.6. Total error budget of learned summaries

Sections 5.3 and 5.5 control two errors separately: how far a summary is from the profile it compresses, and how far a learned profile is from its reference. A reader of a learned model's explanation uses both at once — the minimum-segment summary of the learned boundary values. Write $a_j^L(U)$ and $a_j^S(U)$ for the learned and reference boundary values on a family $\mathcal{F}$, $\hat{a}_j^L$ for the learned summary constructed at tolerance $\varepsilon_j^{\mathrm{sum}}$, and $\delta_j = \max_{U\in\mathcal{F}} |a_j^L(U) - a_j^S(U)|$.

Corollary 13 (total explanation error)**.**

$$\max_{U\in\mathcal{F}} \big|\hat{a}_j^L(U) - a_j^S(U)\big| \le \varepsilon_j^{\mathrm{sum}} + \delta_j, \qquad \delta_j \le \mathcal{K}_j(e_j) \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda. \tag{21}$$

Proof. Triangle inequality for the split $\hat{a}_j^L - a_j^S = (\hat{a}_j^L - a_j^L) + (a_j^L - a_j^S)$; the first term is bounded by construction ($\le \varepsilon_j^{\mathrm{sum}}$), the second by Eq. (19) and Eq. (20). □

The budget combines the two outputs of the framework — a compression and a learned profile — into the error of the final reported artifact. It is a posteriori: the integral bound consumes reference-response information, so Eq. (21) characterizes the worst boundary error when reference derivatives are available, not a deployable guarantee computable from the learned model alone. Section 7 evaluates all three quantities on the test paths.

Scope and interpretation. The classification fixes four objects before any model is fitted: the road representation (subdivision changes which crossing-link costs a boundary includes, Example A.8); the outcome definition (its marginal weights are the classified object); the attribution path in demand space (with several ODs changing jointly, the per-query columns are path dependent; the total is not); and the admissible region family. The structural, model-specific, realized and learned-reference levels answer different questions and are reported separately (Section 3.2).

Approximate conservation. For a model with residual $r(q) = Bf(q) - Mq - b$ and constant b, the corresponding flow-attribution residual is

$$R_j := Bg_j - \mu_j \Delta q_j = \Delta q_j \int_0^1 \partial_j r(\gamma(\lambda))\, d\lambda, \qquad \sum_j R_j = r(q^1) - r(q^0). \tag{22}$$

Small endpoint residuals control the sum, but individual columns can cancel. Approximate output conservation controls attribution residuals only when derivative regularity is also available; Appendix A.4 gives the $C^{1,1}$ conditions, a local bound, sharpness examples and an extension to weighted attributions.

The traffic experiments below use exactly conservative models, and Appendix B.10 compares the exact parameterization against a conservation penalty and an output projection on identical supervision, showing that a small output residual need not imply a small attribution residual.

6. Error-controlled spatial explanations in traffic networks

Sioux Falls absolute attributions use benchmark flows in veh/h and time units of 0.01 h; multiply an attribution by 0.01 for the hourly vehicle-hour total. Anaheim uses vehicle-minutes per hour. All experiments use the stated fixed-path logit-SUE assignment models and prespecified demand scenarios; shares reported over boundary families are enumeration shares over those families. The adaptive reporting rule of Section 5 organizes the experiments: every query either receives one error-controlled common value or an explicit spatial profile. The five studies are presented in reading order — the absolute explanation first, its generating mechanisms second, and the spatial resolution it requires last; Table B.1 (Appendix B.1) collects their settings.

6.1. A large-network spatial explanation

The public Anaheim network has 416 nodes, 914 directed links and 1,406 positive OD demands (Transportation Networks for Research Core Team, 2024). Six largest demands change jointly from 0.8 to 1.2 times their published values; other demands remain fixed with endogenous route choices. Ten fixed free-flow shortest paths per OD exclude other zone centroids as intermediate nodes (first through node 39). This calculation uses the existing 41-state attribution columns. To obtain interpretable region locations without requiring coordinates, define network-distance progress $p(v) = [d_s(v) - d_t(v) + d_s(t)]/[2d_s(t)]$, using shortest distances in the underlying undirected network with the smaller free-flow time on parallel roads — a purely geometric ordering of the family (assignment paths additionally exclude other zone centroids, the progress metric does not). The origin has progress zero and destination progress one. Group tied levels, retain prefixes with two connected induced sides that separate the OD, and orient every region toward the origin. This distance-based family depends on the network and query, before inspecting any attribution. It is a prescribed family, not all connected regions, and it is single-direction: the nested prefixes are totally ordered by progress, so no direction grouping is needed here.

The case asks one transport question: for a prespecified demand change, where along the origin–destination direction does the signed crossing-cost response change markedly, and does the variation come mainly from the study traffic's congestion adjustment or from background rerouting? Figure 3 displays the query with the largest early–late attribution difference, $(25, 2)$, chosen to illustrate the method at its richest variation; the aggregate statistics come from all six queries (Table 2). The figure shows the boundary attribution as a function of progress over the complete family with its reported four-segment summary, and the four-part mechanism decomposition along the same axis. The profiles of the remaining five queries stay positive with family ranges spanning roughly one to one and a half orders of magnitude (largest-to-smallest attribution ratios 12.5–36.9); the complete per-query numbers — system contributions, unrestricted and family ranges, region counts and the attributions of the three prespecified positions — are collected in Appendix B.7 (Table B.3), with the early–late mechanism differences in Table B.4.

Where the attribution changes. The Anaheim TNTP files ship no node coordinates, so the location layer is reported through the crossing links of each boundary rather than a map: the three prespecified positions sit at very different network locations — the early boundary crosses 34 links with two dominant contributions, the middle boundary crosses only four links, and the high plateau isolates the destination neighborhood, where the entire attribution crosses on the single link $63 \to 62$ (Table B.5, Appendix B.7, lists the region sizes, crossing links and principal signed contributions).

Which traffic generates the change. The decomposition panel locates the mechanism. At the early position the attribution is dominated by the free-flow study response (669.3 of the total 801.5, congestion study 111.5, background below 11); at the middle position the two study terms are comparable (211.5 free-flow against 212.6 congestion of the total 412.0, background below 11); at the late position the congestion study term reaches 5213.7 while both background terms are below 0.1 in absolute value: the rise is congestion adjustment of the study traffic itself on the destination-side crossing link, not background rerouting. The decomposition

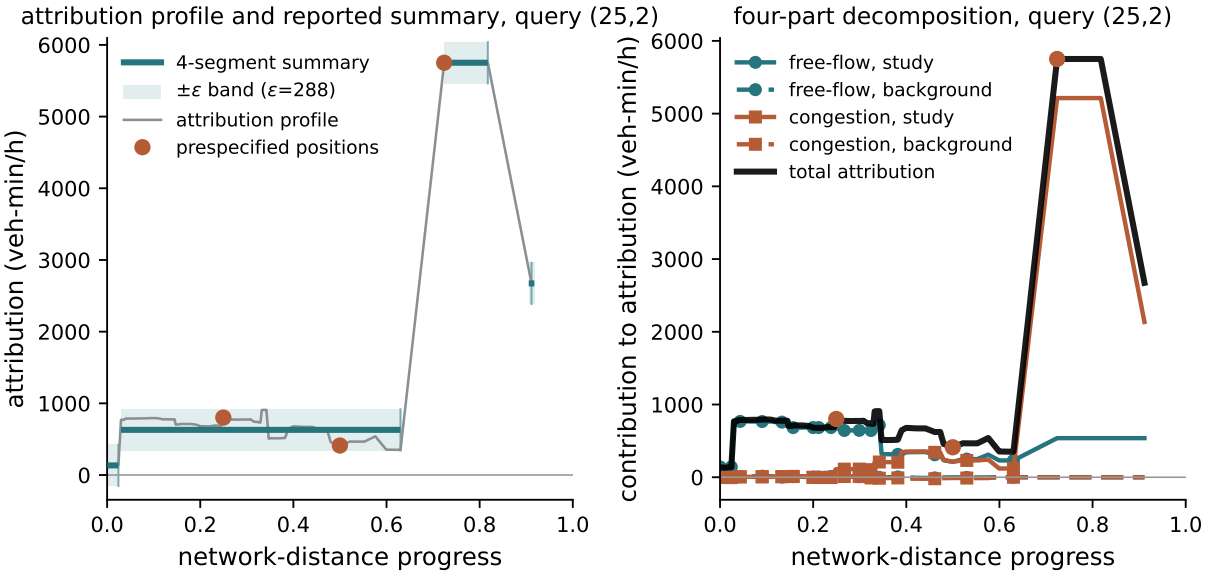


Figure 3: Anaheim boundary-attribution profile for query $(25, 2)$ over the distance-based family, with its reported summary. Left: attribution versus network-distance progress along the OD direction (gray curve); the four-segment minimum summary at the family tolerance (teal step curve with $\pm\epsilon$ band, $\epsilon = 287.6$); ochre dots mark the three prespecified positions. Right: the four-part mechanism decomposition (teal: free-flow weights, ochre: congestion weights; solid: study response, dashed: background); the gray curve is the total attribution.

Table 1
The complete reported summary for query $(25, 2)$ at the family tolerance ($\epsilon = 287.6$; vehicle-minutes per hour). Each row is one segment: its progress range, the number of summarized boundary values, the representative value, and the actual worst error within the segment (at most ϵ by construction).

Segment	Progress range	Boundaries	Representative	Worst error
1 (origin side)	0.000–0.024	6	134.3	2.5
2 (corridor)	0.030–0.630	75	630.2	279.2
3 (plateau)	0.724–0.818	2	5,751.2	0.0
4 (destination)	0.911	1	2,673.1	0.0

is additive at a common equilibrium state and attributes the variation to the corresponding response–weight components; the summary then states, with its error bound, where that variation changes.

The reported summary. Table 1 reports the complete four-segment summary for this query: each segment's position range, the number of boundary values it summarizes, its representative value, and the actual worst error within the segment. An origin-side segment covers the six boundaries nearest the origin at representative value 134.3 (worst error 2.5); one long corridor segment summarizes the 75 boundaries from progress 0.03 to 0.63 at 630.2 (worst error 279.2); the congestion plateau is one segment at 5,751.2 spanning progress 0.724–0.818; and the destination-side boundary at progress 0.911 is reported exactly. Every attribution lies within $\epsilon = 287.6$ of its segment value, the tolerance being 5% of the family maximum 5,751.2; the reader absorbs four numbers in place of 84. Across the six study queries the compression is uniform: Table 2 shows 83–211 boundary values reducing to 4–14 segments under the absolute tolerance and 6–17 under the mixed tolerance, with every boundary value within its stated tolerance by construction; the summary never changes an original attribution — it is the same computed profile at its coarsest resolution that meets the tolerance. The mixed rule controls the relative error at small attributions (median worst small-value relative error 54.9% under the absolute rule against 27.8% under the mixed rule) at a modest segment cost (median

Table 2
Reported minimum-segment summaries on Anaheim under the two tolerance settings of Section 5.3 (Proposition 10): absolute (5% of the largest attribution magnitude per query) and mixed (1% absolute floor plus 10% relative). Worst rel. small: worst relative error at small-magnitude positions (magnitude below one tenth of the family maximum). Every boundary value in a segment lies within its tolerance of the representative value; the complete $(25, 2)$ absolute-setting segmentation appears in Table 1.

		Absolute (5%)		Mixed (1%+10%)	
Query	Regions	Segments	Worst rel. small (%)	Segments	Worst rel. small (%)
$(4, 2)$	211	12	94.1	13	31.9
$(1, 2)$	83	8	53.6	11	45.0
$(2, 4)$	211	14	38.3	17	3.2
$(3, 2)$	157	8	56.3	11	31.4
$(25, 4)$	164	13	0.0	13	0.0
$(25, 2)$	84	4	79.5	6	24.2

10 against 12). Aggregation timings and the original construction costs appear in Appendix B.6; the full tolerance sensitivity, including the mixed absolute–relative rule of Section 5.3, appears in Appendix C.11.

6.2. Mechanisms of spatial variation

We compare three treatments of background traffic: absent, fixed at its baseline link flows, and responsive through endogenous route choice. Eight largest published OD demands, all incident to node 10, change jointly from 0.2 to 1.6 times their base values; every study OD has ten fixed paths. The responsive-background assignment assigns all 528 positive demands (full protocols in Appendix B). Uniform weights provide a compatible control. The active-link outcome $T(F) = F_{10\to16}$ preserves the $(10, 16)$ direction but permits a response-dependent scale; TSTT produces internal imbalance. Figure 4 illustrates the three patterns on the same realized response: compatible weights give one common boundary value on every boundary, the directional outcome balances along the OD but scales with the response, and TSTT weights generate genuine spatial structure. For its seed-zero TSTT response, expanding $\{10\}$ to $\{10, 17\}$ lowers the boundary value from 5.77×10^4 to 4.34×10^4, approximately 92% of the unrestricted spread. The direct-SUE decrease is 24.80% of its own origin-cut value (Appendix C.1). Both sides of these regions are connected. Free-flow weights explain 84.7% of the node-17 imbalance: all incident links have free-flow time 2 except $10 \leftrightarrow 17$ with time 8, giving $[B\,\mathrm{diag}(t_0)g_j]_{17} = 6(g_{10\to17,j} - g_{17\to10,j})$.

Rerouting produces appreciable spatial variation for every query, and connected regions retain 60.41–76.85% of the unrestricted range across the full set (complete per-query records for all three background treatments in Appendix C.2, Tables C.1 and C.2). Normalizations use each setting's own origin boundary value: for $(17, 10)$, the percentage falls from 59.95% under the no-background treatment to 54.77% under the responsive-background treatment while the absolute spread rises from 17,607.8 to 123,202.1, approximately sevenfold.

Response decomposition at common states. Comparing the three background treatments changes both costs and sensitivities. At each common responsive-background state, the mechanism decomposition separates the attribution into study and background response parts, with $BJ_j^{\mathrm{study}} = \mu_j$ and $BJ_j^{\mathrm{bg}} = 0$, both weighted by the same total-flow TSTT gradient:

$$c_j^{\mathrm{tot}} = c_j^{\mathrm{study}} + c_j^{\mathrm{bg}}. \tag{23}$$

At node 15 for $(10, 16)$, study and background divergence contributions are -973.1 and 49,103.4. Yet on the specified expansion $\{10\} \to \{10, 17\}$, the attribution changes only -3361.1 (-1305.0 study, -2056.1 background), or 1.23% of the initial attribution. Large extremal variation can coexist with similar selected boundary values. Component differences add at a fixed region pair; their separate ranges do not add in general.

Free-flow and congestion mechanisms. For BPR costs, $g'_e = t_{0,e} + 0.75t_{0,e}(F_e/c_e)^4$. Crossing these two weight terms with the two response terms gives four additive contributions, evaluated at identical states and

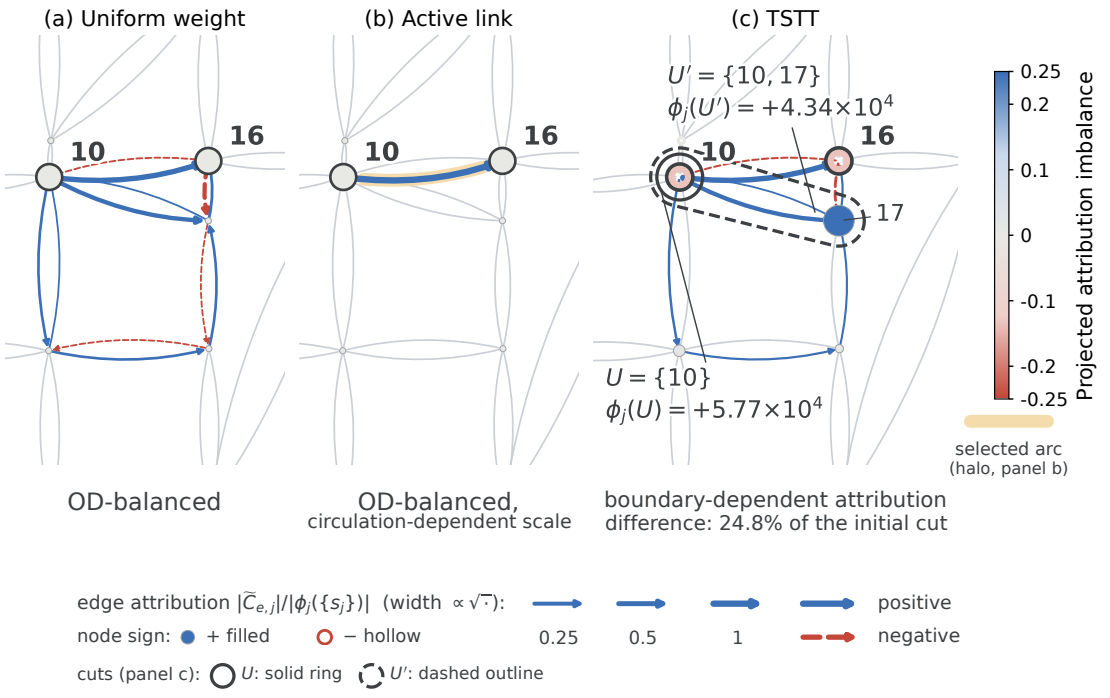


Figure 4: OD $(10, 16)$ endpoint attributions, no-background assignment, learned seed 0. (a) Uniform weights give cut consistency. (b) The signed TSTT boundary attribution decreases by 24.8% when the boundary expands from $\{10\}$ to $\{10, 17\}$; the active-link outcome — which balances in the OD direction but has no response-independent scale — is reported in Appendix C.1. Widths encode attribution magnitude on the shared square-root scale; node colors show the projected attribution imbalance $(d_j)_v$ (Eq. (13)) divided by the absolute origin-cut value.

the same reported boundary:

$$\phi_j(U_-) = \sum_{a\in\{\text{ff},\text{cong}\}} \sum_{b\in\{\text{study},\text{bg}\}} \phi_j^{a,b}, \qquad \phi_j^{a,b} = -\Delta q_j \int_0^1 \mathbf{1}_{U_-}^\top B D_\lambda^a J_j^b(\lambda)\, d\lambda.$$

Table 3 distinguishes the mechanisms behind the sign-crossing queries of Table 4: study congestion dominates $(1, 19)$, both congestion terms are negative for $(1, 15)$, and background congestion dominates $(7, 12)$. The combined free-flow contribution stays positive. The decomposition is additive at a common equilibrium state and identifies the contribution of each response–weight component. The sign pattern itself is a spatial

Table 3
Four parts of the same boundary attribution at the reported U_- (responsive-background six-query protocol; TSTT units; identical 41-point states, responses, and boundaries as Table 4). Study includes all six query ODs; background includes the other 522. Columns cross free-flow/congestion weights with study/background responses. Entries are rounded independently.

OD	Free-flow study	Free-flow background	Congestion study	Congestion background	Total
$(1, 19)$	-361.3	$+759.0$	$-17{,}357.9$	$+3{,}458.6$	$-13{,}501.6$
$(1, 15)$	$+1{,}541.8$	$+333.8$	$-6{,}537.4$	$-3{,}624.2$	$-8{,}286.0$
$(7, 12)$	$+3{,}897.0$	$-1{,}311.1$	$+125.5$	$-20{,}327.6$	$-17{,}616.2$

property, not a magnitude artifact. Negative attributions occupy 17.8%, 2.5% and 7.3% of the 14,477, 15,191 and 16,943 connected regions of the three sign-crossing queries. For $(1, 19)$, the connected minimum $-13{,}501.6$ is attained on a 13-node region, whereas a 13-node free-flow neighborhood has attribution $+10{,}756.0$: region size alone does not determine the attribution. At the minimizing boundary, outgoing attributions sum

to 7,867.3 and incoming attributions to 21,368.8, so the signed crossing-link attribution is negative while $C_j = 50{,}097.7 > 0$; Figure C.2 maps all 26 crossing links. The same contrast appears on the straight screenlines of the prespecified geometric family, whose minimum and maximum attributions for $(1, 19)$ have opposite signs under one positive network-wide contribution (Figure C.1, Appendix C.2). The straight-screenline family uses 36 directions at 5° intervals, gaps between node projections, two-sided connectivity and duplicate removal; construction is independent of attribution. It contains 23–152 distinct regions per query. The $(1, 19)$ and $(7, 12)$ negative minima occur on 2/152 and 10/148 screenlines. Origin-centered hop and free-flow neighborhoods remain positive but can differ in magnitude by factors 8.9, 9.0 and 3.4 for the three sign-crossing queries (Appendix C.5). Sign stability and common-value accuracy are distinct properties.

Stability under alternative attribution paths. With several ODs changing jointly, the per-query columns depend on the demand path; twenty-two piecewise-linear alternatives to the straight-line transition (canonical order, reverse order, twenty fixed-seed random permutations; identical endpoints, quadrature and family) leave every spatial conclusion unchanged: system contributions move by at most 3.4%, sign-crossing status is identical on all orders, common-summary errors shift by at most 0.2 percentage points, and the dominant mechanism is unchanged. The profiles are path-specific in their fine structure (deviations 0.3–4.7%) but not in their traffic interpretation; the complete per-order comparison appears in Table C.10 (Appendix C.12).

6.3. Adaptive spatial resolution

Sioux Falls has 24 nodes and 76 arcs in one block atom. Its structural weight condition is therefore the same for every OD, and the connected and unrestricted classifications coincide; the experiments measure variation due to realized responses and heterogeneous marginal costs. The query-dependent classification is illustrated separately in Figure 2.

Six input-selected queries away from node 10 change jointly from 0.2 to 1.6 times their base values with all other demands fixed and routes responsive; they are the shortest and longest free-flow ODs in each demand tercile (Appendix B.5). Table 4 reports their ranges. The adaptive rule separates them sharply: the geometric accounts for $(7, 18)$ lie in $[642, 702]$, their midpoint 672 has maximum absolute error 30, or 4.3% of the largest attribution, and this family admits one common summary at a 5% tolerance; the $(1, 19)$ and $(7, 12)$ geometric ranges cross zero and require boundary-specific reporting at every tolerance considered below.

Table 4
System contributions and spatial ranges for six prespecified queries, responsive-background assignment (TSTT units). Connected: all admissible connected regions; screenline: the geometric family constructed independently of attribution values. All C_j are positive. Origin-cut attributions, negative shares, and neighborhood ranges are retained in Appendix C.5.

OD	C_j	Connected [min, max]	Screenline [min, max]
$(7, 18)$	672.9	[634, 710]	[642, 702]
$(1, 19)$	50,097.7	[−13,502, 24,260]	[−2,767, 22,124]
$(4, 5)$	2,826.0	[2,396, 3,452]	[2,652, 3,176]
$(1, 15)$	77,403.2	[−8,286, 40,540]	[511, 36,621]
$(6, 8)$	73,964.9	[35,307, 66,695]	[45,230, 65,286]
$(7, 12)$	117,482.3	[−17,616, 60,029]	[−2,967, 57,780]

The input-stratified selection experiment widens the view to an input-only selection of 18 of the 528 positive Sioux Falls OD demands (Appendix C.11 states the rule and reports every query). Each OD changes alone over ±10% and ±20% intervals while other demands remain fixed with endogenous rerouting; the same 36-direction geometric family is used throughout. For each nonempty family, let $\beta = (\phi_{\max} + \phi_{\min})/2$ and $\varepsilon = (\phi_{\max} - \phi_{\min})/2$. This common summary minimizes the largest absolute discrepancy, so every reported boundary satisfies $|\phi(U) - \beta| \le \varepsilon$. The reported normalized error is

$$r_j^{\mathcal{F}} = 100\, \varepsilon_j^{\mathcal{F}} \Big/ \max\big(|\phi_{j,\min}^{\mathcal{F}}|, |\phi_{j,\max}^{\mathcal{F}}|\big), \tag{24}$$

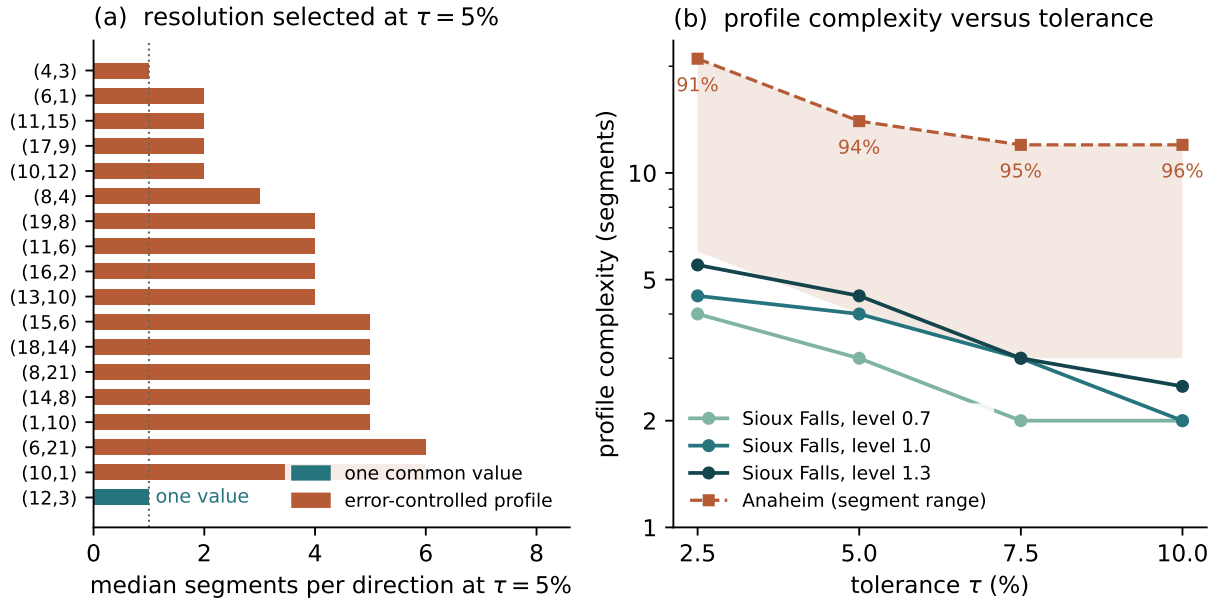


Figure 5: Spatial resolution selected by the error tolerance. Left: the median minimal segments per projection direction that each of the 18 input-stratified queries receives at the 5% family tolerance (baseline load; queries sorted by selected resolution; one query is adequately summarized by a single common value, the remainder by error-controlled profiles). Right: the profile complexity as the tolerance varies from 2.5% to 10% — median segments per projection direction on Sioux Falls (36 normals, distinct chains deduplicated, three load levels) and per-query segments on Anaheim's ordered families; underlying numbers in Table C.9 (Appendix C.11).

the absolute deviation of every attribution from the common value as a percentage of the family's largest attribution magnitude; it does not bound each attribution's own relative error. We apply a fixed descriptive tolerance $r_j^{\mathcal{F}} \leq 5\%$; this is a reporting choice rather than a transport welfare threshold. When all attributions in a family share one sign, $r_j^{\mathcal{F}} \leq 50\%$, so values near 50% indicate that the smallest magnitude is far below the largest, not that the attributions nearly agree. At ±10% and ±20%, respectively, 1 and 1 of 18 queries meet this tolerance, while 1 and 1 have attributions of both signs; representative common summaries at ±20% appear in Appendix C.11.

Figure 5 presents the selection and its cost. A minority of queries admits one common value at a working tolerance; every other query is summarized one direction at a time. Three counts should be distinguished when reading these results: the number of distinct boundary regions N_{unique}, the number of ordered direction chains N_{chain} along which they are reported, and the total number of reported segments L_{total}. On Sioux Falls the 36 projection normals of the geometric family induce 22–35 distinct ordered chains per query over 1–21 regions each, and consecutive entries of one chain are nested along that direction, so the chain — not the merged sequence — is the unit of spatial contiguity; segmenting each chain at the family tolerance needs a median of 2–5.5 segments per direction across the three load levels as the tolerance ranges over 2.5–10% (at the 5% tolerance and baseline load: median 4, maximum 8), with L_{total} at a median of 117 over the distinct chains of the learned-assignment queries. Per-chain minimality therefore does not by itself make the reported representation smaller than the deduplicated set of boundary values; the direction-resolved profiles are the reported representation there, and the multi-direction counts quantify direction-specific spatial resolution. Ordering all directions into one progress sequence yields fewer apparent segments only by grouping non-adjacent cross-sections, so those counts are not spatial summaries and are retained only for comparison (Table C.9). On Anaheim's single-direction ordered families — where the strong compression claims rest — the same relaxation raises the compression of the 83–211 attributions per query from 91.4% (6–21 segments) to 96.4% (3–12 segments). With the same segment count, equal-count partitions have median worst-segment error 1.4–4.0 times the greedy error per direction on Sioux Falls and 4.7–18.8 times the tolerance on Anaheim, so position-aware segmentation is what provides the error control, not the segment count alone.

Table 5
Common-summary statistics across baseline load levels (18 queries, focal ±20%, same 36-direction family). Normalized error is Eq. (24); the ratio column is the median geometric-to-unrestricted spread ratio.

Level	Queries within 5%	Mixed signs	Median error (%)	Median ratio	Max. negative share
0.7	2	0	34.5	0.529	0.00%
1.0	1	1	44.4	0.574	0.67%
1.3	0	3	45.2	0.563	4.03%

The ±10% and ±20% summaries of the same 18 queries are close in normalized terms, so further subdividing the perturbation width adds little information. The congestion state of the baseline is the more informative axis. All 528 demands are scaled jointly to 0.7, 1.0 and 1.3 times their published values; at each level the level equilibrium is solved once, the focal query alone moves by ±20%, other demands stay fixed with responsive routes, and the endpoint identity supplies the attribution column (solver protocol in Appendix B.5). Table 5 shows a monotone pattern: under the light baseline, attributions are flatter across the family — the median normalized error falls to 34.5%, two queries satisfy the 5% tolerance and no family crosses zero — whereas the heavy baseline makes every query exceed the tolerance, three families mix signs and the negative share of $(6, 21)$ reaches 4%. Per-query normalized errors rise from the light to the heavy level for 17 of 18 queries (e.g. $(10, 12)$: 18.6 → 35.2 → 40.3; $(1, 10)$: 31.2 → 44.8 → 48.0). Across the load levels, light-load conditions admit transferable single summaries more often, and heavy-load conditions require boundary-specific reporting. The load-level trend has a theory-side counterpart: the node-imbalance norm $\|d_j\|_2$ rises by a factor of 13.2 from the light to the heavy baseline, and Eq. (14) sums that imbalance into the spatial range (Appendix C.11 reports the rank correlations and the Cauchy–Schwarz comparison).

How much does the model-specific level buy? On the synthetic network of Figure 2, restricting attention to the model's path-feasible response envelope restores boundary consistency despite off-path weight heterogeneity, so the model-specific level certifies a common boundary value that the universal condition denies. On Sioux Falls and Anaheim the path-feasible response envelopes are sufficiently rich that the structural and model-specific tests both reject a single universal boundary value, and the diagnosis is stable across load levels (Appendix C.7 reports the numerical diagnosis of the transferability hierarchy). The realized error-controlled profiles therefore become the operative representation, and the summaries of Table 4 carry the reporting.

7. Response-supervised learning for accurate spatial explanations

Hypothesis. The boundary-error representation of Section 5.5 locates explanation error in the weighted demand-response discrepancy between the learned and reference maps. In the studied TSTT setting the weight-shift component of that discrepancy is small relative to the response component, leading to the testable hypothesis that supervising the flow Jacobian improves spatial explanations even when ordinary flow accuracy is already high. The experiments below compare flow supervision, response supervision and its weighted form under equal budgets and a common validation-only weight selection, examine stability when the training budget is tripled, and measure how many response labels the improvement requires; the full six-objective family, including the endpoint objectives, is reported in Appendix C.8.

Method and setup. The learned map uses the conservative path-share parameterization

$$h_k(q) = q_k \,\mathrm{softmax}\big(z_\theta(q)\big|_{\Pi_k}\big), \qquad f_\theta(q) = \Delta h(q), \tag{25}$$

where Π_k are the fixed path sets of the assignment structure and Δ the path–link incidence: node balance holds identically, and only the shares are learned. The input is the six-dimensional study demand vector with background demands fixed at their published levels; the output covers every path of the structure, so background route choice responds endogenously. The reference model supplies equilibrium flows and sensitivity derivatives. Flow supervision fits predicted flows; response supervision adds the derivative

objective

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{flow}} + \lambda_J \, \mathbb{E}_{q,j} \frac{\|J_j^L(q) - J_j^S(q)\|_2^2}{u_j^2}, \tag{26}$$

with J_j the flow Jacobian column of query j and u_j its RMS training scale; optimizer, batches, validation selection and label costs appear in Appendix B.9. Derivative supervision itself is established (Sobolev training; Czarnecki et al., 2017; Yuan et al., 2022); the target here is network-derived, and the question is what it contributes to spatial explanations. Why the unweighted objective of Eq. (26) acts on the weighted quantity of Eq. (20) is answered by decomposing the weighted response discrepancy at a common demand state:

$$h_j(q) = \underbrace{\big(D^L - D^S\big) J_j^S}_{\text{weight shift}} + \underbrace{D^L \big(J_j^L - J_j^S\big)}_{\text{response error}}, \qquad h_j = D^L J_j^L - D^S J_j^S, \tag{27}$$

and integrating along the demand path yields the column error $e_j = \Delta q_j \int_0^1 h_j(\gamma(\lambda)) \, \mathrm{d}\lambda$. On the 96 test paths the response term dominates overwhelmingly: the path–query median response and weight norms are 206.5 and 3.5 under flow supervision and 147.6 and 3.3 under response supervision, with the response term larger on every path–query pair for both objectives (Eq. (C.1) in Appendix C.8 states the integrated form). Supervising the dominant term is therefore a practical proxy whose effect the boundary metrics measure directly.

Evaluation protocol. All objectives are evaluated on the same 96 test demand paths (Appendix B.9 states the composition); paths are the independent unit, metrics average active queries and seeds within each path, and seed variability is reported separately. Checkpoint and loss-weight selection use validation data only; the test paths enter the final report alone.

Primary comparison. Table 6 reports flow prediction, attribution accuracy, and the end-to-end error of the reported summary on the same test demand paths for both networks. On the 96 Sioux Falls paths, response supervision lowers the median attribution-column error from 21.18% to 14.19% (a relative reduction of one third) and the family-boundary error from 8.74% to 5.75%, while both models retain total-flow errors near 0.03% (medians 0.0322% and 0.0292%). Measured against the demand-change increment rather than total flows, the endpoint flow-prediction error falls from 5.03% to 3.52% — the incremental response that the attributions decompose — so response supervision improves both the attribution column and the response field it integrates; the total-flow columns stay near 0.03% because total flows are dominated by the unchanged background. The improvement holds in every active-query group (Appendix C.9). Path-level paired differences quantify the uncertainty: the median paired column-error difference is -5.03 percentage points with a path-level bootstrap 95% interval of $[-6.01, -4.48]$ (10,000 resamples), improving 92 of 96 paths; the paired family-boundary difference has median -2.06 with interval $[-2.76, -1.65]$, improving 90 of 96, with every group's interval excluding zero and leave-one-path-out medians varying within $[-5.11, -4.94]$ and $[-2.11, -2.00]$. The system-contribution difference is secondary and opposite in sign (median $+0.45$, interval $[0.01, 0.74]$; both methods below 3.5%), confirming that scalar completeness and spatial explanation accuracy measure different properties. The improvement is consistent across all five seeds, with both median error metrics decreasing for every seed and 80–91 of 96 paths improving per seed (Table C.6, Appendix C.9). The same-path decomposition of Eq. (27) attributes the discrepancy to demand-response error on every path–query pair (path–query median response and weight norms 206.5 and 3.5 under flow supervision, 147.6 and 3.3 under response supervision).

Where the flow-only explanation breaks, and what supervision repairs. Figure 6 shows one pre-specified case: among the 96 test paths, the one whose paired family-error improvement is closest to the median improvement (path 22, six active queries; improvement 48.4 against the median 49.2), and within it the query with the largest flow-only family error, OD $(7, 12)$. The panels display the query's longest projection-direction chain (20 nested boundaries, a geometric display choice independent of attribution values), the boundary error against the reference along that chain, and the family-wide worst boundary on the network.

Table 6
Prediction and spatial-explanation accuracy, and the end-to-end error of the reported summary, on the same test demand paths (Sioux Falls: 96 paths, 24 per active-query group; Anaheim: 48 paths). Entries are medians of path-level values (active queries and seeds averaged within each path; flow errors additionally averaged over the path's states). Total flow: relative flow error; incremental: endpoint flow-change error (percent of the reference increment norm); column: attribution-column error; family boundary: maximum boundary error over the query's family as a percentage of the reference family maximum (the Sioux Falls family is the geometric screenline set; Anaheim uses its ordered distance family); final summary: total error of the reported minimum-segment summary (percent of the reference family scale; tolerance 5% of the learned family maximum); Seg./chain: median segments per ordered direction chain (Anaheim's family is one ordered chain); total segments: median over all distinct chains (Sioux Falls, 18–35 chains per query) or over the single family (Anaheim) — the two networks' totals are not comparable one to one.

Network	Method	Total flow (%)	Incremental (%)	Column (%)	Family (%)	Final summary (%)	Seg./chain	Total seg.
Sioux Falls	Flow	0.032	5.03	21.18	8.74	11.35	3.6	117
Sioux Falls	Response	0.029	3.52	14.19	5.75	8.68	3.6	117
Anaheim	Flow	0.024	2.46	1.64	0.94	4.57	9.9	9.9
Anaheim	Response	0.019	2.03	1.31	0.78	4.55	9.9	9.9

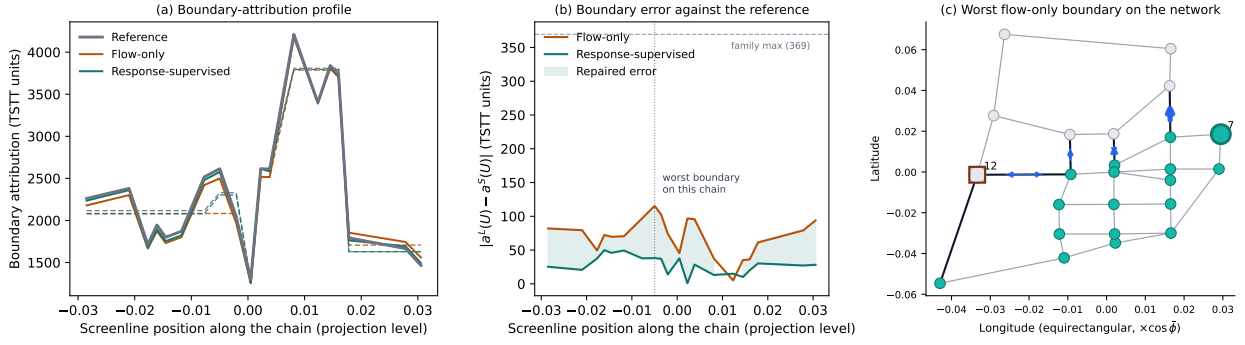


Figure 6: A learned spatial profile repaired by response supervision (pre-specified case: path 22, query $(7, 12)$; five-seed median columns). (a) Boundary-attribution profile along the query's longest projection-direction chain (20 nested boundaries at the 10° direction): reference (gray), flow-only (ochre), response-supervised (teal), each with its minimum-segment summary (dashed). (b) Boundary error against the reference along the same chain; the dotted line marks the worst boundary on the chain and the gray dashed line the family-wide maximum (369.5). (c) The family-wide worst flow-only boundary on the network: teal nodes form U (origin 7 ringed, destination 12 square); arrow width scales with the signed crossing-link contribution (solid blue positive, dashed red negative).

The flow-only model does not merely rescale the profile: it misplaces the attribution across the corridor, with errors concentrated where the reference profile varies. At the worst flow-only boundary (369.5 TSTT units, on a reference value of 5,619.3 that is congestion-dominated: 5,018.7 of it from congestion weights against 600.6 from free-flow weights) the error falls to 80.1 under response supervision — a 78% reduction at that boundary; over the whole family, the worst error falls from 369.5 to 115.2 and the end-to-end error of the reported summary from 679.1 to 463.7 at essentially unchanged segment count (186 to 190 over the family). All quantities come from the same five-seed-median columns the figure displays. The segment summaries of all three columns sit inside their stated tolerance bands, so the repair acts on the profile the summaries compress, not on the compression itself.

The objective family. The boundary-error representation of Eq. (20) defines a family of supervision objectives rather than a single loss; the complete six-objective comparison, including the weighted form identified by Eq. (27) and the boundary-seminorm loss, appears in Appendix C.8. Two of its results organize the present comparison. The unweighted response objective attains the lowest errors on every metric (column 14.19%, family boundary 5.75%, flow 0.029%) with the widest usable weight range, while its weighted form reaches 15.21% — the weight-shift term the unweighted objective ignores is small. And $\mathcal{K}_j$ is a seminorm

whose kernel contains the circulation directions, so optimizing the boundary value alone cannot restore the response field the profile integrates: the boundary-seminorm objective attains the lowest validation boundary error yet a test column error above flow-only training (24.18%, improving only 8 of 96 paths).

Label efficiency, training budget, and wall-clock. The improvement is cheap in labels and stable in budget. Retraining at per-query derivative-center budgets of 6–48 (36–288 centers; full budget 384) reaches half of the full column-error improvement at 36 centers and 90% at 288, with subset interquartile ranges below 0.8 percentage points and flow accuracy unchanged (Figure B.2, Appendix B.11). Tripling the continuation budget to 3,600 updates improves both methods on both networks while the response curve stays below the flow curve at every update count (Figure C.4, Appendix C.9): the median column error falls to 19.46% against 11.41% on Sioux Falls and 1.24% against 0.88% on Anaheim. Derivative supervision costs more per update (1.8× on Sioux Falls, 3.3× on Anaheim); on Sioux Falls the anytime dominance survives the cost difference in wall-clock terms, while on Anaheim the response run reaches the flow method's final error after about 2.2 times its training time.

End-to-end explanation error. The quantities of Table 6 concern the learned profiles; the reported artifact is the learned model's minimum-segment summary, whose total error Table 6 already carries as the budget of Eq. (21) at the working 5% tolerance. Figure 7 traces the same quantities as the tolerance — and with it the number of reported segments — varies from 0.5% to 10%, and Table 7 reports the matched-complexity comparison over that pre-specified grid: at each tolerance the two methods segment their own learned profiles, and the paired difference is evaluated at the matched segment budget — the smaller of the two methods' per-path segment counts, each method interpolated on its own error–segments grid. The two networks occupy different regimes, and the error budget identifies the effective lever on each. On Sioux Falls the model term dominates (8.74% against a 4.76% summary term under flow supervision), the response-supervised curve lies below the flow-supervised curve at every segment count: at the matched segment budget of each tolerance (Table 7) the paired median difference is -1.7 to -2.0 percentage points with bootstrap intervals excluding zero at every tolerance — computed on all 96 paths' data at 0.5–2.5% and on the common-support subset at the two loosest tolerances, where the equal-tolerance comparison over all paths is essentially identical (individual paths can still move the other way; the intervals are for the median) — and the iso-error view quantifies the saving: a final error of 11.0% needs 152 segments under flow supervision but 117 under response supervision, while the 8.8% error that response supervision attains with 117 segments is not reached by flow supervision anywhere in the grid (its best, 8.80%, requires 233 segments). On Anaheim the learned columns are closer to the reference, and at the working 5% tolerance the budget is compression-dominated: the summary term (4.3%) is several times the model term (0.9%), both curves track the reference-compression curve — the error of compressing the reference profile itself at each tolerance — and the two methods are nearly indistinguishable in the final error (4.57 against 4.55), exactly as the budget predicts: the model error is not the active bottleneck there, and tolerance refinement, not a better learned response, is the effective intervention. At the tightest 0.5% tolerance the model term takes over (0.9% and 0.8% against a 0.5% summary term), where the response-supervised final error is again lower (0.91% against 1.06%). Better learned responses matter precisely when model error is the active bottleneck; when compression dominates, the framework correctly identifies tolerance refinement as the relevant lever. On both networks the measured total error of the reported summary stays within the budget on every path. Relative to the baselines, segmentation cuts the total error by a factor of 3.0–3.8 (single common value) and 1.7–2.0 (equal-count) on Sioux Falls, and by 10.5–10.6 and 10.2 on Anaheim — where the strongly ordered profiles make equal-count segmentation nearly as coarse as a single value, so position-aware segmentation is what carries the error control. The final-summary entries of Table 6 are medians of budget components computed on each path–query–seed before aggregation, so they do not add to a guarantee for any single explanation.

Beyond the labeled response domain. Both the derivative centers and the 96 test paths draw base demands from $[0.65, 1.3]$ times the published OD vector, so the improvement could be specific to that labeled band. Two out-of-domain bands — low loads on $[0.40, 0.60]$ and high loads on $[1.35, 1.55]$, 24 paths each under the test generator's rules — separate the two extrapolation directions. On the high-load band, adjacent to the labeled domain, response supervision remains better on every metric (paired medians: column -1.31,

Table 7
Matched-complexity end-to-end comparison over the pre-specified tolerance grid. At each tolerance both methods segment their own learned profiles. L: median total segment count (Sioux Falls: distinct chains; Anaheim: single ordered family). Error: median normalized total error of the reported summary (percent of the reference family scale, as in Table 6). The paired difference is computed at the matched segment budget $L^* = \min(L_{\text{flow}}, L_{\text{response}})$ of each path: each method is interpolated linearly on its own five-point (error, segments) grid, and the median per-path paired difference (response − flow) receives a path-level bootstrap 95% interval (5,000 resamples). Only paths whose L^* lies inside both methods' grid support are compared (n; budgets outside a grid are excluded, not clamped): all 96 Sioux Falls paths at 0.5–2.5%, 95 at 5%, and 14 at 10%, where the two grids barely overlap; all 48 Anaheim paths except 13 at 10%. The equal-tolerance medians (no interpolation, all paths) differ from the tabulated differences by at most 0.11 percentage points and exclude zero at every tolerance on both networks.

	Flow		Response		Matched-budget difference		
Tolerance	L	Error (%)	L	Error (%)	L^*	n	Median (pp) [95% interval]
Sioux Falls (96 paths; model-error dominated)							
0.5%	233	8.80	235	5.82	233	96	-2.02 $[-2.70, -1.60]$
1%	202	9.01	201	6.04	201	96	-1.98 $[-2.68, -1.64]$
2.5%	152	9.63	152	6.91	152	96	-1.93 $[-2.59, -1.49]$
5%	117	11.35	117	8.68	117	95	-1.68 $[-2.41, -1.28]$
10%	77	13.93	77	11.51	25	14	-1.78 $[-3.06, -1.25]$
Anaheim (48 paths; compression dominated at 5%)							
0.5%	28	1.06	27	0.91	27	48	-0.16 $[-0.20, -0.08]$
1%	21	1.37	21	1.24	21	48	-0.11 $[-0.17, -0.07]$
2.5%	12	2.54	12	2.49	12	48	-0.06 $[-0.08, -0.02]$
5%	10	4.57	10	4.55	10	48	-0.02 $[-0.04, -0.01]$
10%	6	8.14	6	8.13	6	35	-0.03 $[-0.05, -0.01]$

family boundary −0.68, unrestricted extrema −5.30 percentage points, all intervals excluding zero). On the low-load band, far from the labeled domain, the advantage disappears: the family-boundary and summary errors are slightly higher under response supervision (paired medians +0.68 and +0.57 percentage points, intervals $[+0.18, +1.06]$ and $[+0.07, +0.90]$), with the column error statistically indistinguishable (+1.25, $[-0.17, +2.99]$). Response supervision therefore provides its gain inside and near the labeled response domain and preserves the error ordering under moderate extrapolation toward higher loads; the low-load band marks where the labeled derivative information stops transferring (Appendix B.9).

Bound tightness. Computed with the same 41-state trapezoidal quadrature for both sides, the exact unrestricted errors $\mathcal{K}_j(e_j)$ lie below the integrated bounds B_j of Eq. (20) (the numerical verification and quadrature accuracy are reported in Appendix C.1). The chain is tight: the median ratio $\mathcal{K}_j(e_j)/B_j$ is 0.95 for flow supervision (quartiles 0.91–0.98) and 0.95 for response supervision (quartiles 0.90–0.97). The integrated bound is therefore a tight *a posteriori* upper bound on the realized worst boundary error — it uses reference-response information, as Section 5.6 notes — and response supervision lowers the family, unrestricted and bound levels in parallel. The geometric family retains about half of the unrestricted error (median ratio $E^{\text{geo}}/\mathcal{K}_j(e_j)$ 0.51 and 0.48), so the single-family errors reported throughout should be read as lower bounds on the worst separating-region error.

Second network: Anaheim. We repeat the comparison on Anaheim by training independently on the second network, using the same conservative parameterization, objective set and evaluation principles, with the network-specific training and validation settings stated in Appendix C.13 (the six largest published OD demands are the study queries; all remaining positive demands stay fixed with endogenous route choice over the same ten-path centroid-restricted structure as Section 6.1; the model outputs all 14,060 path flows; forty-eight test demand paths — twelve per active-query group, same generation rule, fresh seed — are solved at a 21-state quadrature with reference sensitivities); the Anaheim rows of Table 6 report the outcome. Anaheim's attribution columns are easier to learn than Sioux Falls' — the free-flow-weighted response dominates the column and both objectives reach column errors near 1.5% — but the ordering is the same and is consistent across seeds: response supervision lowers the median column error from 1.64% to 1.31% and

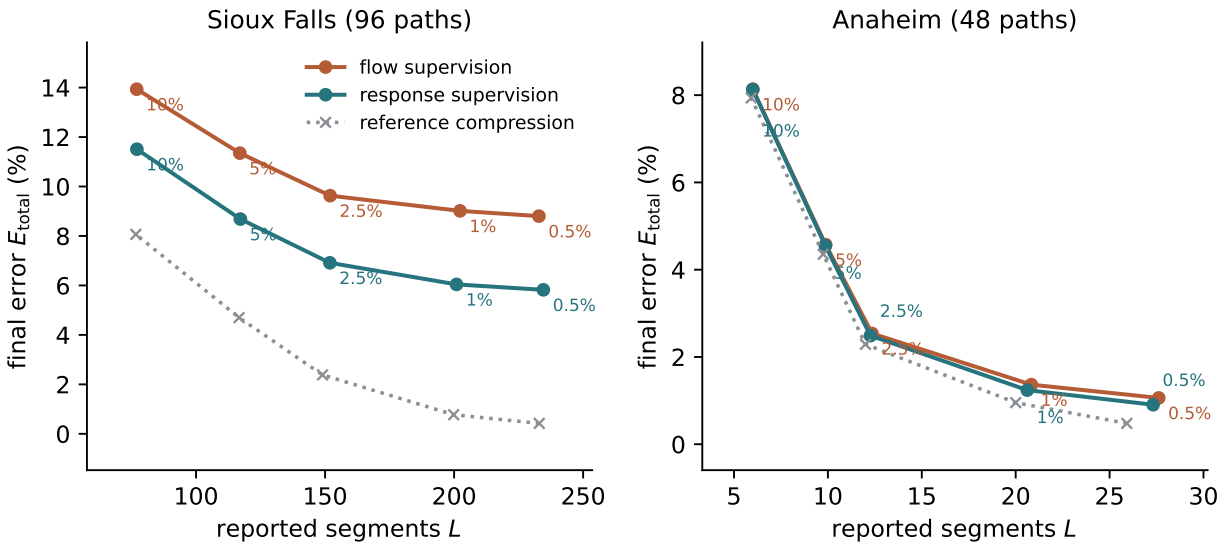


Figure 7: Final error of the reported summary against the number of reported segments, as the family tolerance varies over 0.5–10% (annotated; percent of the learned family maximum). Sioux Falls (left): model-error dominated — the response-supervised curve (teal) lies below the flow-supervised curve (ochre) at every segment count, so better responses convert into either fewer segments at equal accuracy or more accurate summaries at equal complexity. Anaheim (right): compression dominated at the working tolerance — both curves track the gray reference-compression curve, the error of compressing the reference profile itself at each tolerance, plotted at its own segment count (× marks). Per-tolerance segmentation recomputed from the final learned profiles; common per-query normalization as in Table 6.

the family-boundary error from 0.94% to 0.78%, improves 46–48 of 48 paths in column error and 44–47 of 48 in family-boundary error across the five seeds, and does so while flow accuracy is again indistinguishable (0.024% and 0.019%) and the incremental flow error falls from 2.46% to 2.03%. Path-level paired bootstrap quantifies the uncertainty of the smaller Anaheim margins (10,000 resamples, paths as resampling unit, seeds averaged within path): the median paired column-error difference is -0.29 percentage points with a 95% interval of $[-0.33, -0.21]$, improving all 48 paths, and the paired geometric difference is -0.17 with interval $[-0.19, -0.10]$, improving 47 of 48; both intervals exclude zero, and each of the five seeds shows the same direction with its interval excluding zero. The reported summaries inherit the same budget structure as Sioux Falls (absolute path medians: model term 17.0 → 13.1, summary term at the tolerance, 99.7 and 99.3 vehicle-minutes per hour; normalized in Table 6); the total stays within the budget on every path, and the paired total error improves on 73.3% of path–seed units. At the selected 5% tolerance Anaheim's total error is compression-dominated — the model term is below one percent of the family scale — so further reduction of the reported summaries must come primarily from a tighter tolerance, with model improvement raising the ceiling once the tolerance tightens (Figure 7 traces both levers). The direction of every conclusion — better explanations from response supervision, and a dominant response term — therefore does not depend on the network; which term dominates the total error does: at the working tolerance the Anaheim budget is compression-dominated while the Sioux Falls budget carries the larger model term, as the error budget makes explicit.

8. Conclusions

An OD demand effect should be represented neither by one global scalar nor by an unstructured link map, but by the coarsest boundary-attribution profile whose physical and approximation errors are explicitly controlled. The three parts of the paper deliver that representation in sequence.

Topology characterizes when one transferable value is guaranteed uniformly over all flow-conserving responses: for two-sided connected regions, precisely when the marginal outcome weights agree on the block

atoms traversed by the queried OD paths. Multiple queries merge these atoms transitively, so the admissible explanation is a query-dependent structural property rather than a global network attribute, identified before any model is trained. For a learned assignment map this theorem is a structural certificate, checkable at every state along the attribution path.

When one value is insufficient, node divergence determines what is reported: the exact attribution range, the minimax common summary and its error, minimum-segment summaries with explicit approximation errors, and the maximum disagreement between learned and reference explanations. The adaptive reporting rule allocates spatial resolution only where the network response requires it, and the reported summary carries a total error budget that separates model error from compression error. Structural compatibility and compression are certified from the learned model and the reported profile alone; learned-reference fidelity is an offline validation guarantee requiring reference assignments.

For learned assignment, response supervision — the training mechanism the error budget identifies — improves the reported explanation at matched cost. At every segment count on the pre-specified tolerance grid the response-supervised final error is lower on both networks, with paired intervals excluding zero; on Sioux Falls the error that flow supervision cannot reach within the grid (8.8% at 233 segments) is attained by response supervision with 117 segments, and an 11% error needs 35 fewer segments. The mechanism ablation shows the gains are structural rather than an artifact of the pretraining recipe: the exact path-share parameterization dominates direct conservation mechanisms under either pretraining loss, and derivative supervision improves every mechanism. The improvement extends beyond the labeled response domain on the high-load side; on the low-load side the advantage disappears, marking the boundary of the labeled regime. Accurate flow prediction is not sufficient for spatially trustworthy OD explanations; together, these results turn spatial attribution from an ad hoc boundary aggregation into a transport-specific, mathematically controlled and learnable explanation framework.

Limitations. The analysis fixes the road representation, outcome definition, attribution path, and admissible boundary family; subdividing roads changes which crossing-link costs a boundary includes. Per-query attributions under joint demand changes are path-specific, although their signs, summary-error levels, and mechanism rankings were stable across the twenty-two tested alternative orders (Section 6.2). The numerical evidence concerns static fixed-path logit-SUE models on two benchmark networks; extending the framework to dynamic assignment and broader demand domains remains a direction for future work.

Code and data availability

Code, configurations, processed result arrays, and scripts reproducing all tables and figures are available through an anonymized archival repository; its access link accompanies the submission. The archive records software versions, random seeds, TNTP checksums, and the generating script for each reported result; trained checkpoints are regenerated by the released scripts under the fixed seeds. Public access will be enabled upon acceptance. The Sioux Falls and Anaheim networks and demands are available from the public Sioux Falls TNTP repository and Anaheim TNTP repository (Transportation Networks for Research Core Team, 2024); training targets are synthetic fixed-path logit-SUE assignments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Amos, B., Kolter, J.Z., 2017. OptNet: Differentiable optimization as a layer in neural networks, in: Proceedings of the 34th International Conference on Machine Learning, pp. 136–145.

Aumann, R.J., Shapley, L.S., 1974. Values of Non-Atomic Games. Princeton University Press, Princeton, NJ.

Baez, A., Zhang, W., Ma, Z., Das, S., Nguyen, L.M., Daniel, L., 2024. Guaranteeing conservation laws with projection in physics-informed neural networks. doi:10.48550/arXiv.2410.17445, arXiv:2410.17445. arXiv:2410.17445 [cs.LG].

Bar-Gera, H., 2002. Origin-based algorithm for the traffic assignment problem. Transportation Science 36, 398–417. doi:10.1287/trsc.36.4.398.549.

Billera, L.J., Heath, D.C., Raanan, J., 1978. Internal telephone billing rates—a novel application of non-atomic game theory. Operations Research 26, 956–965. doi:10.1287/opre.26.6.956.

Czarnecki, W.M., Osindero, S., Jaderberg, M., Swirszcz, G., Pascanu, R., 2017. Sobolev training for neural networks, in: Advances in Neural Information Processing Systems.

Dhamdhere, K., Sundararajan, M., Yan, Q., 2019. How important is a neuron?, in: International Conference on Learning Representations (ICLR). URL: https://openreview.net/forum?id=SylKoo0cKm.

Even, S., Tarjan, R.E., 1976. Computing an st-numbering. Theoretical Computer Science 2, 339–344. doi:10.1016/0304-3975(76)90086-4.

Guo, X., Zhang, Q., Jiang, J., Peng, M., Zhu, M., Yang, H.F., 2024. Towards explainable traffic flow prediction with large language models. Communications in Transportation Research 4, 100150. doi:10.1016/j.commtr.2024.100150.

Hopcroft, J., Tarjan, R., 1973. Algorithm 447: Efficient algorithms for graph manipulation. Communications of the ACM 16, 372–378. doi:10.1145/362248.362272.

Josefsson, M., Patriksson, M., 2007. Sensitivity analysis of separable traffic equilibrium equilibria with application to bilevel optimization in network design. Transportation Research Part B: Methodological 41, 4–31. doi:10.1016/j.trb.2005.12.004.

Lassen, O.B., Agriesti, S., Eldafrawi, M., Gammelli, D., Cantelmo, G., Gentile, G., Pereira, F.C., 2025. Learning traffic flows: Graph neural networks for metamodelling traffic assignment, in: 2025 9th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS), IEEE. Doi: 10.1109/MT-ITS68460.2025.11223524 (arXiv:2505.11230).

Li, J., Nie, Y.M., 2026. Traffic assignment as a differentiable program. Transportation Research Today 1, 100014. doi:10.1016/j.trt.2026.100014.

Liu, T., Meidani, H., 2024. End-to-end heterogeneous graph neural networks for traffic assignment. Transportation Research Part C: Emerging Technologies 165, 104695. doi:10.1016/j.trc.2024.104695.

Liu, T., Meidani, H., 2025. Multi-class traffic assignment using multi-view heterogeneous graph attention networks. Expert Systems with Applications 286, 128072. doi:10.1016/j.eswa.2025.128072.

Liu, Z., Yin, Y., Bai, F., Grimm, D.K., 2023. End-to-end learning of user equilibrium with implicit neural networks. Transportation Research Part C: Emerging Technologies 150, 104085. doi:10.1016/j.trc.2023.104085.

Liu, Z., Yin, Y., Bai, F., Grimm, D.K., 2025. End-to-end learning of user equilibrium: Expressivity, generalization and learning guarantees. Transportation Science 59, 853–882. doi:10.1287/trsc.2023.0489.

Lundstrom, D.D., Huang, T., Razaviyayn, M., 2022. A rigorous study of integrated gradients method and extensions to internal neuron attributions, in: Proceedings of the 39th International Conference on Machine Learning (ICML), PMLR. pp. 14485–14508. URL: https://proceedings.mlr.press/v162/lundstrom22a.html.

Oxley, J., 2011. Matroid Theory. Number 21 in Oxford Graduate Texts in Mathematics. 2nd ed., Oxford University Press, Oxford. doi:10.1093/acprof:oso/9780198566946.001.0001.

Patriksson, M., Rockafellar, R.T., 2003. Sensitivity analysis of aggregated variational inequality problems, with application to traffic equilibria. Transportation Science 37, 56–68. doi:10.1287/trsc.37.1.56.12821.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics 378, 686–707. doi:10.1016/j.jcp.2018.10.045.

Richter-Powell, J., Lipman, Y., Chen, R.T.Q., 2022. Neural conservation laws: A divergence-free perspective, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022), pp. 38075–38088.

Sarıyer, G., Mangla, S.K., Sözen, M.E., Li, G., Kazançoğlu, Y., 2024. Leveraging explainable artificial intelligence in understanding public transportation usage rates for sustainable development. Omega 127, 103105. doi:10.1016/j.omega.2024.103105.

Seo, T., 2026. End-to-end differentiable network traffic simulation with dynamic route choice. doi:10.48550/arXiv.2604.11380, arXiv:2604.11380. arXiv:2604.11380.

Shadrin, A., 2014. The Landau–Kolmogorov inequality revisited. Discrete and Continuous Dynamical Systems 34, 1183–1210. doi:10.3934/dcds.2014.34.1183.

Sheffi, Y., 1985. Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods. Prentice-Hall, Englewood Cliffs, NJ.

Shi, R., Mo, Z., Di, X., 2021. Physics-informed deep learning for traffic state estimation: A hybrid paradigm informed by second-order traffic models, in: Proceedings of the 35th AAAI Conference on Artificial Intelligence, pp. 540–547. doi:10.1609/aaai.v35i1.16132.

Singal, R., Michailidis, G., Ng, H., 2021. Flow-based attribution in graphical models: A recursive shapley approach, in: Proceedings of the 38th International Conference on Machine Learning (ICML), pp. 9733–9743.

Sundararajan, M., Taly, A., Yan, Q., 2017. Axiomatic attribution for deep networks, in: Proceedings of the 34th International Conference on Machine Learning (ICML), PMLR. pp. 3319–3328. URL: https://proceedings.mlr.press/v70/sundararajan17a.html.

Tobin, R.L., Friesz, T.L., 1988. Sensitivity analysis for equilibrium network flow. Transportation Science 22, 242–250. doi:10.1287/trsc.22.4.242.

Transportation Networks for Research Core Team, 2024. Transportation networks for research. https://github.com/bstabler/TransportationNetworks. Sioux Falls network, trip and node files and Anaheim network and trip files; accessed September–October 2026.

Valente, M., Dias, T.C., Guerra, V., Ventura, R., 2025. Physics-consistent machine learning with output projection onto physical manifolds. Communications Physics 8, 433. doi:10.1038/s42005-025-02329-1.

Wang, J., Wiens, J., Lundberg, S., 2021. Shapley flow: A graph-based approach to interpreting model predictions, in: Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS), pp. 721–729.

Yang, H., Bell, M.G.H., 2007. Sensitivity analysis of network traffic equilibria revisited: the corrected approach, in: Heydecker, B.G. (Ed.), Mathematics in Transport: Selected Proceedings of the Fourth IMA International Conference on Mathematics in Transport. Elsevier, Oxford, pp. 373–395.

Yuan, W., Zhu, Q., Liu, X., Ding, Y., Zhang, H., Zhang, C., 2022. Sobolev training for implicit neural representations with approximated image derivatives. doi:10.48550/arXiv.2207.10395, arXiv:2207.10395.

van Zuylen, H.J., Willumsen, L.G., 1980. The most likely trip matrix estimated from traffic counts. Transportation Research Part B: Methodological 14, 281–293. doi:10.1016/0191-2615(80)90008-9.

A. Mathematical foundations and proofs

A.1. Conventions and conservation transfer

The incidence matrix $B \in \mathbb{R}^{n\times m}$ has entries -1 at an arc's origin and $+1$ at its destination; $\mu_j = e_{t_j} - e_{s_j}$ and $M = [\mu_1, \dots, \mu_K]$. Write $Z = \ker B$ and let $B^\dagger$ be the Moore–Penrose pseudoinverse. All norms below are Euclidean or their induced operator norms unless indicated otherwise. Graphs are loopless multigraphs: parallel and reciprocal arcs remain distinct edges, including in two-edge cycles. Block atoms and the block-cut tree are taken on the underlying undirected multigraph. Formally, $\iota(v)$ is the cut-vertex node when v is an articulation vertex and otherwise its unique block node; $\mathcal{P}(s,t)$ lists the atoms on the unique tree path from $\iota(s)$ to $\iota(t)$. A signed simple s–t path vector p satisfies $Bp = \mu$ and the full solution set is $p + Z$. Such a path uses only atoms of $\mathcal{P}(s,t)$ and meets every atom on it.

Signed responses. The circulation and response vectors are signed first-order sensitivities, not nonnegative commodity flows. At a strictly positive flow, sufficiently small signed circulation perturbations also preserve nonnegativity. A restricted response family is treated explicitly below.

Corollary A.1 (IG exchange)**.** *For an exactly conservative C^1 field on an open neighborhood of the attribution path, $Bg_j = \mu_j \Delta q_j$ (Eq. (4)).*

Proof. Differentiate $Bf(q) = Mq$ to obtain $BJ_f(q) = M$, then integrate the jth column along $\gamma(\lambda) = q^0 + \lambda\Delta q$.

A.2. Cycle structure and the classification proofs

Lemma A.2 (structure of circuit classes)**.** *The non-bridge atoms are the non-singleton circuit classes; each bridge is a singleton class. Moreover $Z = \bigoplus_A Z_A$, where $Z_A = \{z \in Z : \operatorname{supp} z \subseteq A\}$, and the summands have disjoint edge supports.*

Proof. No simple cycle crosses an articulation vertex into another block: it would revisit that vertex. Within a non-bridge block, any two edges lie on a common cycle. One way to see this is to subdivide the two edges; the resulting block is still biconnected, so its two new vertices lie on a common cycle, which descends to the required cycle. This is the usual circuit-class characterization of the graphic matroid (Oxley, 2011). Every signed circulation is a linear combination of signed simple-cycle vectors: subtract the fundamental cycles of a spanning forest to leave a circulation on a forest, necessarily zero by successive leaf elimination. Grouping these cycles by their blocks gives the direct sum, and distinct blocks have disjoint edge sets.

Lemma A.3 (circulation-preserving diagonal maps)**.** *For $D = \operatorname{diag}(w)$, $DZ \subseteq Z$ if and only if w is constant on every non-bridge atom; bridge weights are free.*

Proof. For any signed simple-cycle vector z_C, a circulation supported on that cycle has equal signed amplitudes on successive edges, by node balance. Thus $BDz_C = 0$ forces w constant on C, and Lemma A.2 propagates equality throughout each atom. Conversely, atomwise constancy makes D a scalar on every Z_A, so it preserves their direct sum. Bridges carry no circulation.

Table A.1
Constructive query-dependent compatibility on Figure 2 of the main text, uniform pendant weights. Ranges use all 512 separating sets per query and a conservative unit response; the weighted circulation residual is below 5×10^{-16} in all stages, so these ranges are independent of that response. The intentionally chosen linear weights illustrate compatibility; they are not the marginal costs of the original TSTT outcome.

Stage	$w_{A,b_1,B}$	$w_{b_2,C}$	w_D	Queried ODs	Account ranges
Initial query	1	2	3	(s,u)	$[1,1]$
Add query	1	2	3	$(s,u);(u,t)$	$[1,1];[1,2]$
Adjust merged class	1	1	3	$(s,u);(u,t)$	$[1,1];[1,1]$

Proof (Theorem 5). Add a virtual reverse OD edge (t,s) and set

$$\bar{B} = [B, -\mu], \qquad \bar{D} = \operatorname{diag}(w, \alpha).$$

For a fixed α, property (9) is equivalent to $\bar{D} \ker \bar{B} \subseteq \ker \bar{B}$. Indeed, a kernel vector (x,a) obeys $Bx = a\mu$. If $a \neq 0$, apply (9) to x/a to obtain $BDx = a\alpha\mu$. If $a = 0$, subtract the identities for $p + x$ and p, with any fixed $Bp = \mu$, to obtain $BDx = 0$. These are exactly the two cases of $\bar{B}\bar{D}(x,a) = 0$. Conversely, applying kernel invariance to $(x,1)$ gives (9) for every $Bx = \mu$, at that same α.

By Lemma A.3, this invariance holds precisely when the augmented weights are constant on its block atoms. The virtual edge and a simple s–t path form a cycle meeting every original atom on $\mathcal{P}(s,t)$, so these atoms merge with the virtual edge into one circuit class. No other atoms join it: every new simple cycle contains the virtual edge and a simple s–t path, which uses only those atoms. Other original atoms retain their classes. Thus the augmented criterion is exactly within-atom constancy everywhere and common value α on $\bigcup \mathcal{P}(s,t)$. This proves both directions and identifies $\alpha = w^*$ without changing the universal response family.

Proof (Theorem 6 and Corollary 7). Apply the single-OD theorem to each query at the same w. Necessity requires equality within every atom and among all atoms co-traversed by a queried block path. Their transitive closure is exactly the compatibility-class partition. Conversely, class constancy supplies both single-OD conditions for every query. Along the attribution path, exact conservation gives $BJ_f(\gamma(\lambda))\mathbf{e}_j = \mu_j$; the static theorem therefore yields

$$Bc_j = \Delta q_j \int_0^1 BD_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda = \Delta q_j \left(\int_0^1 w^*_{k(j)}(\lambda) \, d\lambda \right) \mu_j.$$

Corollary A.4 (all-OD class collapse)**.** *Requiring the criterion at one common w for every ordered pair of distinct vertices makes all atoms in a connected component share one constant.*

Proof. For two atoms incident to the same cut vertex, choose endpoints on their opposite sides. Their OD block path contains both atoms, forcing their constants to agree. Connectivity of the block-cut tree propagates the equality throughout the component.

Remark A.1 (a common flow realization). The merging proof requires one common weight vector. Computing the queries at different realized flows does not in general supply this premise.

Example B: query-dependent compatibility under the unrestricted criterion. This example uses the uniform pendant configuration of Figure 2 — all three D weights equal to 3 — and evaluates the unrestricted criterion over all 512 separating sets per query, complementing the connected-region Example A of the main text, whose heterogeneous pendant is invisible to queried connected boundaries. The calculation uses the minimum-norm solution of $Bx = \mu$ and checks $BDZ = 0$ on a basis of $\ker B$, which establishes response independence of the displayed boundary vector. The theorem supplies the uniform common-value guarantee in stages one and three.

A.3. Direction, integration, and network representation

At fixed weights $D = \mathrm{diag}(w)$, a realized model balances when $BDJ_j = \alpha_j \mu_j$ for its actual response. *Universal directional balance* requires $BDx \in \mathrm{span}\{\mu_j\}$ for every $Bx = \mu_j$, allowing its coefficient to depend on x; *structural cut consistency* additionally requires one common coefficient across all such responses, and is the property classified in the main text. For non-collinear queries with shared weights, universal directional balance already implies structural cut consistency because $BD \ker B \subseteq \bigcap_j \mathrm{span}\{\mu_j\} = \{0\}$; a query and its reverse alone do not suffice.

Proposition A.5 (universal directional criterion)**.** *Let $\hat{x}_j = B^\dagger \mu_j$, let N span Z with orthonormal columns, and let P_{μ_j} project orthogonally onto $\mathrm{span}\{\mu_j\}$. At fixed w, universal directional balance is equivalent to*

$$(I - P_{\mu_j})BD\hat{x}_j = 0, \qquad (I - P_{\mu_j})BDN = 0.$$

The scale may depend on the response. Requiring one common boundary value additionally requires $BDN = 0$. If directional balance holds along the whole path for a conservative model, its integrated column balances.

Proof. Every response is $\hat{x}_j + z$, $z \in Z$. Apply $I - P_{\mu_j}$ to $BD(\hat{x}_j + z)$ and vary z. The scale is $\mu_j^\top BD(\hat{x}_j + z)/\|\mu_j\|^2$ and is independent of z precisely when $BDZ = 0$. Integrating the realized directional identity gives the endpoint conclusion. Equivalently, invariance on all incidence fibers, $Bx = By \Rightarrow BDx = BDy$, is exactly $DZ \subseteq Z$, since $x - y$ ranges over Z.

Example A.6 (strict separation of the criteria)**.** *On the diamond $(s,a),(a,t),(s,b),(b,t)$ with weights $(1,1,1,2)$, the upper-path response $(1,1,0,0)$ has divergence μ after weighting, but the lower-path response has weighted divergence $(-1,0,-1,2)$ in node order (s,a,b,t), outside $\mathrm{span}\{\mu\}$. On the triangle $(s,t),(t,u),(u,s)$ with weights $(1,0,0)$, every response $x = (1+c,c,c)$ satisfies $BDx = (1+c)\mu$; direction balances, but the common boundary value fails because the circulation $(1,1,1)$ maps to $\mu \neq 0$.*

If directional balance holds for all queries at common w, differences give $BDZ \subseteq \bigcap_j \mathrm{span}\{\mu_j\}$. For non-collinear queries this intersection is $\{0\}$, so directional balance implies the common-value property. This uses the full response space Z.

Restricted families. For $S \subseteq Z$ with basis N_S, test responses in $x_j^{\mathrm{ref}} + S$, where $Bx_j^{\mathrm{ref}} = \mu_j$. The same proof gives the direction tests with $\hat{x}_j, N$ replaced by x_j^{ref}, N_S; common boundary value replaces the second test by $BDN_S = 0$. Using $\hat{x}_j$ requires $\hat{x}_j - x_j^{\mathrm{ref}} \in S$. For a fixed-path model $f = \Delta h$, $A_{\mathrm{OD}} h(q) = q$ and $B\Delta = MA_{\mathrm{OD}}$. Choose a path vector a_j with $A_{\mathrm{OD}} a_j = \mathbf{e}_j$ and set $x_j^{\mathrm{ref}} = \Delta a_j$. Differentiation yields

$$J_f \mathbf{e}_j - x_j^{\mathrm{ref}} = \Delta(\partial_j h - a_j) \in \Delta \ker A_{\mathrm{OD}} =: T_{\mathrm{swap}}.$$

This space includes OD path exchanges of every OD: background ODs may reroute even when their demands remain fixed.

Example A.7 (endpoint cancellation)**.** *For two series bridges, $f(q) = (q,q)$ and $T(F) = F_1 + F_2 + [\sin(2\pi F_1) - \sin(2\pi F_2)]/(2\pi)$ give*

$$c_1 - c_2 = \frac{\sin(2\pi q^1) - \sin(2\pi q^0)}{\pi}.$$

For $q^0 = 0$, the nonzero balanced endpoints in $[0,1]$ are $q^1 = 1/2, 1$, although the two marginal weights differ almost everywhere. At $q^1 = 0.3$ balance fails. Endpoint cancellation therefore does not establish the pointwise classification conditions.

Example A.8 (network subdivision)**.** *A single edge with $f(q) = q$, $T(F) = \tau F$, $\tau > 0$, has cut attribution $\tau \Delta q$. Subdivide it into $(s,v),(v,t)$ and set $f'(q) = (q,q)$, $T'(F_1,F_2) = \tau[\alpha F_1 + (1-\alpha)F_2]$, $0 < \alpha < 1$. The composite demand-to-outcome map is unchanged, but $c' = \tau \Delta q(\alpha, 1-\alpha)$ and $(Bc')_v = \tau(2\alpha - 1)\Delta q$. The two cuts take values $\tau\alpha\Delta q$ and $\tau(1-\alpha)\Delta q$. They agree only for $\alpha = 1/2$, when each is half the undivided attribution; their sum remains the scalar contribution $\tau \Delta q$. The invariance studied here thus concerns conservative responses on a fixed network with a fixed edgewise outcome definition, rather than invariance under subdivision.*

A.4. Approximate conservation and attribution balance

The residual identity (22) shows that output residuals alone do not control attribution columns. For unit $v \in \operatorname{im} B$ and $r_\varepsilon(q) = \varepsilon v \tanh((q_1 - q_2)/\varepsilon)$, the field $f_\varepsilon(q) = B^\dagger(Mq + r_\varepsilon(q))$ has residual exactly r_ε, globally bounded by ε and zero on $q_1 = q_2$; yet on $q(\lambda) = \lambda(\delta, \delta)$ its partial derivatives are $v, -v$, so $R_1 = \delta v$ and $R_2 = -\delta v$, and its residual Jacobian Lipschitz constant grows as ε^{-1}. Added to an exactly conservative field with minimum flow $\varrho > 0$ on a compact positive-demand box, it preserves strictly positive flows when $\varepsilon \|B^\dagger\|_2 < \varrho$; the traffic experiments use exactly conservative models, so these bounds delimit scope rather than enter the numerics.

Theorem A.9 (local neighborhood bound)**.** *Let $r \in C^{1,1}(\mathcal{N}; \mathbb{R}^n)$ on an open neighborhood of the attribution path, with $\operatorname{Lip}(J_r) \le H$. Suppose, for some $\rho > 0$ and every j, λ, the coordinate segment $\gamma(\lambda) + [-\rho, \rho]\mathbf{e}_j$ lies in $\mathcal{N}$ and its two endpoint residual norms are at most ε. Then*

$$\|R_j\| \le |\Delta q_j|(\varepsilon/\rho + H\rho/2).$$

Proof. At $q = \gamma(\lambda)$, Taylor's formula gives $r(q \pm \rho\mathbf{e}_j) = r(q) \pm \rho\partial_j r(q) + \xi_\pm$ with $\|\xi_\pm\| \le H\rho^2/2$. Subtraction bounds $\|\partial_j r(q)\|$ by $\varepsilon/\rho + H\rho/2$; integration gives the claim.

The optimized square-root rate is sharp. Indeed $r_\varepsilon = \varepsilon v \sin((q_1 - q_2)/\sqrt{\varepsilon})$ has supremum norm ε and $\operatorname{Lip}(J_{r_\varepsilon}) \le 2$, while the same diagonal path gives $R_1 = \delta\sqrt{\varepsilon}v = -R_2$. Thus a uniform $o(\sqrt{\varepsilon})$ rate is impossible for bounded Jacobian Lipschitz constant.

Lemma A.10 (minimum-norm support)**.** *For a signed s–t path p, $\hat{x} = B^\dagger\mu = (I - P_Z)p$ is supported on $\bigcup \mathcal{P}(s, t)$.*

Proof. The minimum-norm point of $p + Z$ is its orthogonal projection onto $Z^\perp$. Lemma A.2 gives $P_Z = \sum_A P_{Z_A}$ with disjoint supports; both p and every $P_{Z_A}p$ vanish on off-path atoms.

Proposition A.11 (combined dispersion–residual bound)**.** *Let $f \in C^1$ near the attribution path, $r = Bf - Mq - b$ with the constant background balance b of Section 5.6 ($b = 0$ without background), and let $w(\lambda) = \nabla T(f(\gamma(\lambda)))$ be continuous. Let $\bar{w}(\lambda)$ be the compatibility-class mean vector and $\eta(\lambda) = \|w(\lambda) - \bar{w}(\lambda)\|_\infty$. Writing $J_j(\lambda) = J_f(\gamma(\lambda))\mathbf{e}_j$ and $d_j = (I - P_{\mu_j})Bc_j$, we have*

$$\|d_j\|_2 \le |\Delta q_j| \|B\|_2 \int_0^1 \left[\eta(\lambda)\|J_j(\lambda)\|_2 + \|\bar{w}(\lambda)\|_\infty \|B^\dagger\|_2 \|\partial_j r(\gamma(\lambda))\|_2\right] d\lambda.$$

Proof. Differentiation gives $BJ_j = \mu_j + \partial_j r$ — b is constant and drops out of the derivative — so $J_j = \hat{x}_j + z_j + B^\dagger\partial_j r$, $z_j \in Z$. Let $\bar{D} = \operatorname{diag}(\bar{w})$. Class constancy gives $B\bar{D}z_j = 0$ by Lemma A.3. By Lemma A.10, $\hat{x}_j$ is supported on the OD block path, whose atoms lie in one compatibility class $k(j)$; hence $B\bar{D}\hat{x}_j = \bar{w}_{k(j)}\mu_j$. Substituting leaves $(I - P_{\mu_j})B[\operatorname{diag}(w - \bar{w})J_j + \bar{D}B^\dagger\partial_j r]$ inside the projector; the projector is non-expansive, the two diagonal operator norms are η and $\|\bar{w}\|_\infty$, and integration gives the bound. For exact conservation only the dispersion term remains; for $\eta = 0$ only the derivative residual remains. Theorem A.9 can bound that derivative term under its local hypotheses.

A.5. Exact boundary error and its path bound

Proof. Set $b = Be_j$ and $I = V \setminus \{s_j, t_j\}$. Since $\mathbf{1}^\top b = 0$, Eq. (15) applied to e_j gives an interval of extrema with midpoint

$$-b_{s_j} - \frac{1}{2}\sum_{v \in I} b_v = \frac{1}{2}(b_{t_j} - b_{s_j})$$

and half-width $\frac{1}{2}\sum_{v \in I} |b_v|$. The larger absolute endpoint is the absolute midpoint plus this half-width, proving Eq. (19). This argument concerns the attained extrema; it does not require every value between them to be attained. The map $\mathcal{K}_j$ is a seminorm because it is a sum of absolute linear functionals. Moreover $e_j = \Delta q_j \int_0^1 h_j(\gamma(\lambda))\, d\lambda$. Its homogeneity and triangle inequality therefore yield

$$\epsilon_j^{\mathcal{U}} \le \mathcal{K}_j(e_j) \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda,$$

proving the path statement. Its kernel is $\ker B$: if the seminorm vanishes, all interior divergences vanish and $b_{s_j} = b_{t_j}$; the zero sum then makes both endpoints zero.

A.6. Connected-region classification

We retain the loopless multigraph conventions of Appendix A.1; reciprocal arcs are distinct undirected edges when defining atoms and connectivity.

Lemma A.12 (localization of a separating bond)**.** *If $\mathcal{G}[U]$ and $\mathcal{G}[V \setminus U]$ are connected, all edges of $\delta(U)$ lie in one block atom (or its singleton bridge). If $s \in U$ and $t \notin U$, that atom lies on $\mathcal{P}(s,t)$.*

Proof. For two distinct crossing edges, connect their endpoints by a simple path inside U and by another inside its complement. The two paths have disjoint vertex sets; together with the crossing edges they form a simple cycle, including the possible two-edge cycle. Hence all crossing edges share a circuit class. A singleton crossing edge is a bridge. A simple s–t path crosses this boundary, so it meets its atom; every atom met by such a path lies on $\mathcal{P}(s,t)$.

Lemma A.13 (connected prefixes in an atom)**.** *Let A be a non-bridge atom and let $a \neq b$ be two of its vertices. Its vertices admit an ordering $v_1 = a, \ldots, v_k = b$ such that every proper nonempty prefix and its suffix induce connected subgraphs. The assertion includes two-vertex parallel-edge atoms.*

Proof. For two vertices the assertion is immediate. Otherwise add an a–b edge temporarily if needed. The connected-prefix property is the classical *st*-numbering property (Even and Tarjan, 1976); we give a construction here. A biconnected graph with this edge admits an open-ear construction starting from a cycle containing that edge. For completeness, this construction follows directly from vertex connectivity. If vertices remain outside the current subgraph, any component of those vertices has at least two distinct attachment vertices in the current subgraph: a unique attachment would be an articulation of the original graph. A path through the component between two attachments supplies the next ear. Repeat until all vertices have entered. Edges left over are irrelevant to the ordering. The initial cycle exists because the added edge is not a bridge.

Order the other arc of the initial cycle from a to b. For each subsequent ear, let u precede v among its two attachment vertices; insert its internal vertices, in their order from u to v, immediately after u. Every new internal vertex has an earlier and a later neighbor. Existing vertices keep these neighbors. The first and last vertices remain a, b. Every vertex except a therefore has an earlier neighbor and every vertex except b a later neighbor. Repeated steps along these neighbors connect every prefix to a and every suffix to b. Removing the temporary edge does not affect this argument: it is never internal to a proper prefix or suffix.

Lemma A.14 (external components of a traversed atom)**.** *Let A be an atom of a connected loopless multigraph. Delete all edges of A while keeping its vertices. The remaining graph has exactly one connected component T_v for each $v \in V(A)$, containing v and no other vertex of A, and these components partition V. This holds with parallel edges and reciprocal arcs; for a bridge atom $\{e\}$, deleting e leaves exactly the two connected components of $G - e$, one containing each endpoint of e.*

Proof. No two distinct vertices $u, v \in V(A)$ are joined by a path that avoids the edges of A. Suppose such a path P existed. For a bridge atom $\{e\}$, the path P together with e would contain a simple cycle through the bridge e, but a bridge lies on no cycle. For a non-bridge atom (a block), pick any u–v path Q inside A; it uses only edges of A, possibly the single edge, or the two-edge cycle formed by parallel or reciprocal arcs. Walking along P from u until its first vertex w that lies on Q and returning to u along Q exhibits a simple cycle containing at least one edge of $Q \subseteq A$ and at least one edge of P outside A. Every simple cycle is contained in one block, so this cycle's block would contain edges of A and edges outside A, contradicting that a block's edge set is exactly the non-bridge atom A. Hence each $v \in V(A)$ reaches no other vertex of A without using A's edges, so the components $\{T_v\}_{v \in V(A)}$ are pairwise distinct; vertices outside A attach to exactly one of them, again because any path from such a vertex into A ends at a first vertex of A, and two different endpoints would contradict the claim just proved. The T_v therefore partition V.

Proof (Theorem 2). For sufficiency, Lemma A.12 places every crossing edge in one traversed atom. Its weight is α, so $-\mathbf{1}_U^\top BDx = -\alpha \mathbf{1}_U^\top Bx = -\alpha \mathbf{1}_U^\top (e_t - e_s) = \alpha$. No other edge enters this boundary sum.

For necessity, fix a traversed atom A, delete its edges, and form the components T_v of Lemma A.14. There are distinct vertices a, b with $s \in T_a$ and $t \in T_b$; these are the entry and exit of the OD block path in A. For a bridge atom, take the cut consisting of its s-side component. Since its unit conservative crossing response is fixed, the assumed attribution immediately gives that bridge's weight α.

For a non-bridge atom, choose the ordering of Lemma A.13. Set $U_i = \bigcup_{\ell \le i} T_{v_\ell}$ for $1 \le i < k$. The bags and the prefix/suffix connectivity show $U_i \in \mathcal{U}_{st}^{\mathrm{conn}}$. Its crossing edges are exactly those of the corresponding prefix in A. Now choose any signed local response y with $B_A y = e_b - e_a$, where B_A is the incidence matrix on A. Extend it by a signed unit path from s to a inside T_a and from b to t inside T_b, with zero on other edges. The result satisfies $Bx = \mu$. Thus the assumption gives

$$-\sum_{\ell \le i} (B_A D_A y)_{v_\ell} = \alpha \quad (1 \le i < k).$$

The first prefix gives $(B_A D_A y)_a = -\alpha$, differences of consecutive prefixes give zero at every interior vertex, and zero total incidence sum gives $(B_A D_A y)_b = \alpha$. Consequently $B_A D_A y = \alpha(e_b - e_a)$ for every local conservative y. The single-OD classification (Theorem 5), applied to the one-atom graph A, forces every edge in A to have weight α. This holds for every traversed atom. The sufficiency argument already shows that no off-path weight is constrained.

Proof (Corollary 3). Apply Theorem 2 at the common w to each query. Each queried path requires one weight on all its edges; intersecting paths propagate equality transitively. Conversely these transitive equalities satisfy every single-query condition. Atoms not on any queried path enter none of these equalities.

A.7. Finite-change supervision and joint-path implication

Proof (Proposition B.1). Write $r \in \{L, S\}$ and $\kappa^r(q) = g(f^r(q))$ componentwise. Along a single-coordinate segment $q(u) = q^- + u(q^+ - q^-)$, the chain rule gives

$$\frac{d}{du}\kappa^r(q(u)) = (q_j^+ - q_j^-) D^r(q(u)) J_j^r(q(u)).$$

Integration gives the attribution column (Eq. (2)) exactly as $c_j^r = \kappa^r(q^+) - \kappa^r(q^-) = z_j^r$. Corollary 12 applied to their difference proves the stated maximum-error identity. No reference input derivative is needed to obtain this label.

For the local-to-joint implication set $A(q) = B[\kappa^L(q) - \kappa^S(q)]$. Equality of every separating boundary attribution of a sufficiently small single-coordinate change implies $\mathcal{K}_j(\kappa^L(q + \epsilon e_j) - \kappa^L(q) - \kappa^S(q + \epsilon e_j) + \kappa^S(q)) = 0$. The seminorm's kernel is $\ker B$: its expression forces all internal entries of Bh to vanish and its two endpoint entries to agree; their zero total then forces both to vanish. Hence $A(q + \epsilon e_j) = A(q)$. Differentiability and division by ϵ yield

$$\partial_j A(q) = B[D^L(q) J_j^L(q) - D^S(q) J_j^S(q)] = 0.$$

On any valid joint attribution path, integration of this identity gives zero divergence of the corresponding attribution-column error, and therefore matching attributions on every separating boundary. This needs the local matching premise throughout the domain containing the path; a finite training sample supplies no such premise.

B. Experimental methods

B.1. Network, path sets, and demand settings

The public Sioux Falls network has 24 nodes and 76 directed arcs (Transportation Networks for Research Core Team, 2024). All arcs belong to one block atom, hence one compatibility class. The eight queries are the positive-demand OD pairs with the largest published demands, with ties broken by ascending origin then destination node ID. All happen to involve node 10. The queries are $(10, 16)$, $(16, 10)$, $(10, 11)$, $(10, 15)$, $(15, 10)$,

Table B.1
Settings of the five experimental studies. Background treatments: none (all other demands zero); frozen (background link flows fixed at the joint baseline equilibrium); responsive (all demands at published values, all routes re-equilibrated). The learned-assignment study is Section 7; full protocols in Appendices B.1 and B.7.

Study	Network, study ODs	Demand change	Background / boundary family	Grid
Large-network profile	Anaheim, 6 largest demands	joint $0.8 \to 1.2\times$ base	responsive / ordered distance family	41 states
Background comparison	Sioux Falls, 8 largest demands	joint $0.2 \to 1.6\times$ base	none, frozen, responsive / connected families	41 states
Mechanism decomposition	Sioux Falls, 6 prespecified ODs	joint $0.2 \to 1.6\times$ base	responsive / connected families, screenlines	41 states
Adaptive representation	Sioux Falls, 18 input-stratified ODs	focal $\pm 10\%$, $\pm 20\%$; 3 load levels	responsive / 36-direction screenlines	endpoints
Learned explanation	Sioux Falls + Anaheim	96 / 48 test paths	responsive / screenlines; ordered distance	41 / 21

$(10, 17)$, $(11, 10)$, $(17, 10)$, with published demands $q_{\text{base}} = (4400, 4400, 4000, 4000, 4000, 3900, 3900, 3900)$ veh/h. The attribution path changes these demands from $0.2q_{\text{base}}$ to $1.6q_{\text{base}}$. Node IDs in the traffic experiments are one-based. Flows retain the benchmark's approximate-hourly convention; one time unit is 0.01 h. A reported TSTT number is $\sum_e F_e t_e(F_e)$ with F_e in veh/h and t_e in these time units: multiplying by 0.01 gives the hourly vehicle-hour total, without extrapolating to a day. All absolute attribution and boundary-value numbers use the same convention.

For each OD we retain every loopless directed path if enumeration finishes within 200 paths, otherwise the ten shortest by free-flow cost (Yen's algorithm, lexicographic edge-sequence tie breaking); no OD in this network has a fully enumerated set under the threshold. All eight query ODs use ten paths. The link–path incidence Δ and the matrix A summing paths within each OD stay fixed, so $B\Delta = MA$ with the appropriate full or study OD incidence matrix. The full-background ten-path set has 5,280 paths (ten per OD; the expanded set 10,560, twenty per OD); Appendix C.4 compares these candidate-path models.

The path-exchange response space $T_{\text{swap}} = \Delta \ker A$ (Appendix A.3) has dimension 32 in A with ten paths for each study OD, and dimension 53 in C with either ten or twenty paths for every OD, equal to the full circulation-space dimension $m - n + 1 = 53$ (ranks computed from within-OD differences of link–path incidence columns, cross-checked against singular-value bases of $\Delta \ker A_{\text{OD}}$). On Sioux Falls, full background therefore spans the complete circulation space through allowable path exchanges, while the eight-OD path model spans a proper subspace. On Anaheim, whose centroid-restricted ten-path sets cannot use nodes 2–38 in transit, T_{swap} has dimension 404 of a circulation-space dimension of 499: a proper subspace, although still large enough that the model-specific weight tests fail decisively there ($\rho_{\text{swap}} = 2.2$).

The three demand settings share the same query endpoints. In the no-background setting, all other demands are zero. In the frozen-background setting, assign all positive background demands jointly with the study demands fixed at $q^0 = 0.2q_{\text{base}} = (880, 880, 800, 800, 800, 780, 780, 780)$ veh/h in the listed query order, freeze the resulting background link flows, and re-equilibrate only the eight query ODs. In the responsive-background assignment, the other 520 positive OD demands stay at published values while all 528 ODs re-equilibrate jointly at every path point and perturbation. TSTT and its gradient use total link flows in the frozen- and responsive-background assignments; the latter is a full-background assignment with eight attribution queries, not an attribution evaluation of all 528 ODs.

Topology-release distribution. Table B.2 quantifies, on four public networks, how much weight freedom the connected condition releases relative to the unrestricted condition. For every positive-demand OD pair, the constrained-edge share is the fraction of edges lying in the block atoms traversed by that OD's paths; the connected criterion constrains exactly those weights, whereas the unrestricted criterion constrains every edge (100%). The distribution is tightly concentrated on Sioux Falls (one block atom), Anaheim and Winnipeg; Eastern-Massachusetts carries a peripheral tail, with 2.2% of OD pairs constraining fewer than 75% of edges and a minimum of 0.8% reached by OD pairs localized in a single peripheral block atom.

Table B.2
Distribution over positive-demand OD pairs of the constrained-edge share of the connected condition (percent of edges in the queried block atoms; the unrestricted condition constrains 100%). Quantiles over OD pairs; the final column is the minimum over OD pairs.

Network	Nodes	Edges	OD pairs	10%	50%	90%	Min.
Sioux Falls	24	76	528	100	100	100	100
Anaheim	416	914	1,406	95.4	95.4	96.1	95.4
Winnipeg	1,052	2,836	4,344	95.3	95.3	95.3	95.3
Eastern-Mass.	74	258	1,113	89.1	89.1	92.2	0.8

B.2. Equilibrium and its sensitivities

Unless stated otherwise, the fixed-path logit stochastic user equilibrium (SUE) uses $\theta = 1$ and BPR costs $t_e(F_e) = t_{0,e}[1 + 0.15(F_e/c_e)^4]$. With Π_j the path set,

$$h_p = q_j p_p(F), \quad p_p(F) = \frac{\exp[-\theta(\Delta^\top t(F))_p]}{\sum_{r\in\Pi_j}\exp[-\theta(\Delta^\top t(F))_r]}, \quad F = \Delta h. \tag{B.1}$$

This is the solution of

$$\min_{h\geq 0,\ Ah=q}\left\{\sum_e \int_0^{(\Delta h)_e} t_e(s)ds + \theta^{-1}\sum_p h_p(\log h_p - 1)\right\}.$$

For positive demands the feasible set is nonempty and compact, the objective strictly convex, and the minimizer positive: transferring mass to a zero path flow from a positive path of the same OD improves the objective, with entropy change $\theta^{-1}\epsilon\log\epsilon + O(\epsilon) < 0$. Its Hessian and sensitivity equations are

$$H = \Delta^\top \operatorname{diag}(t_e')\Delta + \theta^{-1}\operatorname{diag}(1/h_p) \succ 0, \qquad \begin{bmatrix} H & A^\top \\ A & 0 \end{bmatrix}\begin{bmatrix} \partial_j h \\ \partial_j \nu \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{e}_j \end{bmatrix}. \tag{B.2}$$

A has full row rank, so the system is nonsingular and the implicit function theorem gives a C^1 demand-to-flow map.

For the responsive-background assignment we solve the equivalent total-link-flow fixed point $F = L(F,q) = \Delta\{q_j p_{\Pi_j}(F)\}_j$ by Newton steps with residual line search. For the frozen-background setting the corresponding equation is $F = F^0_{\text{bg}} + \Delta_{\text{study}}\{q_j p_{\Pi_j}(F)\}_{j\in\text{study}}$, where F^0_{bg} is fixed at the joint q^0 equilibrium. Costs and gradients use this total F in both equations. Every solution is checked against its path-loading equation, with maximum absolute path-flow gap below 10^{-7} veh/h. The 76-dimensional implicit system avoids forming a dense path Hessian. For an OD group R, define

$$S_R = \theta\sum_{k\in R}\Delta_k\left[\operatorname{diag}(h_k) - h_k h_k^\top/q_k\right]\Delta_k^\top, \quad W = \operatorname{diag}(t_e'(F_e)), \quad S = S_{\text{all}}.$$

Differentiating the logit probabilities gives $\partial_F L_R = -S_R W$. Consequently the total sensitivity satisfies

$$(I + SW)J_j = \Delta_j p_j, \qquad J_j^{\text{study}} = \Delta_j p_j - S_{\text{study}} W J_j, \qquad J_j^{\text{bg}} = -S_{\text{bg}} W J_j. \tag{B.3}$$

Here "study" includes all eight queried ODs and "background" the other 520: $J_j = J_j^{\text{study}} + J_j^{\text{bg}}$, $BJ_j^{\text{study}} = \mu_j$, $BJ_j^{\text{bg}} = 0$, the study term including responses of the other seven study ODs. The decomposition uses one joint equilibrium, rather than sensitivities at different congestion states. In the frozen-background setting the background flows have no derivative, so the same implicit system uses $S = S_{\text{study}}$, with $J_j^{\text{study}} = J_j$ and $J_j^{\text{bg}} = 0$.

Central finite differences independently re-solve Eq. (B.1) at $q \pm a_j\mathbf{e}_j$, $a_j = 0.005 q_{\text{base},j}$ and $a_j/2$; both steps are checked at every $\theta = 1$ ten-path responsive-background integration point, the path-set and θ variations add half-step checks for all eight columns at the path midpoint, and the link Newton solver is compared against the successive-averages background equilibria at their common eleven quadrature points.

B.3. Conservative learned surrogate

The zero-background surrogate parameterizes $h_j = q_j \operatorname{softmax}(z(q)|_{\Pi_j})$, $f = \Delta h$. The three hidden layers each have 128 tanh units; the final layer is linear. The logit head uses training-standardized demands while the raw q_j multiplies the shares, preserving nonnegative flow and exact conservation. Computations use double precision. Training targets are fixed-path logit-SUE assignments ($\theta = 1$, path fixed-point tolerance 10^{-6}). The 2500/857/857 training/validation/test split uses seed 20260820, with demands $q_j = q_{\mathrm{base},j} s_j$ and independent s_j drawn from $\mathrm{LogNormal}(0, 0.5^2)$ clipped to $[0.2, 2]$. Adam uses learning rate 0.003, full-batch link-flow MSE, at most 12,000 steps, and validation every 50 steps. The minimum-validation-error state of each seed $0, \ldots, 4$ supplies the reported results, with automatic differentiation supplying their Jacobians; the test split is used only for final reporting.

B.4. Integration grids, connected boundaries, and cancellation diagnostics

All numerical integrations use composite trapezoids. The principal learned zero-background results use 201 quadrature points; direct zero-background SUE and both background settings use 41 in the main table. The frozen-background setting compares 11 with 41 points; the responsive-background setting compares 11, 21, and 41 nested points; the two near-balanced queries add 81 and 161 nested points; the learned-versus-SUE sensitivity comparison uses identical 41-point grids, kept separate from the 201-point learned results (Appendix C.1).

Connected boundaries require both $\mathcal{G}[U]$ and $\mathcal{G}[V \setminus U]$ to be connected on the underlying undirected graph, with $s_j \in U$, $t_j \notin U$. Exact enumeration yields 10,270–13,253 admissible sets for each of the eight main study ODs, and 4,621–16,943 for the six additional queries of Appendix B.5. Extrema of $-\mathbf{1}_U^\top B c_j$ are evaluated by exact enumeration, cross-checked against independent subset enumeration on restricted neighborhoods; attaining sets are verified for membership with values recomputed by direct summation.

For cancellation diagnostics define $A_j = |\Delta q_j| \int_0^1 \sum_{v \notin \{s_j, t_j\}} |(B D_\lambda J_j)_v| d\lambda$ and $\kappa_j = 1 - S_j / A_j$ when $A_j > 0$. The triangle inequality gives $0 \le S_j \le A_j$. Both quantities use the specified 201-point learned grid and are summarized over the five seeds; checks on additional path-set or dispersion models sample the demand-path midpoint and endpoints.

B.5. Six pre-specified queries away from node 10

The selection rule uses only published demands and free-flow shortest-path times. Candidates are the positive-demand ODs whose origin and destination both differ from node 10; ranking them by demand and splitting into three equal-count terciles, the minimum- and maximum-free-flow-shortest-path OD within each tercile is selected, with ties broken by ascending (origin, destination). The six queries are $(7, 18)$, $(1, 19)$, $(4, 5)$, $(1, 15)$, $(6, 8)$, and $(7, 12)$, with published demands $(200, 300, 500, 500, 800, 700)$ veh/h and free-flow shortest-path times $(2, 22, 2, 23, 2, 19)$ time units; the joint protocol, sensitivities, and boundary enumeration are as in Section 6.2, with conservation checks at responsive-background floating-point levels (maximum path fixed-point residual 1.6×10^{-10} veh/h, midpoint implicit-versus-finite-difference column error 2.6×10^{-8}, completeness 4.3×10^{-7}).

For the single-OD checks, let $\bar{q} = q^{\mathrm{base}}$ be the full published demand vector over all 528 positive-demand ODs. For each focal query $j \in \{(1, 19), (1, 15), (7, 12), (6, 8)\}$, define

$$q^{(a)} = \bar{q} + (a q_j^{\mathrm{base}} - \bar{q}_j) e_j, \qquad q^{(b)} = \bar{q} + (b q_j^{\mathrm{base}} - \bar{q}_j) e_j, \qquad (a, b) \in \{(0.2, 1.6), (0.9, 1.1), (0.8, 1.2)\}, \quad \text{(B.4)}$$

where e_j is the coordinate unit vector. All other 527 demands, including the other five selected queries, remain at their published values $\bar{q}_{-j}$ along the segment. Direct equilibria and the learned small-band checks use these same reference states. The responsive-background solver, path sets, and boundary definitions are unchanged; checks and interval results are in Appendix C.5. For each interval, Eq. (10) gives the attribution column from the two endpoint equilibria alone.

B.6. Same-task computation timings

All timings describe the same fixed tasks and the same computational environment that produced the reported attributions (Section 5). For the separable single-OD task on query $(1, 19)$ (only its demand changes, with endogenous rerouting; shared path set, solver tolerance and 152-region family; medians of three uncached runs), the endpoint identity computes the attribution column in 0.088 s against 0.510,

0.948 and 1.813 s for 21-, 41- and 81-state sensitivity integration, whose attribution columns agree with the endpoint values to 6.6×10^{-5}, 1.6×10^{-5} and 4.1×10^{-6} in maximum absolute geometric-family boundary error; the identity eliminates sensitivity integration for separable single-OD tasks but does not separate OD columns when several demands change jointly. For spatial aggregation of a fixed region family once reference columns are available, the three methods of Section 5.4 agree to 2.2×10^{-11} TSTT units on the 1,642 oriented Sioux Falls regions: the prefix scan takes 4.16 ms per six columns (87.92 ms per 120), the vectorized one-sparse-product $A_{\mathcal{F}}C$ takes 0.24 ms per six (3.46 ms per 120) after a one-time 69.5 ms matrix build, and direct crossing-edge summation is slowest (28.51/1,176.15 ms); on the 910 Anaheim query–region pairs the same ordering holds (0.10/0.30/13.31 ms), with agreement to 5.9×10^{-12} against node sums, and the 21-state reference construction, including paths and sensitivities, took 1,278.7 s.

B.7. Anaheim network, equilibrium and predefined regions

Both network and demand files are the public Transportation Networks for Research Core Team Anaheim TNTP data (Transportation Networks for Research Core Team, 2024). The network header supplies free-flow times in minutes and capacities in vehicles per hour; all roads use the published BPR coefficient 0.15 and power 4. We use $t_e(F_e) = t_e^0[1 + 0.15(F_e/c_e)^4]$ and TSTT marginal weight $t_e^0[1 + 0.75(F_e/c_e)^4]$. The six study OD demands, ranked by descending base demand and then lexicographic OD, are $(4, 2)$: 2106.7, $(1, 2)$: 1365.9, $(2, 4)$: 1271.4, $(3, 2)$: 1237.9, $(25, 4)$: 1232.4, $(25, 2)$: 1232.3 vehicles per hour. They move jointly from 0.8 to 1.2 times base demand; all remaining positive demands stay at their published base values and retain endogenous route choice. Every OD uses ten loopless free-flow shortest routes from Yen's algorithm. Ties within 10^{-12} minutes are resolved by lexicographic edge-index sequence in the TNTP road order. Nodes 1–38 are centroids and cannot be intermediate route nodes; the first through node is 39. Logit scale $\theta = 1$ minute^{-1} applies to route travel time in minutes. The edge Newton solver targets maximum loading residual below 10^{-8} vehicles per hour, uses at most 80 iterations and backtracking up to 30 halvings per step, and checks the original path loading residual below 10^{-7}. Implicit derivatives solve the same fixed-point equation. Trapezoidal integration uses equally spaced states, with warm starts from a neighboring saved state. The 41-state refinement reuses all 21 old states and computes only the twenty new midpoints. The distance-based region construction and fixed progress targets are specified in Section 6.1. The reported range uses every admissible prefix in that family. Same-state mechanisms compare regions chosen nearest the progress targets, using free-flow weights t^0 and congestion weights equal to the total-flow TSTT marginal weight minus t^0. Figure B.1 plots the five profiles not shown in the main text.

Table B.3
Anaheim attributions over the complete distance-based family, with the three prespecified positions (vehicle-minutes per hour). Column heads are the prespecified targets; the exact selected levels are (row order) 0.250/0.501/0.752, 0.252/0.500/0.740, 0.248/0.499/0.750, 0.251/0.498/0.751, 0.251/0.504/0.768 and 0.250/0.500/0.724.

OD	System	Unrestricted range	Family range	Regions	$\phi(0.25)$	$\phi(0.50)$	$\phi(0.75)$
(4,2)	34664.1	[-12712.6,20514.0]	[307.5,9832.1]	211	2323.5	1441.8	692.3
(1,2)	16773.6	[-6032.2,9762.0]	[172.7,6374.8]	83	259.0	818.0	360.5
(2,4)	12143.7	[-4374.6,6733.0]	[98.7,1905.9]	211	257.0	714.6	1061.8
(3,2)	14706.7	[-5613.8,9052.9]	[202.8,5777.4]	157	236.2	1087.3	434.6
(25,4)	6919.3	[-1805.3,3091.8]	[96.8,1212.2]	164	545.6	540.7	1212.2
(25,2)	11445.9	[-4157.3,6904.5]	[131.8,5751.2]	84	801.5	412.0	5751.2

B.8. Learning objectives and exact endpoint targets

Equation (20) suggests learning the reference's weighted demand response where it affects spatial attributions. With training-only positive scales a_j, the boundary term is

$$\mathcal{L} = \mathcal{L}_{\text{flow}} + \lambda_J \mathcal{L}_{\text{response}} + \lambda_{\text{cut}} \mathcal{L}_{\text{boundary}}, \qquad \mathcal{L}_{\text{boundary}} = \frac{1}{NK} \sum_{q,j} \left[\frac{\mathcal{K}_j(h_j(q))}{a_j} \right]^2. \tag{B.5}$$

The reference Jacobian is obtained from the existing implicit SUE sensitivity system (Appendix B.2). The empirical study compares six objectives under a shared continuation protocol (Appendix B.9): flow only; the

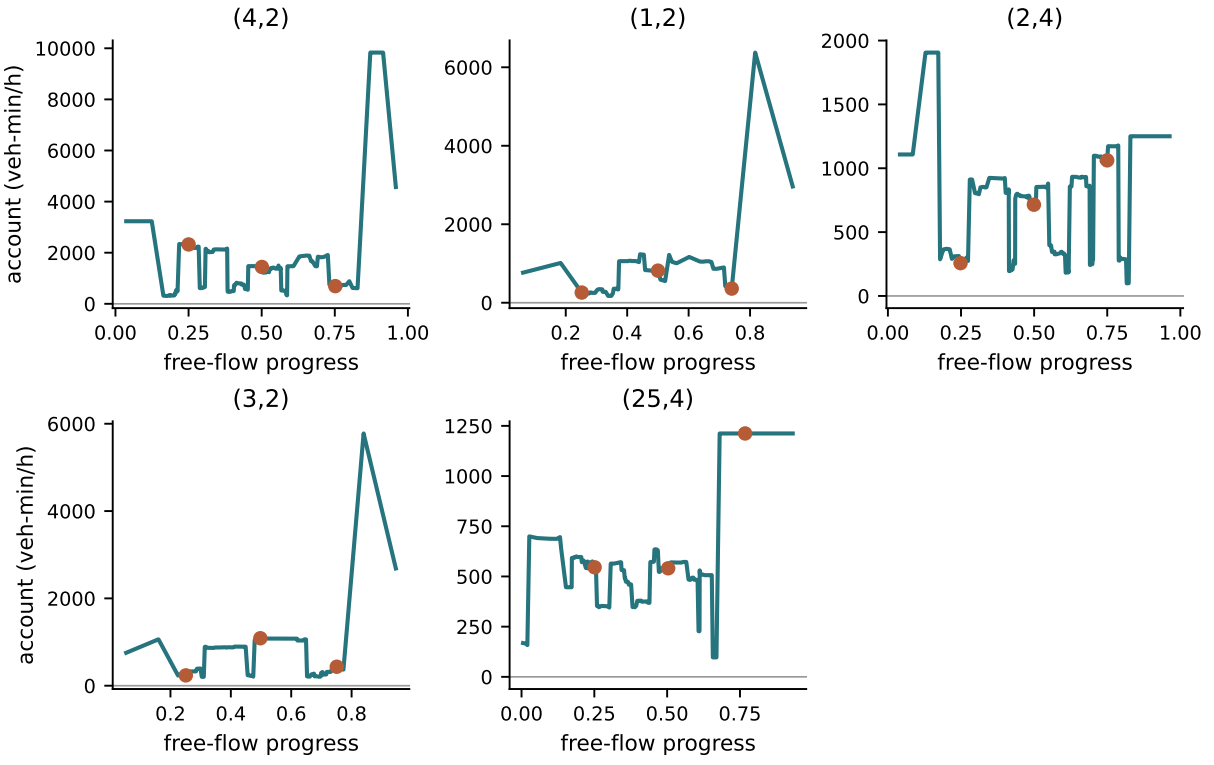


Figure B.1: Anaheim spatial response profiles of the remaining five study queries over the distance-based families: the attribution versus network-distance progress along each query's own OD direction, with ochre dots marking the three prespecified positions. All stay positive; family ranges span roughly one to one and a half orders of magnitude.

Table B.4
Anaheim differences between preselected early and late regions, split by free-flow and congestion weights and study and background responses (vehicle-minutes per hour).

OD	Free-flow study	Free-flow background	Congestion study	Congestion background	Total
(4,2)	-416.91	-8.53	-1499.94	294.16	-1631.21
(1,2)	9.85	6.96	83.01	1.71	101.54
(2,4)	312.07	-65.41	667.06	-108.90	804.83
(3,2)	50.04	11.80	136.55	0.04	198.43
(25,4)	47.28	-40.11	666.28	-6.89	666.56
(25,2)	-131.77	-10.52	5102.23	-10.15	4949.79

Table B.5
Location layer of the $(25, 2)$ case: the boundary at each reported position, its crossing links, and the principal signed attribution contributions (vehicle-minutes per hour; positive entries add to the origin-side attribution). Regions contain the origin; the two plateau endpoints share the attribution $5{,}751.2$ on link $63 \to 62$, while their second crossing link differs ($87 \to 86$ at the start, $2 \to 87$ at the end, contributing -0.0 at both).

Position (progress)	Region (nodes)	Crossing links	Principal links (from→to, contribution)	Attribution
Early 0.250	231	34	$269 \to 261$ $+550.5$; $271 \to 192$ $+163.9$	801.5
Middle 0.500	405	4	$191 \to 190$ $+297.0$; $66 \to 65$ $+105.4$	412.0
Plateau 0.724–0.818	413–414	2	$63 \to 62$ $+5{,}751.2$; ($87 \to 86 / 2 \to 87$ -0.0)	$5{,}751.2$
Destination 0.911	415	2	$62 \to 2$ $+2{,}673.1$; ($2 \to 87$ -0.0)	$2{,}673.1$

unweighted response $\|J_j^L - J_j^S\|$; the weighted response $\|D^L J_j^L - D^S J_j^S\|$ of Eq. (27); the pointwise boundary seminorm $\mathcal{K}_j(D^L J_j^L - D^S J_j^S)$ of Eq. (B.5); and the two endpoint objectives below. Derivative supervision itself is established in Sobolev training (Czarnecki et al., 2017), including finite-difference derivative targets (Yuan et al., 2022); the network-derived target here selects the part of weighted response error that changes boundary attributions. A circulation in $\ker B$ is invisible to $\mathcal{K}_j$, so flow and response losses remain necessary for the other attribution objectives. The target matches the reference's actual response; it does not force its true internal attribution divergence to zero or replace TSTT weights with uniform weights. Equation (20) is a path inequality; finite training-point losses alone do not certify an entire demand domain.

For the following finite-change statement, the outcome is explicitly separable:

$$g(F) = (g_e(F_e))_{e\in E}, \qquad T(F) = \sum_{e\in E} g_e(F_e).$$

Proposition B.1 (exact finite-change boundary target)**.** *Let f^L, f^S and the edgewise outcome g be continuously differentiable on an open demand domain, and let $q^{\pm} = q \pm h e_j$ with their entire segment in that domain. Put*

$$z_j^r = g(f^r(q^+)) - g(f^r(q^-)), \qquad r \in \{L, S\}.$$

For the attribution of this single-OD change,

$$\max_{U:\, s_j\in U,\ t_j\notin U} |\phi_j^L(U) - \phi_j^S(U)| = \mathcal{K}_j(z_j^L - z_j^S). \tag{B.6}$$

If all sufficiently small single-coordinate changes have matching one common value on every separating boundary at every point of the domain, then $BD^L J_j^L = BD^S J_j^S$ throughout the domain. Integrating yields matching boundary attribution columns for any joint demand path in that domain (with the same columnwise path convention).

Appendix A.7 proves both statements. The first is an exact finite-change target, whereas the pointwise response objective optimizes the upper bound in Eq. (20); integrating before taking $\mathcal{K}_j$ can retain cancellations. The second statement requires a domain-wide local premise and is not a guarantee from finite training data. With several ODs changing, the endpoint outcome difference only identifies the sum of their attribution columns.

For sampled single-OD endpoint pairs, a direct training objective is

$$\mathcal{L}_{\text{pair}} = \mathcal{L}_{\text{flow}} + \lambda_{\text{edge}} \frac{1}{N} \sum_{(q,h,j)} \frac{\|z_j^L - z_j^S\|_2^2}{b_j^2} + \lambda_{\text{cut}} \frac{1}{N} \sum_{(q,h,j)} \left[\frac{\mathcal{K}_j(z_j^L - z_j^S)}{a_j}\right]^2, \tag{B.7}$$

with positive training-only scales a_j, b_j. The edge term controls changes invisible to the boundary seminorm, whose kernel is $\ker B$. Two reference endpoint equilibria supply z_j^S; training uses ordinary parameter backpropagation through two predictions, without reference Jacobian labels or input-Jacobian differentiation.

B.9. Learned-model evaluation protocol

Six objectives continue each of five common conservative models for 1,200 Adam updates, with learning rate 3×10^{-5}, flow batch 32 and query batch eight. The flow data and seed-specific minibatch sequences are shared. Each of six queries has 64 training and 16 validation endpoint pairs. Centers are sampled coordinatewise from $[0.65, 1.3]$ times baseline demand (seed 102111); half-widths 2.5% and 15% occur equally often. Only one demand changes per pair, while background routes remain responsive.

The flow loss averages $[(f_e^L - F_e^S)/s_e]^2$ over minibatch states and edges, with $s_e = 0.01\sqrt{\text{mean}_{\text{train}}[(F_e^S)^2]}$ from the 2,500 training states. Each auxiliary objective adds one term $\lambda \mathcal{L}_{\text{aux}}$ to this flow loss, with the weight grid $\lambda \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ searched identically for every objective. Each normalizer carries the units of its own target, so that the common grid explores comparable regularization ranges: rescaling the outcome $T \mapsto cT$ rescales the weighted terms' normalizers with them and leaves every effective

weight unchanged. With $\langle\cdot\rangle_j$ denoting the average over query j's training centers or endpoint labels, the training-only scales are

$$u_j = \max\{\sqrt{\langle\|J_j^S\|_2^2\rangle_j}, 10^{-6}\}, \qquad v_j = \max\{\sqrt{\langle\|D^S J_j^S\|_2^2\rangle_j}, 10^{-6}\}, \qquad b_j = \max\{\sqrt{\langle\|z_j^S\|_2^2\rangle_j}, 10^{-6}\},$$
$$a_j = \max\{\sqrt{\langle\mathcal{K}_j(z_j^S)^2\rangle_j}, 10^{-6}\}, \quad \bar{a}_j = \max\{\sqrt{\langle\mathcal{K}_j(D^S J_j^S)^2\rangle_j}, 10^{-6}\}.$$

The auxiliary terms are: unweighted response $\langle\|J_j^L - J_j^S\|_2^2/u_j^2\rangle$; weighted response $\langle\|D^L J_j^L - D^S J_j^S\|_2^2/v_j^2\rangle$, with $D^r = \mathrm{diag}(\nabla T(F^r))$ the outcome-gradient weights of Eq. (2) evaluated at each model's own flows (TSTT: $\nabla_e T = t_e + F_e t_e'(F_e) = t_{0,e}[1 + 0.75(F_e/c_e)^4]$); the pointwise boundary seminorm $\langle[\mathcal{K}_j(D^L J_j^L - D^S J_j^S)/\bar{a}_j]^2\rangle$; the endpoint-edge term $\langle\|z_j^L - z_j^S\|_2^2/b_j^2\rangle$; and the endpoint-cut bundle, which adds 0.01 times $\langle[\mathcal{K}_j(z_j^L - z_j^S)/a_j]^2\rangle$ to the endpoint-edge term (a fixed 0.01 inner ratio, not an independently tuned objective). The three response-family objectives share one set of derivative labels; the two endpoint objectives share one set of endpoint labels. The weighted objective is run at every grid weight under this own-scale normalization from the shared initializations.

Checkpoint selection uses one uniform rule for every objective and weight: among the common initial model (update zero) and every 100th continuation update whose validation flow error is at most 1.1 times the initial model's value, take the minimum validation boundary mean-square error, using the same 96 validation pairs throughout. The initial model ensures the pool is never empty. The weight λ^* of each objective is then chosen on validation data only: among the grid values whose seed-median selected validation flow error stays within 1.1 times the seed-median initial error, minimize the seed-median validation boundary error. The flow-only control reuses the flow continuation under the same uniform rule. Test paths never enter selection or weight choice. The extended-budget replication of Section 7 repeats the response and flow continuations with 3,600 updates under this unchanged protocol (per-network costs in Appendix C.13).

The final evaluation uses 96 fixed test paths, all generated by one rule: for each active-query group $k \in \{1, 2, 3, 6\}$, draw a base state uniformly from $[0.65, 1.3]$ times baseline demand, choose the k active queries in round-robin order, add independent changes of 15–30% of their baseline demands with alternating same-direction and mixed-direction signs, and solve all 41 states of the resulting straight demand path. Twenty-four paths (six per group, seed 102112) and seventy-two paths (eighteen per group, seed 102113) were generated; models and validation choices were fixed before evaluation on the additional paths. For every path, flow error first averages relative Euclidean errors across its states. Attribution statistics average active queries and seeds within each path and then take medians across the 96 paths; seeds do not add independent paths. For column error e_j and reference column c_j, relative column, system and unrestricted-boundary errors divide $\|e_j\|_2$, $|\mathbf{1}^\top e_j|$ and $\mathcal{K}_j(e_j)$ by the corresponding reference values, with absolute floor 10^{-6}. Geometric error uses $\max_{U \in \mathcal{F}_j^{\mathrm{geo}}} |\mathbf{1}_U^\top B e_j|$ and the same maximum for c_j. Percentages multiply ratios by 100. The incremental-flow error at the path endpoints is $\|[f^L(q^1) - f^L(q^0)] - [f^S(q^1) - f^S(q^0)]\|_2 / \|f^S(q^1) - f^S(q^0)\|_2$, reported with the absolute numerator because the reference increment is small relative to total flows (per path–seed medians of both, and the smallest reference increment, in Appendix C.9).

The decomposition in Eq. (C.1) is evaluated at the same 41 states with the same quadrature. Component norms describe error mechanisms; cancellation prevents treating their norms as additive shares. The fixed-screenline sign comparison in Appendix C.6 uses the separate pre-specified six-query baseline segment and is not pooled with these 24 paths. Table B.6 reports the incremental computation costs of training-label generation, common validation and explanation timing under the shared CPU protocol. Training labels comprise 768 endpoint states for either endpoint objective, or 384 center states and their sensitivities for the three response-family objectives (unweighted, weighted, boundary-seminorm), which share one label set. Every method additionally uses the same 192 endpoint states for validation; unused center-validation derivatives are excluded. Existing flow data, pretrained models and the common test reference are excluded from these incremental costs. The 984 test states and sensitivities took 74.4 s in the reference run. Explanation timing covers a 24-path timing subset of the 96 test paths (the seed-102112 paths, six per active-query group; forward evaluation, six directional derivatives per state, quadrature and node aggregation, excluding model loading): the table entries are totals over that subset, about 0.3 s per path, not the full 96-path evaluation. Continuation times include initialization, validation and saving under concurrent jobs, pooled across the

Table B.6
Incremental computation costs (seconds) under the shared continuation protocol. Training-label and common-validation costs are measured separately on the same CPU protocol; training times are continuation-time medians over the weight grid and seeds, and explanation times are sequential medians for the selected models over the 24-path timing subset defined below (totals for that subset).

Labels	Training labels	Common validation	Continuation	Explanation
Flow only	0.00	6.45	246.0	7.34
Response family	16.12	6.45	155–425	7.13
Endpoint (edge, cut)	25.75	6.45	441–497	7.27–7.59

original run and the renormalized-weight re-run under lighter load; they describe the actual training expense and do not establish a method speed advantage.

Out-of-domain evaluation. The two out-of-domain bands of Section 7 (low loads on $[0.40, 0.60]$ and high loads on $[1.35, 1.55]$ times the base vector) reuse the test-path generator unchanged — 24 paths per band, active-query groups $k \in \{1, 2, 3, 6\}$ with six replicates, deltas drawn from $U[0.15, 0.30]$ times the base values with odd replicates flipping every even-indexed delta, fresh seeds — with reference SUE flows and sensitivities at the same 41-state quadrature; the frozen flow and response checkpoints are evaluated without retraining. The unrestricted metric compares the closed-form node-divergence extrema of the learned and reference columns, normalized by the reference unrestricted span.

B.10. Conservation mechanism ablation

This experiment compares ways of obtaining flow conservation on Sioux Falls under otherwise identical supervision, asking what the conservative parameterization of Eq. (25) contributes beyond conservation itself: (i) *exact* — the path-share parameterization, pretrained on the unscaled flow loss (the published five-seed models of Section 7); (ii) *exact (scaled)* — the same architecture pretrained on the per-link scaled loss, the one recipe difference of the earlier three-way comparison, removed by training the exact parameterization on the direct models' objective; (iii) *penalty* — a direct MLP on link flows with a squared node-balance penalty (weight 1, normalized by the nodal-throughput RMS over the training states, floored at the network average); (iv) *projection* — the same direct MLP whose output is projected each step onto $\{f : Bf = Mq+b\}$ by the differentiable least-norm map $f \mapsto f - B^\top(BB^\top)^\dagger(Bf - Mq - b)$ (Moore–Penrose inverse; $BB^\top$ is singular for a connected network). The direct-MLP width is solved from the parameter-count equation so that all four models have 714,979 parameters against the exact model's 715,040. Pretraining shares the data, optimizer (Adam 3×10^{-3}, batch 256, at most 12,000 steps) and the min-validation checkpoint rule (validation metric: unscaled relative L2 flow error) across all architectures; the loss differs only as stated. Continuation follows the protocol of Appendix B.9 unchanged (1,200 updates, shared labels and minibatch sequences, the uniform selection rule including the pre-trained initial model in the candidate pool), with flow-only and response supervision for each mechanism. The exact rows reproduce the published five-seed medians exactly (21.18% and 14.19% column error), and every projection output satisfies $Bf = Mq + b$ to 2×10^{-11}; the path-share construction holds node balance to the same scale under both pretraining losses.

Table B.7 supports three readings. First, the pretraining loss is not what separates the exact parameterization from the direct mechanisms: under the same scaled loss the exact architecture attains 21.60% and 14.35% column error against the projection's 104.1% and 99.4%, and training the exact architecture on the scaled loss leaves its column errors within half a percentage point of the published unscaled values (21.18%, 14.19%) while slightly lowering its end-to-end summary error. The exact-versus-direct gap is therefore a property of the parameterization, not of its training recipe. Second, conservation itself still matters: the penalty model matches the flow accuracy of the exact model (0.040%) with a small output residual (42 veh/h against nodal throughputs in the thousands), yet its integrated attribution residual — the quantity R_j of Eq. (22) — reaches 0.66 in units of the demand-conservation scale $|\Delta q_j| \|\mu_j\|_2$, its derivative balance residual is 1.16 in the same normalization, and the downstream outcome columns inherit the damage; the chain of Eq. (22) is exactly what fails, because the penalty controls $r(q)$ but not $\partial_j r$. The projection enforces output conservation exactly and transfers it to derivatives and to the integrated residual, cutting the column error to 104.1% while remaining far above the exact parameterization, because

Table B.7
Conservation mechanism ablation on the 96 Sioux Falls test paths (medians of path-level values; active queries and seeds averaged within path). Mechanisms: path-share parameterization under the two pretraining losses, and the two direct mechanisms under the scaled loss. Residuals are per-query maxima over the path states, maximized over queries: the node-balance residual $\max_q \|Bf(q) - Mq - b\|_\infty$ (veh/h); the derivative residual $\max_{q,j} \|\partial_j[Bf](q) - \mu_j\|_\infty / \|\mu_j\|_2$. Column: attribution column error; incremental: endpoint flow-change error; E_{total}: normalized end-to-end summary error.

Mechanism	Method	Flow (%)	Incremental (%)	Node resid.	Deriv. resid.	Column (%)	E_{total}
Exact	flow	0.032	5.03	3×10^{-11}	9×10^{-16}	21.18	11.35
Exact	response	0.029	3.52	3×10^{-11}	8×10^{-16}	14.19	8.68
Exact (scaled)	flow	0.033	5.53	3×10^{-11}	8×10^{-16}	21.60	10.33
Exact (scaled)	response	0.032	3.81	3×10^{-11}	8×10^{-16}	14.35	8.19
Penalty	flow	0.040	22.76	42.1	1.16	183.7	148.8
Penalty	response	0.039	22.24	42.1	1.17	177.9	144.9
Projection	flow	0.026	14.54	2×10^{-11}	2×10^{-15}	104.1	46.4
Projection	response	0.026	14.42	2×10^{-11}	2×10^{-15}	99.4	45.6

projecting an unconstrained response field onto node balance does not reproduce the path-share structure that the assignment imposes. Third, response supervision improves every mechanism (penalty 183.7 → 177.9, projection 104.1 → 99.4, exact 21.18 → 14.19 and 21.60 → 14.35), with by far the largest effect inside the exact parameterization: the inductive benefit of conservation-by-construction and the statistical benefit of derivative supervision are complementary rather than substitutes.

B.11. Label-budget repetitions

The label-efficiency experiment of Section 7 retrains response supervision at the selected weight with per-query derivative-center budgets of 6, 12, 24 and 48 centers (36–288 in total against the full 384). Each budget uses three independent query-balanced random nested subsets: for replication ρ and query j, a fresh permutation of that query's 64 training centers is drawn from $\text{rng}(20261101 + \rho + 97j)$, and the budget-$b$ subset is the first b entries, so the subsets of different budgets nest within a replication while different replications are independent. Training follows the continuation protocol of Appendix B.9 unchanged (shared flow data and minibatch sequence, 1,200 updates, uniform selection rule); only the derivative-label pool shrinks. The reported curves aggregate hierarchically: within each subset replication, the median over the 96 test paths; across the three replications, the median of those three medians, with the shaded band spanning their interquartile range. The full-budget point (384 centers) is the single run with the complete label set and carries no subset variation. Across all budgets the interquartile range stays within 0.8 percentage points for the column error and within 0.4 for the family-boundary error, and flow accuracy stays within 0.027–0.031%.

C. Numerical accuracy and supporting results

C.1. Grid refinement, derivative, and solver checks

All differences here are observed numerical comparisons, not certified bounds. Direct attribution-column differences are at most 0.178% for 11 versus 41 states and 1.1×10^{-4}% for six-query/single-OD 41 versus 81 states. Near-zero attributions on 41/81/161 grids change by less than 5×10^{-6} percentage points (absolute < 0.01); the learned 201/801-state relative difference is 4.97×10^{-6}. Implicit versus re-solved central-difference Jacobian errors are at most 7.6×10^{-5} (halved step 2.5×10^{-5}), or 3.0×10^{-8} for six-query/single-OD columns. Newton versus successive-averages differences over 1,353 solves are at most 1.31×10^{-11}. Maximum fixed-point and balance residuals are 3.10×10^{-8} and 1.82×10^{-11} veh/h. Scalar completeness relative discrepancies are at most 3.36×10^{-5} for frozen- and responsive-background settings and 1.9×10^{-7} for single-OD checks. Per-edge endpoint checks give relative column errors 7.7×10^{-8}–1.1×10^{-6} and maximum entry residuals 5.4×10^{-4}–1.6×10^{-2} TSTT units. Per-query records remain in the arrays. The direct $(10, 16)$ attribution change is 24.80% on its 41-state grid; Figure 4 uses the seed-zero learned model on 201 states (24.8%). On the same realized response, the active-link outcome $T(F) = F_{10 \to 16}$ balances exactly in the OD direction on the single selected arc while its scale depends on the circulation — the directional balance of Section 3.2 without a response-independent boundary value.

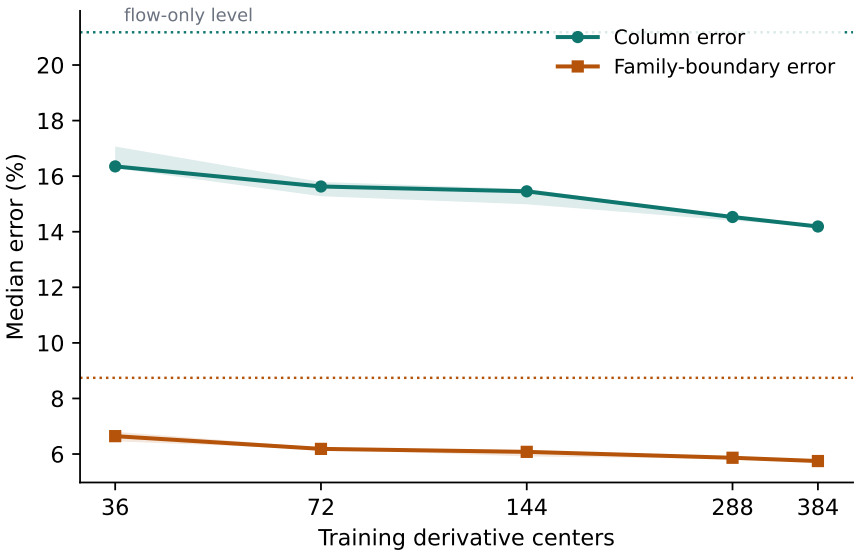


Figure B.2: Response supervision under reduced label budgets on the 96 test paths, at the validation-selected weight: median attribution column and family-boundary errors against the number of training derivative centers (36–384; logarithmic axis), aggregated as stated above (shaded: interquartile range over the three subset replications; the 384-center point is the single full-label run), with the flow-only levels marked (dotted).

C.2. Normalization, cancellation, and sensitivity

Every OD of the background-treatment comparison of Section 6.2 appears in Tables C.1 and C.2: the learned zero-background unrestricted ratios of the three most divergent queries $(17, 10)$, $(16, 10)$ and $(11, 10)$ are 59.71% (58.92–61.23 over seeds), 36.80% (36.41–38.40) and 0.18% (0.03–0.27), closely tracking the direct-SUE values.

Complete cut-range records. Table C.2 reports the complete eight-OD records of the responsive-background treatment (all demands at published values, all routes re-equilibrated; 41 quadrature points, implicit sensitivities): connected regions retain 60.41–76.85% of the unrestricted range on every OD, and the per-query ordering of the learned zero-background ratios above is retained. The two auxiliary treatments of Section 6.2 are complete in Table C.1: zero background removes most of the spread of the three queries that rerouting later amplifies, and the frozen background sits between the two.

Table C.1
Complete eight-OD unrestricted cut ranges (percent of each attribution's absolute origin-cut value) under the two auxiliary background treatments. A: zero background (direct logit-SUE, 41 quadrature points); B: background link flows frozen at the joint baseline equilibrium (study-only re-solve, 11-point chain, the protocol of the frozen-flow check).

	OD							
Treatment	10→16	16→10	10→11	10→15	15→10	10→17	11→10	17→10
A: zero background (%)	26.94	36.86	0.0075	0.77	0.77	37.48	0.0073	59.95
B: frozen background (%)	34.18	40.24	1.26	19.01	22.13	43.71	0.87	56.52

Error-metric definitions. The surrogate test flow error is, within each seed, the mean over the 857 test demands of $\|f(q) - F^{\mathrm{SUE}}(q)\|_2 / \|F^{\mathrm{SUE}}(q)\|_2$, reported as the median over the five seeds; Jacobian column errors $\|J_j^{\mathrm{learned}} - J_j^{\mathrm{SUE}}\|_2 / \|J_j^{\mathrm{SUE}}\|_2$ are pooled over the $5 \times 41 \times 8$ seed–point–OD columns; attribution-column errors use the per-column norm over the 40 seed–OD columns on the common 41-point grid.

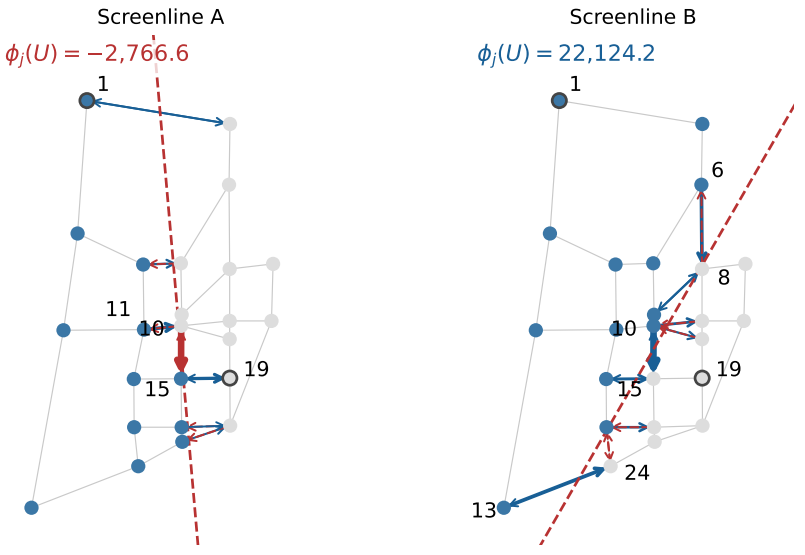


Figure C.1: Contrasting screenline attributions for the same positive OD contribution. Two straight screenlines for OD $(1, 19)$ under the responsive-background assignment. Blue nodes belong to U; gray nodes are outside; dashed lines separate the two sides. Node 1 is inside and node 19 outside in both panels. The two lines have different normal directions, so they are labeled Screenline A (left) and Screenline B (right); the crossing progress of each with the straight origin–destination segment is 0.61 and 0.78, an ordering of the crossing positions that does not by itself make the two enclosed regions nested. The two boundary attributions have opposite signs while the network-wide contribution is positive; arrows encode the signed crossing contributions (solid blue = positive, dashed red = negative, width = magnitude). Principal crossing contributions, Screenline A: $10 \to 15$ -9195.2, $11 \to 10$ $+3169.6$, $15 \to 19$ $+2689.9$; Screenline B: $10 \to 15$ $+9195.2$, $13 \to 24$ $+3602.1$, $6 \to 8$ $+3508.8$ (TSTT units). The lines are the minimum and maximum of the prespecified geometric family; the complete irregular connected minimum and its crossing-link decomposition appear in Figure C.2.

Table C.2
Complete eight-OD cut ranges under the responsive-background treatment: unrestricted and connected-family ranges as percentages of each attribution's absolute origin-cut value, the connected-to-unrestricted ratio, and the number of enumerated connected regions. Direct SUE, 41 quadrature points, implicit sensitivities.

OD	Unrestricted range (%)	Connected (%)	Ratio (%)	Regions
$10 \to 16$	109.04	83.80	76.85	13,253
$16 \to 10$	166.26	127.72	76.82	13,253
$10 \to 11$	84.67	51.27	60.56	10,270
$11 \to 10$	81.09	48.98	60.41	10,270
$10 \to 15$	71.88	49.90	69.42	11,182
$15 \to 10$	84.05	58.24	69.29	11,182
$10 \to 17$	49.11	31.57	64.27	11,482
$17 \to 10$	54.77	35.27	64.39	11,482

Same-path flow, sensitivity, and boundary-attribution errors. The test-split flow error is not paired with the Jacobian comparison; re-summarized on the same 41 zero-background quadrature points for each seed, flow error is 0.136% (median; 1.331% maximum), Jacobian column error 0.516% (9.562% maximum), attribution-column error 0.349% (1.80% maximum), and the connected-boundary error E_j^{cut} 0.204% (1.739% maximum; largest errors at OD $(16, 10)$, seed 3). The per-query absolute origin-cut denominators behind Tables C.1 and C.2, the responsive-background link-sum comparison (C_j spanning 2.078–4.282×10^5 TSTT units, 1.01–2.37 times the origin-cut attributions, endpoint completeness within 2.3×10^{-5} relative) and the node-level study/background divergence breakdown behind Section 6.2 accompany the result arrays in the code package.

For the learned zero-background low-range group, five-seed median pointwise imbalance budgets $100A_j/|\phi_j(\{s_j\})|$ lie between 0.22% and 0.95%, versus 26.85–59.71% for the high group; the low-group median cancellation factors are at most 0.313 (one seed reaches $\kappa = 0.852$), so small pointwise imbalance mainly limits the low-group range, while the high group's median κ is below 8×10^{-5}.

Zero-background route-choice dispersion checks at $\theta \in \{0.5, 1, 2\}$ (eleven-point direct-SUE calculations) retain the large/small separation — large ranges 27–64%, the others below 9.4%.

C.3. Full-background six-query surrogate validation

The six-query study model shares the conservative parameterization of Appendix B.3 with a full 5,280-path head: the input is the six-dimensional study demand vector (the other 522 ODs stay at published demand), z is the same 3×128 tanh MLP on standardized demands, and $f = \Delta h$ conserves exactly by construction. It is pretrained flow-only under the same sampling principle as the zero-background surrogate (the six study demands scaled by i.i.d. LogNormal$(0, 0.5^2)$ factors clipped to $[0.2, 2]$, 2500/857/857 split; unscaled MSE; Adam at 3×10^{-3}, minibatch 256, at most 12,000 steps, validation every 50), and the minimum-validation checkpoint of each seed $0, \ldots, 4$ supplies the five common initial models continued by every objective of Section 7; the test split enters only final reporting. This flow-only model already reaches held-out flow errors of 0.040% (seed median) while its attribution-column errors remain large (medians 7.4–81.2% across the six queries): accurate flows do not by themselves yield accurate spatial explanations — the observation that motivates the response-supervised protocol. The two-term split of Eq. (C.1) locates this error in the response term; complete validation records, conservation residuals and across-seed envelopes are included in the released result arrays.

C.4. Sensitivity to candidate paths and route-choice dispersion

The path-set comparison expands all 528 OD sets to twenty candidates each (the first ten unchanged), with demands, BPR functions, $\theta = 1$, and 41 quadrature points fixed: across the eight queries, absolute unrestricted and connected ranges change by at most 0.977% and 1.011%, with the three largest internal-node terms keeping their ordered node IDs (moving at most 1.43%). The $\theta \in \{0.5, 2\}$ check splits the specified $(10, 16)$ boundary difference into study/background components $-7{,}404.9/-2{,}371.5$ at $\theta = 0.5$ and $+115.3/+235.7$ at $\theta = 2$; these are different fixed-path models, not convergence to unrestricted SUE. For all three additional models, implicit derivative columns match re-solved half-step central differences at the demand-path midpoint (maximum 0.00249%, completeness gap 2.85×10^{-5}).

C.5. Six-query interval, mechanism, boundary-family, and scenario checks

Extremal ranges and negative shares. Unrestricted extrema follow the closed form (15), connected extrema the same enumeration as the connected ranges; the negative shares in Section 6.2 are exact counts over the connected families (4,621–16,943 admissible sets per query), and all displayed values come from the 41-point columns. Scenario robustness: under twenty candidate paths, $\theta \in \{0.5, 2\}$ and the single-OD $0.2 \to 1.6$ re-run, the connected ranges of the three sign-crossing queries move by at most a few percent (e.g. for $(1, 19)$, the connected range moves from the ten-path baseline $[-13{,}502,\ 24{,}260]$ to $[-13{,}451,\ 24{,}276]$ under twenty paths, and to $[-13{,}024,\ 24{,}064]$ at ten paths under $\theta = 0.5$; unrestricted intervals within 1% of the baseline row), and the single-OD small-interval bands narrow to $[-1{,}947,\ 3{,}494]$, $[-1{,}201,\ 5{,}838]$ and $[-2{,}517,\ 8{,}608]$ at $\pm 10\%$ with every C_j positive ($+7{,}202$ to $+33{,}734$; the control $(6, 8)$ stays sign-consistent throughout). The complete range table accompanies the result arrays. Table C.3 reports, for each sign-crossing query, the connected minimizing boundary U_- at the baseline setting: its attribution, and the interior divergence sum $\phi_j(\{s_j\}) - \phi_j(U_-)$, decomposed as the table caption specifies, at the same joint state and gradient as the reported columns (additivity $\leq 1.2 \times 10^{-10}$). By Eq. (16), the negative divergence captured inside U_- (4,664.9, 6,960.1, and 19,811.5 for the three queries) covers 83%, 88%, and 72% of the gaps $\phi_j(U_-) - \phi_j^{\min}$, while the captured positive divergence (19,938.0, 18,194.2, 46,901.4) is 95.5%, 94.8%, and 85.7% of the query totals. The sign change arises differently across queries: study response dominates for $(1, 19)$ (largest node 15: own $+14{,}464.0$), study- and background-dominated nodes share the sign for $(1, 15)$, and both contribute about equally for $(7, 12)$ (node 8). The extremal node sets U_- and U_+ (one-based node IDs; $U_-(1, 19)$ is the shaded region of Figure C.2) accompany the result arrays.

Neighborhood families. Origin-centered families built from undirected hop distance or shortest free-flow distance (attained radii, destination excluded, two-sided connected) contain 4–14 admissible sets per

Table C.3
Connected minimizing boundary U_- at the baseline setting for the three sign-crossing queries, in TSTT units. Interior is the divergence sum over $U_- \setminus \{s_j\}$, decomposed into the query's own response, the other five query ODs, and the background response; the study group of the main text is the sum of the first two columns. Entries are rounded independently.

OD	$\phi_j(\{s_j\})$	$\phi_j(U_-)$	Interior	Own OD	Other query	Background
$(1, 19)$	1,771.6	−13,501.6	15,273.1	19,480.9	−59.4	−4,148.3
$(1, 15)$	2,948.1	−8,286.0	11,234.1	7,856.9	−26.7	3,404.0
$(7, 12)$	9,473.8	−17,616.2	27,089.9	14,002.3	−3.2	13,090.8

query and have positive ranges with minima at the origin cut (1,771.6 to 15,803.5 for $(1, 19)$, 2,948.1 to 26,602.0 for $(1, 15)$, 9,473.8 to 32,139.4 for $(7, 12)$); for short-path queries $(7, 18)$, $(4, 5)$ and $(6, 8)$ only the origin cut is admissible.

Straight screenlines. Coordinates use an equirectangular frame (longitude scaled by the cosine of mean latitude). Section 5.4's 36-direction construction, with duplicate regions removed, gives 35, 152, 32, 123, 23 and 148 regions in Table 4 order. All attributions agree with direct incidence aggregation below 10^{-9}. Refining directions from 5° to 1° leaves both negative minima unchanged; negative shares change from 1.3% to 1.8% for $(1, 19)$ and 6.8% to 6.2% for $(7, 12)$. Other queries stay positive; only $(7, 18)$'s minimum changes, from 642.4 to 641.4.

Model-setting and single-OD checks. Re-solving the protocol of Appendix B.5 under twenty candidate paths per OD, $\theta \in \{0.5, 2\}$, and the single-OD demand paths ($0.2 \to 1.6$ of published demand, one moving coordinate; C_j within 3.9%, staying positive at 50,322.2, 77,672.6, 117,476.8 for the three crossing queries) leaves every crossing query's connected range straddling zero, with extrema attained by exactly the joint-scenario sets U_- and U_+ (changes at most 0.9% and 0.5%) and the control $(6, 8)$ sign-consistent throughout; solver and finite-difference checks match Appendix B.5 levels, and the 41-point check ($(1, 19)$, $0.8 \to 1.2$) reproduces the endpoint column to 1.3×10^{-5} TSTT units per edge.

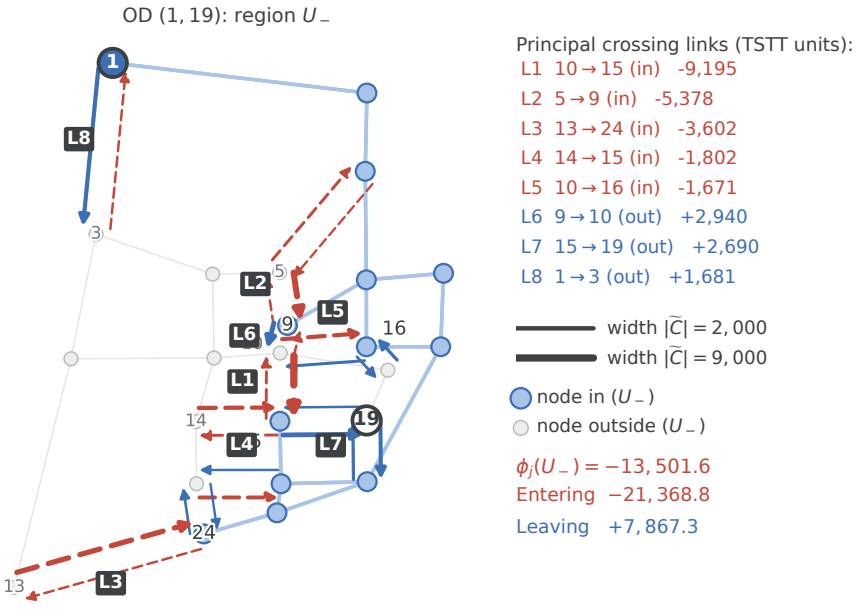


Figure C.2: The connected minimizing region U_- for OD $(1, 19)$ (responsive-background six-query protocol). Shaded nodes form U_- (origin node 1 ringed; destination 19 outside). Crossing-link arrows keep the link direction; width scales with $|c_{e,j}|$, and color and style encode the signed contribution to $\phi_j(U_-)$, $a_{e,j} = \sigma_e c_{e,j}$ ($\sigma_e = +1$ leaving, -1 entering; solid blue positive, dashed red negative), so a positive attribution on an entering link appears red. The side panel lists, in TSTT units, the five largest negative and three largest positive contributions among the 26 crossing links (L1–L5 negative, L6–L8 positive, "(in)"/"(out)" marking entering and leaving links), with the width scale and the membership legend.

C.6. Sign preservation on the six-query baseline segment

On the six-query connected families, the five flow-supervised pretrained models (Section B.3) show five-seed median largest-boundary disagreements of 4.2–6.9% of the direct connected span for four queries, rising to 99.5% for the small-response query $(4, 5)$ and beyond the span for $(7, 18)$ — flow-only supervision does not control boundary error. Sign reliability, by contrast, is boundary specific: all five seeds retain the three connected-family sign crossings and positive system contributions, and at the fixed direct-model minimizing screenline every seed keeps the negative sign for $(1, 19)$ and $(7, 12)$ (seed ranges $[-3886.6, -3365.1]$ and $[-2810.4, -1916.6]$ against references -2766.6 and -2967.3, with largest family-wide errors below the reference sign margins). These are screenline-specific certificates of sign preservation, not of column magnitudes; the per-query screenline table and across-seed envelopes are included in the released result arrays.

C.7. Numerical diagnosis of the transferability hierarchy

The universal structural condition quantifies over all conservative responses; Section 3.2 introduced the model-specific level that reasons on the path-feasible response envelope of the assignment model's fixed path sets. Table C.4 evaluates the hierarchy numerically. On the Figure-1 network with its published post-merge weights, the universal condition fails because the off-path pendant atom is heterogeneous, yet the model-specific tests hold exactly for the all-simple-paths structure: every rerouting that model can make preserves the common boundary value, so restricting attention to the path-feasible envelope restores transferability that the universal condition denies. On Sioux Falls the two levels coincide — the full-background fixed-path model spans the complete circulation space through path exchanges, so its path-feasible envelope contains every conservative response (Appendix B.1) — and both fail at realistic TSTT weights, leaving the realized summaries of Table 4 as the operative output. The diagnosis is stable across load levels: the normalized universal violation stays within 2.08–2.21 from the 0.7 to the 1.3 baseline while ρ_{swap} falls from 1.12 to 0.80 as congestion grows. On Anaheim the universal test fails more decisively (normalized violation 10.08) and

Table C.4
Numerical diagnosis of the transferability hierarchy of the four levels of Section 3.2. η_{all} and η_{conn}: weight spread within all atoms / within the queried atoms, normalized by the mean weight. ρ_{swap}: weighted-divergence leakage of unit rerouting directions relative to a unit reference response; $\rho_{\text{swap}} = 0$ certifies $BDN = 0$, and together with zero off-span residuals $\rho_{\text{ref},j}$ it certifies both conditions of the model-specific level. On Sioux Falls (level 1.0) three of the six queries have $\rho_{\text{ref},j} = 0$; the other three lie in 0.83–0.97. Realized: median normalized geometric common-summary error of the responsive-background six queries. Synthetic: published post-merge weights; Sioux Falls: TSTT weights at the level-1.0 equilibrium, where the path-exchange space attains the full circulation dimension 53.

Network	η_{all}	η_{conn}	ρ_{swap}	Realized (%)
Figure-1 synthetic	0.61	0	0	0 (model responses)
Sioux Falls	2.08	=	0.88	32.3

the model-specific test fails with it ($\rho_{\text{swap}} = 2.18$ over the 404-dimensional exchange space), so there too the realized layer carries the reporting.

C.8. Additional learning comparisons

The attribution-column error of a learned model splits exactly into a weight-shift and a response term (L/S: learned/direct in Eq. (2)):

$$c_j^L - c_j^S = \Delta q_j \int_0^1 (D_\lambda^L - D_\lambda^S) J_j^S \, \mathrm{d}\lambda + \Delta q_j \int_0^1 D_\lambda^L (J_j^L - J_j^S) \, \mathrm{d}\lambda, \tag{C.1}$$

satisfied entrywise within 10^{-10} TSTT units on the 96 test paths; the weight-shift term has querywise median norm at most 0.2% of the reference column and the response term is larger in every seed–query unit.

Six objectives under a fair weight search. Table C.5 completes the objective comparison under the shared continuation budget and the common validation-only weight search (Appendix B.9). Each objective's auxiliary term is normalized by a scale carrying that objective's own units — in particular the weighted-response term by its reference scale $v_j^2 = \langle \|D^S J_j^S\|_2^2 \rangle_j$ rather than the unweighted scale — so that a common weight grid explores comparable regularization ranges; under this normalization every objective trains at a properly selected weight. Weighted-response supervision reaches a column error of 15.21% at its selected weight, improves 84 of 96 paths, and trains across the whole low-to-mid weight range; the unweighted form is the simplest member of the family and attains the lowest errors on every metric. With validation-selected weights, both endpoint objectives improve attribution accuracy relative to flow-only training (21.18% → 16.2% in column error), although their errors remain above those of the response-supervised objectives. The boundary-seminorm objective shows the largest validation–test gap: its validation boundary error decreases monotonically with the weight up to the selected $\lambda^* = 0.3$ and reaches the lowest validation value of any objective, yet its test column error at that weight stays above flow-only training (24.18%, improved on only 8 of 96 paths). As Section 7 discusses through the seminorm kernel, optimizing the boundary value alone does not repair the underlying response field, whereas supervising that field improves every spatial error metric. †The endpoint-cut bundle is the endpoint-edge objective plus a boundary term at a fixed ratio of 0.01; it is reported as an ablation of that fixed mixture ratio, not as an independently tuned competitor.

C.9. Group-level uncertainty for the 96-path comparison

Path-level bootstrap intervals for the paired median differences (10,000 resamples; column error / family-boundary error, in percentage points): single-query group $[-4.90, -2.95]$ / $[-1.99, -0.88]$ improving 24/24 and 23/24 paths; two-query group $[-6.48, -4.31]$ / $[-3.15, -1.52]$, 22/24 and 21/24; three-query group $[-9.26, -3.36]$ / $[-4.11, -1.36]$, 22/24 and 22/24; six-query group $[-8.59, -5.22]$ / $[-3.86, -1.73]$, 24/24 and 24/24. Leave-one-path-out medians of the paired column difference vary within $[-5.11, -4.94]$ and of the paired family-boundary difference within $[-2.11, -2.00]$, so no single path carries the result. The system-contribution difference has median $+0.45$ percentage points with interval $[0.01, 0.74]$, improving 39 of 96 paths, while both methods retain system errors below 3.5% (medians 2.52% and 3.13%).

Table C.5
Six supervision objectives on the same 96 test demand paths, each at its validation-selected weight λ^* under the shared continuation budget and uniform selection rule (Appendix B.9); the weighted-response objective is normalized by its own reference scale v_j. Entries are medians of path-level values; improved: paths on which the objective lowers the column (family-boundary) error relative to flow-only training.

Objective	λ^*	Flow pred. (%)	Column (%)	Family boundary (%)	Improved vs. flow	
					col.	fam.
Flow only	—	0.032	21.18	8.74	—	—
Response supervision	1	0.029	14.19	5.75	92/96	90/96
Weighted response	0.1	0.032	15.21	6.93	84/96	79/96
Boundary seminorm	0.3	0.037	24.18	7.21	8/96	45/96
Endpoint-edge	0.3	0.035	16.23	7.46	72/96	66/96
+ cut term†	0.3	0.035	16.26	7.47	72/96	67/96

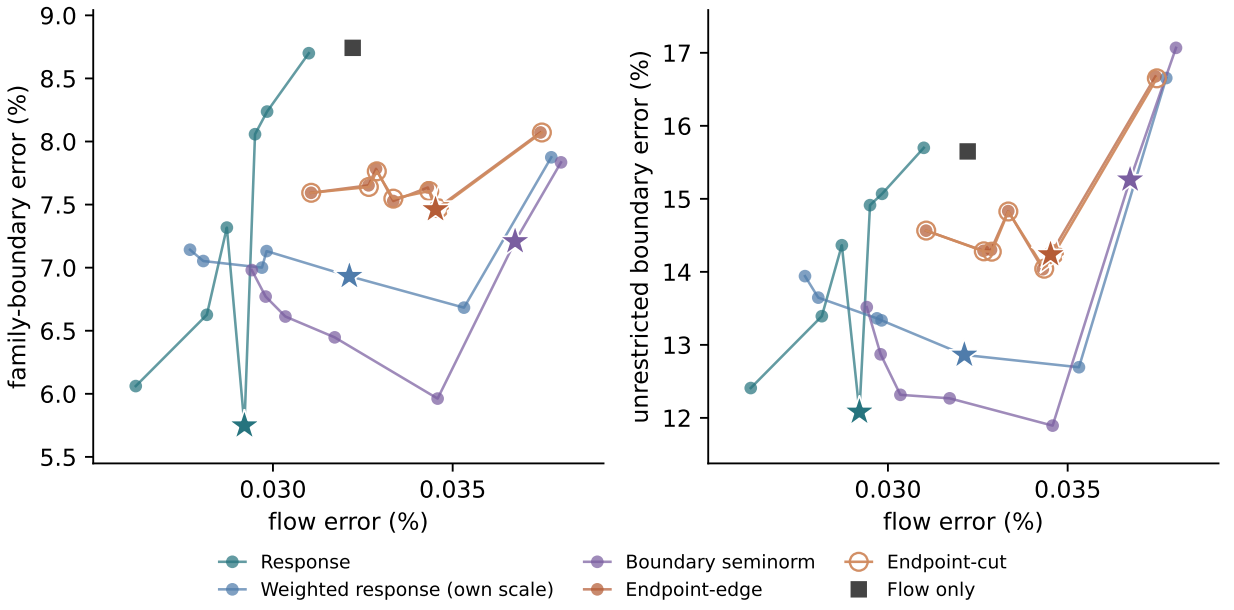


Figure C.3: Explanation–flow trade-off of the six supervision objectives across the weight grid, on the 96 Sioux Falls test paths. Each point is one (objective, weight) configuration: median flow error against median family-boundary error (left) and unrestricted boundary error (right); the validation-selected weight of each objective is starred. Response supervision sweeps the efficient frontier in both panels, interpolating between flow-only behavior and the lowest boundary error. Under its own scale normalization the weighted-response objective tracks the same band across the low-to-mid weights; only the largest weight violates the validation flow-accuracy constraint and selects the initial checkpoint. The boundary-seminorm objective reaches its best errors only at unselected intermediate weights, and the endpoint objectives (nearly coincident; the cut term is a fixed-ratio minor addition) trace interior curves.

The incremental-flow error of Table 6 uses the same aggregation as the other columns (medians of path-level values, active queries and seeds averaged within each path), and the reference increments are well conditioned: the smallest reference increment norm is 45 veh/h on Sioux Falls and 508 veh/h on Anaheim against medians of 258 and 1,672. Response supervision therefore improves not only the boundary attributions but also the incremental flow response from which they are integrated, while the total-flow errors stay near 0.03% because total flows are dominated by the unchanged background.

Figure C.4 reports the validation curves behind the extended-budget comparison of Section 7: on both networks the seed-median response curve stays below the flow curve at every validation point, and both continue to decrease at the 3,600-update cap.

Table C.6
Per-seed medians of path-level values on the 96 paths and path-level improvement counts (response vs. flow supervision).

	Column (%)		Geometric (%)		Improved paths	
Seed	flow	resp.	flow	resp.	col.	geo.
0	19.55	15.93	8.40	5.67	85/96	84/96
1	15.66	12.44	7.07	4.57	80/96	84/96
2	22.81	14.91	9.08	6.52	88/96	89/96
3	22.33	14.95	7.40	5.98	87/96	84/96
4	16.65	12.02	6.57	4.23	87/96	91/96

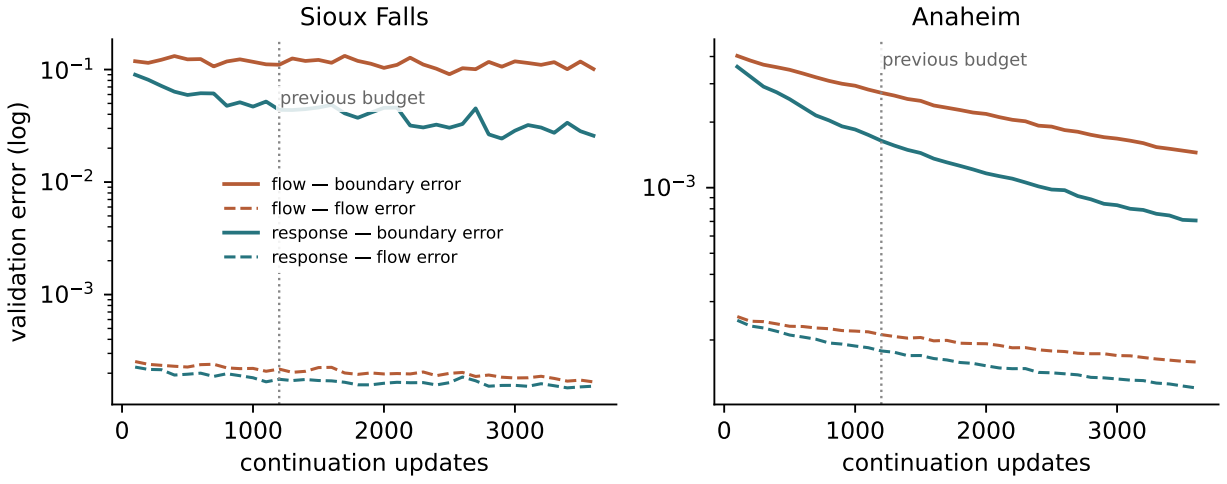


Figure C.4: Validation convergence under the extended continuation budget (seed-median validation boundary error, solid, and relative flow error, dashed; log scale): Sioux Falls (left) and Anaheim (right), flow-only (ochre) versus response supervision (teal). The dotted line marks the 1,200-update budget of the main comparison; both objectives keep improving beyond it, and the response curves stay below the corresponding flow curves throughout.

C.10. Unrestricted and integrated-bound errors by group

Table C.7 completes Table 6 with the unrestricted and integrated-bound columns of Eq. (20), as percentages of the reference column's boundary seminorm; the geometric and unrestricted columns use different denominators and are not one inequality, and the bound chain $\mathcal{K}_j(e_j) \le B_j$ holds in absolute units for the shared quadrature (Section 7).

Table C.7
Unrestricted (exact $\mathcal{K}_j(e_j)$ over all separating regions) and bound (integrated bound of Eq. (20)) errors, medians of path-level values on the 96 test paths, as percentages of the reference column's boundary seminorm.

	Unrestricted (%)		Bound (%)	
Active	flow	resp.	flow	resp.
1	4.54	2.95	4.62	3.00
2	15.48	12.65	15.67	12.80
3	16.36	12.39	16.70	12.81
6	19.43	12.28	20.61	13.48
All 96	15.65	12.08	16.34	12.83

C.11. Complete input-stratified common-summary results

Positive non-diagonal published demands are stratified by demand and directed free-flow shortest-path time using empirical tertiles, with ties kept in the lower bin. Within each of nine cells, lexicographically sorted OD indices at zero-based positions $\lfloor N/3 \rfloor$ and $\lfloor 2N/3 \rfloor$ are retained. No attribution values enter selection. Endpoint differences identify total edge attribution, without identifying the same-state study/background response split. The two perturbation widths share every field of the selection protocol and differ only in the demand interval; their normalized errors are close for every query, so Table C.8 reports both in one table. Each query's common value and error follow from the tabulated range by the midpoint rule $\beta = (\phi_{\min} + \phi_{\max})/2$; absolute minima, maxima, midpoints and half-ranges for both widths, the complete normalized interval set, and the per-position boundary-value sequences behind the segmentation of Proposition 10 accompany the result arrays in the code package.

Table C.8
Complete geometric common summaries for the 18 input-stratified queries at both widths. Normalized error is Eq. (24) at $\pm 10\%$ and $\pm 20\%$; ranges are $[\phi_{\min}, \phi_{\max}]$ at $\pm 20\%$ in Sioux Falls cost units; P and M mean positive and mixed signs at $\pm 20\%$.

OD	System	Range ($\pm 20\%$)	Norm. err. ($\pm 10\%$)	Norm. err. ($\pm 20\%$)	Sign
(4,3)	507.2	[429.0, 535.0]	9.9	9.9	P
(12,3)	337.2	[330.7, 340.2]	1.4	1.4	P
(6,1)	2360.6	[658.3, 2290.4]	35.6	35.6	P
(15,6)	9390.2	[415.4, 4433.5]	45.3	45.3	P
(6,21)	4577.8	[-3.4, 2349.4]	50.1	50.1	M
(18,14)	3422.3	[150.8, 1929.7]	46.1	46.1	P
(8,4)	25097.5	[1270.7, 15741.3]	46.0	46.0	P
(19,8)	16939.7	[733.2, 7266.4]	45.0	45.0	P
(11,6)	6165.9	[1236.4, 6106.4]	39.9	39.9	P
(16,2)	17763.1	[896.3, 9628.6]	45.3	45.3	P
(8,21)	7605.5	[752.6, 4214.1]	41.1	41.1	P
(14,8)	17445.3	[890.3, 7527.1]	44.1	44.1	P
(11,15)	47925.5	[17628.9, 27737.8]	18.2	18.2	P
(17,9)	24040.9	[5482.3, 18519.3]	35.2	35.2	P
(10,12)	55214.5	[9822.4, 33248.6]	35.2	35.2	P
(13,10)	54411.9	[2918.8, 31345.2]	45.3	45.3	P
(1,10)	33917.2	[2182.9, 20856.4]	44.8	44.8	P
(10,1)	34264.5	[2182.7, 21039.5]	44.8	44.8	P

Table C.9
Tolerance sensitivity of the reported summaries, within fixed projection directions (the input-stratified selection experiment for Sioux Falls, 18 queries per level, 36 normals with duplicate chains removed; the Anaheim ordered-family experiment for Anaheim, six single-direction queries). Single: number of queries whose whole family admits one common value at the stated tolerance; segments: median per-direction count on Sioux Falls at load levels 0.7/1.0/1.3 (maximum over directions in parentheses) or per-query range on Anaheim; compression: $1-$segments/regions, median per direction (Sioux Falls) or per query (Anaheim).

Tolerance τ	Sioux single (0.7/1.0/1.3)	Sioux med. seg./dir.	Anaheim segments	Anaheim compression
2.5%	2/1/0	4.0 (10)/4.5 (12)/5.5 (13)	6–21	91.5%
5%	2/1/0	3.0 (7)/4.0 (8)/4.5 (10)	4–14	93.8%
7.5%	2/1/1	2.0 (7)/3.0 (8)/3.0 (8)	3–12	95.5%
10%	2/2/1	2.0 (6)/2.0 (6)/2.5 (7)	3–12	96.4%

Tolerance rule and small-magnitude detail. The absolute rule is absolute: $\epsilon_j = 5\%$ of the family's largest attribution magnitude, so it permits errors much larger than 5% of the small attributions. The mixed rule $|\hat{a}_i - a_i| \le \epsilon_{\text{abs}} + \tau |a_i|$ keeps relative control at small magnitudes; a candidate segment admits a common representative iff $\max_{i \in I}(a_i - t_i) \le \min_{i \in I}(a_i + t_i)$ with $t_i = \epsilon_{\text{abs}} + \tau |a_i|$, feasibility is inherited by subintervals, and the same greedy returns the minimum count. On Anaheim, at the absolute rule the worst relative error at small-magnitude positions (attributions below 10% of the family maximum) has median 0.55 and maximum

Table C.10
Attribution-path robustness on the responsive-background six-query protocol: straight-line joint path versus 22 one-at-a-time orders (canonical, reverse, 20 seeded permutations). C_j range: minimum and maximum over the orders; deviation relative to the straight-line value. ε shift: median normalized common-summary error over the orders against the straight-line value. Sign: orders whose family range keeps the straight-line sign-crossing status; mechanism: orders whose largest-magnitude component at U_- matches the straight line's.

OD	C_j range	Max $\lvert C_j\rvert$ dev.	ε shift	Sign	Mechanism
$(7, 18)$	[666.5, 678.4]	1.0%	$4.3\% \to 4.3\%$	22/22	—
$(1, 19)$	[48,451.9, 51,817.7]	3.4%	$56.3\% \to 56.4\%$	22/22	22/22
$(4, 5)$	[2,780.8, 2,911.4]	3.0%	$8.2\% \to 8.2\%$	22/22	—
$(1, 15)$	[75,306.6, 79,553.2]	2.8%	$49.3\% \to 49.3\%$	22/22	22/22
$(6, 8)$	[72,157.8, 75,713.7]	2.4%	$15.4\% \to 15.5\%$	22/22	—
$(7, 12)$	[117,367.2, 117,573.5]	0.1%	$52.6\% \to 52.5\%$	22/22	22/22

0.94; the mixed rule with $\epsilon_{\text{abs}} = 1\%$ of the family maximum and $\tau = 10\%$ lowers these to 0.28 and 0.45 while the median segment count rises from 10 to 12 (per-query range 6–17), i.e. the compression is essentially retained and the detail at small-magnitude positions doubles in accuracy. Enlarging the relative part of the absolute 5% absolute rule without shrinking the absolute floor ($\epsilon_{\text{abs}} = 5\%$, $\tau \geq 5\%$) instead reduces segment counts by merging large-magnitude runs while the small-position error worsens — confirming that the absolute floor, not the relative part, controls the small-magnitude detail.

Load-level protocol and node-imbalance diagnostics. The load-level experiment solves one equilibrium per level (0.7, 1.0, 1.3 times the published demands), the focal query moving alone over $\pm 20\%$ under the identical geometric family. Across the 54 query–level pairs, the node-imbalance norm $\|d_j\|_2$ of Eq. (13) rises 13.2-fold from the light to the heavy baseline (medians 1196.4 to 15763.3), tracks the geometric-family spread and the unrestricted range S_j with rank correlation 0.99, and the realized ratio $S_j/\|d_j\|_2$ has median 2.53 against the Cauchy–Schwarz bound $\sqrt{n-2} = 4.69$ of Eq. (16) — mechanism-consistent readings of Eq. (14) in which the load levels change realized weights, responses and demand increments together.

C.12. Computational protocol for alternative attribution orders

Each of the 22 alternative orders of Section 6.2 changes the six study ODs one at a time (41 quadrature states per segment, identical solver tolerance, endpoints and boundary family as the straight-line run; canonical order, reverse order, and twenty permutations under seed 20261031). Numerical checks: straight-line columns reproduce the stored reference arrays to 1.7×10^{-15} relative; every order's endpoint equilibria match the straight-line endpoints to 6.1×10^{-9}; the path-independent total outcome change reproduces to 1.5×10^{-7} relative. Each order's record stores the attribution column, family range, common summary and normalized error, sign-crossing flag, maximum profile deviation against the straight-line attributions, and mechanism components at the fixed regions U_- (arrays in the code package). The dominant mechanism at U_- coincides with the straight line's on all 22 orders for each of the three mechanism-tracked queries.

C.13. Anaheim learned-assignment protocol

The Anaheim experiment mirrors the Sioux Falls protocol in structure; differences are stated here once. The study queries are the six largest published OD demands of Section 6.1; all remaining positive demands are fixed with endogenous route choice over the same ten-path centroid-restricted structure. The model is the conservative parameterization of Eq. (25) with $K = 1{,}406$ OD softmaxes over $P = 14{,}060$ paths and $m = 914$ link outputs; the input is the six-dimensional study demand vector. Reference labels: 1,000 flow states ($q \sim U(0.65, 1.3)\, q_{\text{base}}$; 800 train, 200 validation), 480 derivative centers (384 train, 96 validation) with equilibrium sensitivities of the six study columns, and 48 test demand paths (twelve per active-query group $k \in \{1, 2, 3, 6\}$, mixed and same signs, generation rule identical to the Sioux Falls test paths under a fresh seed) solved at a 21-state quadrature with sensitivities at every state. Training: stage A pretrains flow-only from scratch for up to 4,000 steps (Adam, learning rate 3×10^{-3}, batch 128, per-link scaled MSE, best-validation selection every 50 steps); stage B continues for 1,200 steps with the method objective exactly as in the Sioux Falls protocol (learning rate 3×10^{-5}, batch 32 flow samples plus 8 derivative centers, rotation over the six study columns), methods *flow* and *response*, five seeds. Validation: flow relative error on the 200

held-out flow states; K-MSE, the boundary-seminorm of the weighted response discrepancy normalized per query, on the 96 held-out centers. Selection: minimum validation K-MSE among states with validation flow error at most 1.1 times the stage-A checkpoint, else minimum flow error; test paths enter neither training nor selection. Selected checkpoints cluster at the training cap, as in Sioux Falls, so the operating point is budget-limited rather than validation-plateaued; the extended-budget replication below triples the cap under the identical rule. Numerical checks: conservation of every reference state (study plus background supply) holds to 9.1×10^{-12} vehicle-units; reference-column completeness relative to the endpoint outcome change is at most 1.2×10^{-5}; the total error of every reported summary stays within its budget. Costs: reference data 2,488 equilibrium solves with sensitivities ($\approx$35 s per state single-threaded, parallelized over twelve workers), pretraining $\approx$40 minutes per seed, fine-tuning $\approx$5–9 minutes per (method, seed).

Extended budget. The convergence check of Section 7 repeats the stage-B continuation for both methods and five seeds with the budget tripled to 3,600 updates: identical initialization, learning rate, batches, label sets, validation cadence and selection rule, and the same test paths for the final report. The extension is an exact continuation of the shared protocol on Sioux Falls, where the 3,600-update runs repeat the 1,200-update stage with the same initialization and minibatch sequence. Every Anaheim run selects beyond the original cap (all five response seeds at 3,600 updates); the 3,600-update fine-tuning takes $\approx$30–105 minutes per (method, seed) single-threaded on Anaheim and $\approx$4–6 minutes per run on Sioux Falls.